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The A1C Test & Diabetes

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What is the A1C test?

The A1C test is a blood test that provides information about your average levels of blood glucose, also called blood sugar, over the past 3 months. The A1C test can be used to diagnose type 2 diabetes and prediabetes.¹ The A1C test is also the primary test used for diabetes management.

A health care professional draws blood.

An A1C test is a blood test that reflects your average blood glucose levels over the past 3 months.

The A1C test is sometimes called the hemoglobin A1C, HbA1c, glycated hemoglobin, or glycohemoglobin test. Hemoglobin is the part of a red blood cell that carries oxygen to the cells. Glucose attaches to or binds with hemoglobin in your blood cells, and the A1C test is based on this attachment of glucose to hemoglobin.

The higher the glucose level in your bloodstream, the more glucose will attach to the hemoglobin. The A1C test measures the amount of hemoglobin with attached glucose and reflects your average blood glucose levels over the past 3 months.

The A1C test result is reported as a percentage. The higher the percentage, the higher your blood glucose levels have been. A normal A1C level is below 5.7 percent.

Why should a person get the A1C test?

Testing can help health care professionals

find prediabetes and counsel you about lifestyle changes to help you delay or prevent type 2 diabetes

find type 2 diabetes

work with you to monitor the disease and help make treatment decisions to prevent complications

If you have risk factors for prediabetes or diabetes, talk with your doctor about whether you should be tested.

Couple walking with their dog.

You may be able to prevent or delay type 2 diabetes with lifestyle changes such as weight loss or being physically active most days of the week.

How is the A1C test used to diagnose type 2 diabetes and prediabetes?

Health care professionals can use the A1C test alone or in combination with other diabetes tests to diagnose type 2 diabetes and prediabetes. You don't have to fast before having your blood drawn for an A1C test, which means that blood can be drawn for the test at any time of the day.

If you don't have symptoms but the A1C test shows you have diabetes or prediabetes, you should have a repeat test on a different day using the A1C test or one of the other diabetes tests to confirm the diagnosis.²

A1C results and what the numbers mean

*Any test used to diagnose diabetes requires confirmation with a second measurement, unless there are clear symptoms of diabetes.

Diagnosis*	A1C Level
Normal	below 5.7 percent
Prediabetes	5.7 to 6.4 percent
Diabetes	6.5 percent or above

When using the A1C test for diagnosis, your doctor will send your blood sample taken from a vein to a lab that uses an NGSP-certified method. The NGSP External link, formerly called the National Glycohemoglobin Standardization Program, certifies that makers of A1C tests provide results that are consistent and comparable with those used in the Diabetes Control and Complications Trial.

Blood samples analyzed in a doctor's office or clinic, known as point-of-care tests, should not be used for diagnosis.

The A1C test should not be used to diagnose type 1 diabetes, gestational diabetes, or cystic fibrosis NIH external link-related diabetes. The A1C test may give false results in people with certain conditions.

Having prediabetes is a risk factor for developing type 2 diabetes. Within the prediabetes A1C range of 5.7 to 6.4 percent, the higher the A1C, the greater the risk of diabetes.

Is the A1C test used during pregnancy?

Health care professionals may use the A1C test early in pregnancy to see if a woman with risk factors had undiagnosed diabetes before becoming pregnant. Since the A1C test reflects your average blood glucose levels over the past 3 months, testing early in pregnancy may include values reflecting time before you were pregnant. The glucose challenge test or the oral glucose tolerance test (OGTT) are used to check for gestational diabetes, usually between 24 and 28 weeks of pregnancy. If you had gestational diabetes, you should be tested for diabetes no later than 12 weeks after your baby is born. If your blood glucose is still high, you may have type 2 diabetes. Even if your blood glucose is normal, you still have a greater chance of developing type 2 diabetes in the future and should get tested every 3 years.

Can other blood glucose tests be used to diagnose type 2 diabetes and prediabetes?

Yes. Health care professionals also use the fasting plasma glucose (FPG) test and the OGTT to diagnose type 2 diabetes and prediabetes. For these blood glucose tests used to diagnose diabetes, you must fast at least 8 hours before you have your blood drawn. If you have symptoms of diabetes, your doctor may use the random plasma glucose test, which doesn't require fasting. In some cases, health care professionals use the A1C test to help confirm the results of another blood glucose test.

Can the A1C test result in a different diagnosis than the blood glucose tests?

Yes. In some people, a blood glucose test may show diabetes when an A1C test does not. The reverse can also occur—an A1C test may indicate diabetes even though a blood glucose test

does not. Because of these differences in test results, health care professionals repeat tests before making a diagnosis.

People with differing test results may be in an early stage of the disease, when blood glucose levels have not risen high enough to show up on every test. In this case, health care professionals may choose to follow the person closely and repeat the test in several months.

Why do diabetes blood test results vary?

Lab test results can vary from day to day and from test to test. This can be a result of the following factors:

Blood glucose levels move up and down

Your results can vary because of natural changes in your blood glucose level. For example, your blood glucose level moves up and down when you eat or exercise. Sickness and stress also can affect your blood glucose test results. A1C tests are less likely to be affected by short-term changes than FPG or OGTT tests.

The following chart shows how multiple blood glucose measurements over 4 days compare with an A1C measurement.

Blood Glucose Measurements Compared with A1C Measurements over 4 Days

View full-sized imageGraph shows blood glucose in mg/dL on the y-axis and day of the week on the x-axis. A black line straight across the center shows an A1C of 7.0%. A blue line starts at 145 mg/dL and rises and falls, with the lowest level at 140 mg/dL and the highest at 175 mg/dL pre-lunch on Monday. Each day, the fasting or pre-breakfast level is the lowest level.

Blood glucose (mg/dL) measurements were taken four times per day (fasting or pre-breakfast, pre-lunch, pre-dinner, and bedtime).

The straight black line shows an A1C measurement of 7.0 percent. The blue line shows an example of how blood glucose test results might look from self-monitoring four times a day over a 4-day period.

A1C tests can be affected by changes in red blood cells or hemoglobin

Conditions that change the life span of red blood cells, such as recent blood loss, sickle cell disease [NIH external link](#), erythropoietin treatment, hemodialysis, or transfusion, can change A1C levels.

A falsely high A1C result can occur in people who are very low in iron; for example, those with iron-deficiency anemia [NIH external link](#). Other causes of false A1C results include kidney failure or liver disease.

If you're of African, Mediterranean, or Southeast Asian descent or have family members with sickle cell anemia or a thalassemia [NIH external link](#), an A1C test can be unreliable for diagnosing or monitoring diabetes and prediabetes. People in these groups may have a different type of hemoglobin, known as a hemoglobin variant, which can interfere with some A1C tests. Most people with a hemoglobin variant have no symptoms and may not know that they carry this type of hemoglobin. Health care professionals may suspect interference—a falsely high or low result—when your A1C and blood glucose test results don't match.

Doctor sharing test results with patient.

If you're of African, Mediterranean, or Southeast Asian descent, you could have a different type of hemoglobin that affects your diabetes care.

Not all A1C tests are unreliable for people with a hemoglobin variant. People with false results from one type of A1C test may need a different type of A1C test to measure their average blood glucose level. The NGSP provides information for health care professionals about which A1C tests are appropriate to use for specific hemoglobin variants [External link](#).

Read about diabetes blood tests for people of African, Mediterranean, or Southeast Asian descent. The NIDDK has information for health care providers on Sickle Cell Trait & Other Hemoglobinopathies & Diabetes.

Small changes in temperature, equipment, or sample handling

Even when the same blood sample is repeatedly measured in the same lab, the results may vary because of small changes in temperature, equipment, or sample handling. These factors tend to affect glucose measurements—fasting and OGTT—more than the A1C test.

Health care professional shares test results with woman.

Your health care professional can help you understand your test results.

Health care professionals understand these variations and repeat lab tests for confirmation. Diabetes develops over time, so even with variations in test results, health care professionals can tell when overall blood glucose levels are becoming too high.

How precise is the A1C test?

When repeated, the A1C test result can be slightly higher or lower than the first measurement. This means, for example, an A1C reported as 6.8 percent on one test could be reported in a range from 6.4 to 7.2 percent on a repeat test from the same blood sample.³ In the past, this range was larger but new, stricter quality-control standards mean more precise A1C test results.

Health care professionals can visit ngsp.org External link to find information about the precision of the A1C test used by their lab.

How is the A1C test used after diagnosis of diabetes?

Your health care professional may use the A1C test to set your treatment goals, modify therapy, and monitor your diabetes management.

Experts recommend that people with diabetes have an A1C test at least twice a year.⁴ Health care professionals may check your A1C more often if you aren't meeting your treatment goals.⁴

What A1C goal should I have?

People will have different A1C targets, depending on their diabetes history and their general health. You should discuss your A1C target with your health care professional. Studies have

shown that some people with diabetes can reduce the risk of diabetes complications by keeping A1C levels below 7 percent.

Managing blood glucose early in the course of diabetes may provide benefits for many years to come. However, an A1C level that is safe for one person may not be safe for another. For example, keeping an A1C level below 7 percent may not be safe if it leads to problems with hypoglycemia, also called low blood glucose.

Less strict blood glucose control, or an A1C between 7 and 8 percent—or even higher in some circumstances—may be appropriate in people who have

limited life expectancy

long-standing diabetes and trouble reaching a lower goal

severe hypoglycemia or inability to sense hypoglycemia (also called hypoglycemia unawareness)

advanced diabetes complications such as chronic kidney disease, nerve problems, or cardiovascular disease

How does A1C relate to estimated average glucose?

Estimated average glucose (eAG) is calculated from your A1C. Some laboratories report eAG with A1C test results. The eAG number helps you relate your A1C to daily glucose monitoring levels. The eAG calculation converts the A1C percentage to the same units used by home glucose meters—milligrams per deciliter (mg/dL).

The eAG number will not match daily glucose readings because it's a long-term average—rather than your blood glucose level at a single time, as is measured with a home glucose meter.

Will the A1C test show short-term changes in blood glucose levels?

Large changes in your blood glucose levels over the past month will show up in your A1C test result, but the A1C test doesn't show sudden, temporary increases or decreases in blood glucose levels. Even though A1C results represent a long-term average, blood glucose levels within the past 30 days have a greater effect on the A1C reading than those in previous months.

Clinical Trials for the A1C Test and Diabetes

The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and other components of the National Institutes of Health (NIH) conduct and support research into many diseases and conditions.

What are clinical trials and are they right for you?

Clinical trials are part of clinical research and at the heart of all medical advances. Clinical trials look at new ways to prevent, detect, or treat disease. Scientists are conducting research to learn more about diabetes, including studies about A1C. For example

how the relationship between A1C and blood glucose may vary in different racial and ethnic groups

to find other tests that may be better than A1C for some people

to look for ways to further improve A1C test results. Because the A1C value depends on the average life span of your red blood cells, knowing whether the life span of your red blood cells is longer or shorter may give your doctor helpful information.

Researchers also use clinical trials to look at other aspects of care, such as improving the quality of life for people with chronic illnesses. Find out if clinical trials are right for you [NIH external link](#).

What clinical trials are open?

Clinical trials that are currently open and are recruiting can be viewed at [ClinicalTrials.gov](#) [NIH external link](#).

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Related Topics

English

Artificial Pancreas

On this page:

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Clinical Trials for the Artificial Pancreas

What is an artificial pancreas?

An artificial pancreas is a system made of three parts that work together to mimic how a healthy pancreas controls blood glucose, also called blood sugar, in the body. An artificial pancreas is mainly used to help people with type 1 diabetes.

In type 1 diabetes, the pancreas does not produce insulin. People with type 1 diabetes control their blood glucose level by checking it and taking insulin, either by injection or through an insulin infusion pump, several times a day. An artificial pancreas automatically monitors your blood glucose level, calculates the amount of insulin you need at different points during the day, and delivers it.

Most artificial pancreas systems require you to count and enter the amount of carbohydrates you consume at mealtime. These are called “hybrid” artificial pancreas systems, because some of the insulin is given automatically and some is given based on the information you enter. These systems help control blood glucose levels throughout the day and night, making it easier for people with type 1 diabetes to keep their blood glucose level in range. Keeping blood glucose levels in range will prevent other health problems from developing and may improve daily life for people with type 1 diabetes.

How do artificial pancreas systems work?

Three devices make up an artificial pancreas system.

A continuous glucose monitor (CGM) tracks blood glucose levels every few minutes using a tiny sensor that is inserted under the skin. The sensor wirelessly sends the information to a program stored on a smartphone or on an insulin infusion pump.

The program calculates how much insulin is needed and signals the insulin infusion pump when insulin needs to be delivered.

The insulin infusion pump will deliver small doses of insulin throughout the day when blood glucose levels are not in your target range. There are different types of insulin pumps.

One type of pump is worn outside the body on a belt or in a pocket or pouch. Insulin flows from the pump through a plastic tube that connects to a smaller tube, called a catheter, which has a needle that is inserted under the skin and stays in place for several days.

Another type of pump attaches directly to the skin with an adhesive pad and gives insulin through a catheter inserted under the skin. This kind of pump is replaced every few days.

A woman is seated at a table holding a smartphone and a continuous glucose monitor, which is connected to a needle inserted at the back of her upper arm.

A woman wears a continuous glucose monitor on the waistband of her jeans. The sensor and needle are attached to her abdomen.

An artificial pancreas system uses a continuous glucose monitor, an insulin pump, and a program stored on the pump or a smartphone (top). The insulin pump can be worn on a belt, stored in a pocket, or attached directly to the skin (bottom).

Are there other names for the artificial pancreas?

Other names for the artificial pancreas include

automated insulin-delivery system

closed-loop system

bionic pancreas

What are the different types of artificial pancreas systems?

There are several types of artificial pancreas systems. They include

threshold suspend and predictive suspend systems

insulin-only systems

dual hormone systems

Threshold suspend and predictive suspend systems

The threshold suspend and predictive suspend systems can temporarily stop or “suspend” delivering insulin if your blood glucose level gets low.¹

The threshold suspend system stops delivering insulin when your blood glucose level drops to a pre-set level. The predictive suspend system calculates your blood glucose level and will stop delivering insulin before your blood glucose level gets too low. Neither system automatically increases insulin doses.

Stopping insulin delivery at the right moment can help a person with type 1 diabetes avoid low blood sugar, or hypoglycemia, a condition when a person’s blood glucose level is lower than their target range. These systems may help people with type 1 diabetes who develop hypoglycemia overnight, particularly children.

Insulin-only systems

Insulin-only systems keep your blood glucose level within your target range by automatically increasing or decreasing the amount of insulin delivered to your body based on your CGM values. Insulin-only systems can increase insulin doses if your blood glucose level is higher than your target range.

One type of insulin-only system is the hybrid system. The hybrid insulin-only system automatically adjusts insulin doses in response to your CGM values. But you must still count carbohydrate levels and calculate insulin doses for all meals and snacks.

Dual hormone systems

Researchers are currently developing and testing systems that use two hormones—insulin to lower glucose levels and glucagon to raise blood glucose levels. Using two hormones to control blood glucose is similar to the way the pancreas works in people who do not have diabetes. These systems may be able to tightly control glucose levels without causing hypoglycemia. Researchers are also testing how well other combinations, such as rapid-acting insulin and pramlintide [NIH external link](#), can control blood glucose levels.²

Who can use an artificial pancreas?

You need a prescription from your doctor to get an artificial pancreas. Together, you and your doctor will consider which system can best control your blood glucose levels and fit your lifestyle. Other factors to consider are

age—some models are approved for children as young as 2 years old, while others are approved for people ages 6 years and older

whether you can place the devices in the correct position without help, check that they are working properly, and adjust them or enter carbohydrate data as needed

whether you prefer an insulin pump with tubing or one that attaches directly to your skin

whether you have access to a Wi-Fi network so that information can flow between the parts of the system

Parents or guardians of children with type 1 diabetes should talk about appropriate artificial pancreas systems with the child's doctor.

Researchers continue to develop new and different types of artificial pancreas systems that will meet the needs of people with type 1 diabetes and other types of diabetes.

How much an artificial pancreas will cost you depends on your health insurance provider. Private insurance plans or government plans may not cover all artificial pancreas systems available for people with type 1 diabetes. More information on how to pay for type 1 diabetes devices and medicines is found in Financial Help for Diabetes Care.

What are the benefits of an artificial pancreas?

An artificial pancreas may help people with type 1 diabetes reach their target blood glucose levels and improve their quality of life.

With an artificial pancreas system, your glucose levels will be monitored continuously. The computer program improves blood glucose control by automatically adjusting the amount of insulin it delivers to keep your blood glucose levels in range. The system helps you avoid hypoglycemia and hyperglycemia.

Doctors can monitor insulin doses remotely and recommend dosage adjustments for people who need closer supervision. Also, parents or guardians can monitor blood glucose levels from their own smartphones throughout the day and night.

What are the limits of an artificial pancreas?

Artificial pancreas systems are not completely “hands off.” You have to regularly maintain the devices to be sure they are working properly and enter meal sizes into the system every time you eat. You will also need to

check the CGM and infusion pump catheter to be sure they are in place and change them when needed

check the CGM for accuracy and replace the CGM sensor from time to time

count the amount of mealtime carbohydrates and enter them into the system

adjust the computer program settings to make sure you get the right amount of insulin to keep your blood glucose level in your target range

reboot or reconnect the CGM, infusion pump, and computer program if there are problems

manage high or low blood glucose levels if the system is not able to keep your blood glucose in range

The adhesive patches used with these systems may cause skin redness or irritation. Some medicines you take might also interfere with the glucose monitor.

How does the NIDDK support artificial pancreas research?

For more than 20 years, the NIDDK has funded technology to improve the lives of people with type 1 diabetes. Together with other federal government agencies and partner organizations, the NIDDK works to advance blood glucose monitoring, automate insulin delivery, and decrease complications of living with type 1 diabetes.

The NIDDK supports clinical trials to test artificial pancreas systems for children,³ adolescents,^{2,4} and adults⁴ with type 1 diabetes. New trials will test artificial pancreas systems in larger groups of people over longer periods of time.

Recent clinical trials funded by the NIDDK have shown that artificial pancreas systems control blood glucose levels better than other methods, especially through the night—a challenge for many people with type 1 diabetes—and in children as young as 6 years old. Researchers continue to improve the design and components of artificial pancreas systems and to expand their use in different age groups and in real-life situations, such as during exercise.

Clinical Trials for the Artificial Pancreas

The NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for artificial pancreas?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of artificial pancreas systems, such as

computer programs that improve blood glucose monitoring and insulin delivery during daily activities and mealtimes

different types of artificial pancreas systems (insulin-only or dual hormone)

different types of insulin that can be used in these systems

artificial pancreas systems that can be used by different age groups, such as older adults and very young children

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

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Related Topics

English

Continuous Glucose Monitoring

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What issues could you have while using a continuous glucose monitor?

What is an artificial pancreas?

How does NIDDK support research on continuous glucose monitors?

Clinical Trials on Continuous Glucose Monitors

What is continuous glucose monitoring?

Continuous glucose monitoring means using a device to automatically estimate your blood glucose level, also called blood sugar, throughout the day and night. You can see what your blood glucose level is at any time. You can also review how your blood glucose level changes over a few hours or days and spot trends.

Seeing your blood glucose levels in real time can help you make more informed decisions about the food and beverages you consume, the physical activity you do, and the medicines you take. Keeping your blood glucose level in your target range can help prevent other health problems caused by diabetes.

How does a continuous glucose monitor work?

A continuous glucose monitor (CGM) estimates what your glucose level is every few minutes and keeps track of it over time.

A CGM has three parts. First, there is a tiny sensor that can be inserted under your skin, often the skin on your belly or arm, with a sticky patch that helps it stay there. These sensors are called disposable sensors. Another type of CGM sensor—called an implantable sensor—may be placed inside your body. CGM sensors estimate the glucose level in the fluid between your cells, which is very similar to the glucose level in your blood. Sensors must be replaced at specific times, such as every few weeks, depending on the type of sensor you have.

The second part of the CGM is a transmitter. The transmitter sends the information, without using wires, to the third part, a software program that is stored on a smartphone, on an insulin pump, or on a separate device called a receiver.

A woman with a continuous glucose monitor on her arm looks at her smartphone to check her glucose level.

Most CGMs send information without using wires to an app on a smartphone.

Who can use a continuous glucose monitor?

Your doctor may recommend that you use a CGM if you need insulin to manage type 1 diabetes, type 2 diabetes, or another form of diabetes. Talk with your doctor about whether using a CGM could help you manage your diabetes.

Doctors can prescribe CGMs for adults and children. Some models can be used for children as young as 2 years old.

Your doctor may suggest using a CGM all the time or only for a few days to help adjust your diabetes care.

What are the different types of continuous glucose monitors?

All CGMs estimate blood glucose levels, but they store and display the information in different ways.

Some CGMs send and display information to your smartphone or receiver automatically. These CGMs are called “real-time” CGM devices. Another type of CGM, called “intermittent-scan,” estimates glucose levels continuously. But you will need to scan the CGM with a separate receiver or smartphone every few hours to view and store the data. A third type of CGM collects data about your blood glucose level for your doctor to download and review later. Doctors provide this type of CGM to check on your diabetes care, and you wear it for a limited time.

Other differences between CGM models include

whether the sensor is placed on the skin or is implanted

how often the sensor has to be replaced

how long it takes the CGM to warm up

how you adjust the program settings

For some CGM models, you may need to do a finger-stick test with a standard blood glucose monitor to calibrate the system and make sure the CGM readings are correct.

What are some features of continuous glucose monitors?

When worn, CGMs are always on and recording glucose levels—whether you’re showering, working, exercising, or sleeping. Many CGMs work with apps that have special features, such as

ways to track the food and beverages you consume, your physical activity level, and the medicines you take

the ability to download data onto a computer or smart device so you can easily see trends in your glucose levels

an alarm that goes off when your glucose level is too low or too high, helping you prevent emergencies

For safety, it is important to act quickly if a CGM alarm sounds when your glucose level is too low or too high. You should get help or follow your treatment plan to bring your glucose level into a healthy range. Some CGM models can also send information to a second person’s smartphone—such as a parent, partner, or caregiver. For example, if a child’s glucose level drops dangerously low overnight, the CGM could be set to wake a parent in the next room.

What are the benefits of a continuous glucose monitor?

Compared with a standard blood glucose meter, using a CGM can help you

better manage your glucose levels every day

have fewer low blood glucose emergencies

need fewer finger sticks

The CGM will create an alert and might display a graphic that shows whether your glucose level is rising or dropping—and how quickly—so you can choose the best way to reach your target range.

Over time, keeping your glucose levels in the healthy range can help you stay well and prevent diabetes complications. The people who benefit the most from a CGM are those who use it every day or nearly every day.

What issues could you have while using a continuous glucose monitor?

Researchers are working to make CGMs more accurate and easier to use. However, you may experience some issues while using a CGM.

For safety, you may sometimes need to compare your CGM glucose readings with a finger-stick test and a standard blood glucose meter. This could be needed if you doubt the accuracy of your CGM readings, if you are changing your insulin dose, or if your CGM gives a warning alert.¹

You might have to replace parts of your CGM over time. Disposable CGM sensors should be replaced every 7 to 14 days, depending on the model. Some implantable sensors can last up to 180 days. You may have to replace the transmitters of some CGMs. You may also need to reconnect the CGM, transmitter, and receiver or smartphone if your CGM is not working correctly.

Skin redness or irritation from the sticky patches used to attach the sensor may occur for some people.¹ Your doctor can suggest techniques or medicines to help relieve skin problems.

A CGM costs more than using a standard glucose meter, but it may be covered by your health insurance. You might be able to get financial help for diabetes care from your health insurance or other resources. Check with your health insurance plan or Medicare [External link](#) to see if the costs will be covered.

A woman compares her blood glucose reading from a standard glucose monitor with a CGM reading on a smartphone. A lancet used to test blood glucose is on the table in front of her.

For safety, you may sometimes need to compare your CGM glucose readings with a finger-stick test and a standard blood glucose meter.

What is an artificial pancreas?

An artificial pancreas, also called an automated insulin delivery system (AID), mimics how a healthy pancreas controls blood glucose in the body. A CGM, an insulin pump, and a software program that shares information between the CGM and insulin pump make up the artificial pancreas.

The CGM estimates glucose levels and wirelessly sends the information to a software program on a smartphone or insulin pump. The program calculates how much insulin your body needs, and the insulin pump delivers the insulin when glucose levels rise higher than your target range. On the other hand, if your glucose levels fall lower than your target range, the artificial pancreas can lower or stop the amount of insulin given by the insulin pump.

The artificial pancreas is mainly used to help people with type 1 diabetes keep their glucose levels in their target range.

How does NIDDK support research on continuous glucose monitors?

NIDDK has a long-standing commitment to funding research to better understand diabetes and improve the lives of people with the disease.

NIDDK-funded research helped scientists learn that glucose levels in the fluid between cells could be used to estimate blood glucose levels.² This discovery helped create the CGMs that people with diabetes use today.

NIDDK also supported the Diabetes Control and Complications Trial, which showed that people with diabetes could use blood glucose monitors at home to closely control their blood glucose levels and reduce their risk of health problems.³

Clinical Trials on Continuous Glucose Monitors

NIDDK conducts and supports clinical trials for many diseases and conditions, including diabetes. Trials look for new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for continuous glucose monitors?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future. Researchers are studying many aspects of CGMs, such as how CGMs could be made more sensitive, reliable, and comfortable to wear. Researchers are also studying how they might be used to manage different types of diabetes or other medical conditions.

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Related Topics

English

Diabetes & Foot Problems

In this section:

How can diabetes affect my feet?

What can I do to keep my feet healthy?

When should I see my health care provider about foot problems?

Clinical Trials

Foot problems are common in people with diabetes. You might be afraid you'll lose a toe, foot, or leg to diabetes, or know someone who has, but you can lower your chances of having

diabetes-related foot problems by taking care of your feet every day. Managing your blood glucose levels, also called blood sugar, can also help keep your feet healthy.

How can diabetes affect my feet?

Over time, diabetes may cause nerve damage, also called diabetic neuropathy, that can cause tingling and pain, and can make you lose feeling in your feet. When you lose feeling in your feet, you may not feel a pebble inside your sock or a blister on your foot, which can lead to cuts and sores. Cuts and sores can become infected.

Diabetes also can lower the amount of blood flow in your feet. Not having enough blood flowing to your legs and feet can make it hard for a sore or an infection to heal. Sometimes, a bad infection never heals. The infection might lead to gangrene.

Gangrene and foot ulcers that do not get better with treatment can lead to an amputation of your toe, foot, or part of your leg. A surgeon may perform an amputation to prevent a bad infection from spreading to the rest of your body, and to save your life. Good foot care is very important to prevent serious infections and gangrene.

Although rare, nerve damage from diabetes can lead to changes in the shape of your feet, such as Charcot's foot. Charcot's foot may start with redness, warmth, and swelling. Later, bones in your feet and toes can shift or break, which can cause your feet to have an odd shape, such as a "rocker bottom."

Illustration of Charcot's foot showing an enlarged sole of the foot with a rounded shape.

Charcot's foot can cause your feet to have an odd shape, such as a "rocker bottom."

What can I do to keep my feet healthy?

Work with your health care team to make a diabetes self-care plan, which is an action plan for how you will manage your diabetes. Your plan should include foot care. A foot doctor, also called a podiatrist, and other specialists may be part of your health care team.

Include these steps in your foot care plan:

Tips to Take Care of Your Feet

Check your feet every day.

Wash your feet every day.

Smooth corns and calluses gently.

Trim your toenails straight across.

Wear shoes and socks at all times.

Protect your feet from hot and cold.

Keep the blood flowing to your feet.

Get a foot check at every health care visit.

Check your feet every day

You may have foot problems, but feel no pain in your feet. Checking your feet each day will help you spot problems early before they get worse. A good way to remember is to check your feet each evening when you take off your shoes. Also check between your toes. If you have trouble bending over to see your feet, try using a mirror to see them, or ask someone else to look at your feet.

Look for problems such as

cuts, sores, or red spots

swelling or fluid-filled blisters

ingrown toenails, in which the edge of your nail grows into your skin

corns or calluses, which are spots of rough skin caused by too much rubbing or pressure on the same spot

plantar warts, which are flesh-colored growths on the bottom of the feet

athlete's foot

warm spots

Someone checking the bottom of their bare foot.

If you have certain foot problems that make it more likely you will develop a sore on your foot, your doctor may recommend taking the temperature of the skin on different parts of your feet. A “hot spot” can be the first sign that a blister or an ulcer is starting.

Cover a blister, cut, or sore with a bandage. Smooth corns and calluses as explained below.

Wash your feet every day

Wash your feet with soap in warm, not hot, water. Test the water to make sure it is not too hot. You can use a thermometer (90° to 95° F is safe) or your elbow to test the warmth of the water. Do not soak your feet because your skin will get too dry.

Photo of a person’s bare foot with hands examining the toes.

After washing and drying your feet, put talcum powder or cornstarch between your toes. Skin between the toes tends to stay moist. Powder will keep the skin dry to help prevent an infection.

Smooth corns and calluses gently

Thick patches of skin called corns or calluses can grow on the feet. If you have corns or calluses, talk with your foot doctor about the best way to care for these foot problems. If you have nerve damage, these patches can become ulcers.

Photo of a woman smoothing the bottom of her foot with a pumice stone.

If your doctor tells you to, use a pumice stone to smooth corns and calluses after bathing or showering. A pumice stone is a type of rock used to smooth the skin. Rub gently, only in one direction, to avoid tearing the skin.

Do NOT

cut corns and calluses

use corn plasters, which are medicated pads

use liquid corn and callus removers

Cutting and over-the counter corn removal products can damage your skin and cause an infection.

To keep your skin smooth and soft, rub a thin coat of lotion, cream, or petroleum jelly on the tops and bottoms of your feet. Do not put lotion or cream between your toes because moistness might cause an infection.

Trim your toenails straight across

Trim your toenails, when needed, after you wash and dry your feet. Using toenail clippers, trim your toenails straight across. Do not cut into the corners of your toenail. Gently smooth each nail with an emery board or nonsharp nail file. Trimming this way helps prevent cutting your skin and keeps the nails from growing into your skin.

Photo of someone clipping their toenails.

Have a foot doctor trim your toenails if

you cannot see, feel, or reach your feet

your toenails are thick or yellowed

your nails curve and grow into the skin

If you want to get a pedicure at a salon, you should bring your own nail tools to prevent getting an infection. You can ask your health care provider what other steps you can take at the salon to prevent infection.

Wear shoes and socks at all times

Wear shoes and socks at all times. Do not walk barefoot or in just socks – even when you are indoors. You could step on something and hurt your feet. You may not feel any pain and may not know that you hurt yourself.

Check the inside of your shoes before putting them on, to make sure the lining is smooth and free of pebbles or other objects.

Photo of a man feeling inside his shoe.

Make sure you wear socks, stockings, or nylons with your shoes to keep from getting blisters and sores. Choose clean, lightly padded socks that fit well. Socks with no seams are best.

Wear shoes that fit well and protect your feet. Here are some tips for finding the right type of shoes:

Walking shoes and athletic shoes are good for daily wear. They support your feet and allow them to “breathe.”

Do not wear vinyl or plastic shoes, because they do not stretch or “breathe.”

When buying shoes, make sure they feel good and have enough room for your toes. Buy shoes at the end of the day, when your feet are the largest, so that you can find the best fit.

If you have a bunion, or hammertoes, which are toes that curl under your feet, you may need extra-wide or deep shoes.¹ Do not wear shoes with pointed toes or high heels, because they put too much pressure on your toes.

If your feet have changed shape, such as from Charcot’s foot, you may need special shoes or shoe inserts, called orthotics. You also may need inserts if you have bunions, hammertoes, or other foot problems.

When breaking in new shoes, only wear them for a few hours at first and then check your feet for areas of soreness.

Medicare Part B insurance [External link](#) and other health insurance programs may help pay for these special shoes or inserts. Ask your insurance plan if it covers your special shoes or inserts.

Protect your feet from hot and cold

If you have nerve damage from diabetes, you may burn your feet and not know you did. Take the following steps to protect your feet from heat:

Wear shoes at the beach and on hot pavement.

Put sunscreen on the tops of your feet to prevent sunburn.

Keep your feet away from heaters and open fires.

Do not put a hot water bottle or heating pad on your feet.

Photo of someone's feet wearing shoes and walking on the sand.

Wear socks in bed if your feet get cold. In the winter, wear lined, waterproof boots to keep your feet warm and dry.

Keep the blood flowing to your feet

Try the following tips to improve blood flow to your feet:

Put your feet up when you are sitting.

Wiggle your toes for a few minutes throughout the day. Move your ankles up and down and in and out to help blood flow in your feet and legs.

Do not wear tight socks or elastic stockings. Do not try to hold up loose socks with rubber bands.

Be more physically active. Choose activities that are easy on your feet, such as walking, dancing, yoga or stretching, swimming, or bike riding.

Stop smoking.

Photo of a man sitting reading a book with his feet resting on footstool.

Smoking can lower the amount of blood flow to your feet. If you smoke, ask for help to stop. You can get help by calling the national quitline at 1-800-QUITNOW or 1-800-784-8669. For tips on quitting, go to SmokeFree.gov External link.

Get a foot check at every health care visit

Ask your health care team to check your feet at each visit. Take off your shoes and socks when you're in the exam room so they will remember to check your feet. At least once a year, get a thorough foot exam, including a check of the feeling and pulses in your feet.

Photo of a doctor examining someone's bare feet.

Get a thorough foot exam at each health care visit if you have

changes in the shape of your feet

loss of feeling in your feet

peripheral artery disease

had foot ulcers or an amputation in the past¹

Ask your health care team to show you how to care for your feet.

When should I see my health care provider about foot problems?

Call your health care provider right away if you have

a cut, blister, or bruise on your foot that does not start to heal after a few days

skin on your foot that becomes red, warm, or painful—signs of a possible infection

a callus with dried blood inside of it, which often can be the first sign of a wound under the callus

a foot infection that becomes black and smelly—signs you might have gangrene

Ask your provider to refer you to a foot doctor, or podiatrist, if needed.

Clinical Trials

The National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and other components of the National Institutes of Health (NIH) conduct and support research into many diseases and conditions.

What are clinical trials, and are they right for you?

Clinical trials are part of clinical research and at the heart of all medical advances. Clinical trials look at new ways to prevent, detect, or treat disease. Researchers also use clinical trials to look at other aspects of care, such as improving the quality of life for people with chronic illnesses. Find out if clinical trials are right for you [NIH external link](#).

What clinical trials are open?

Clinical trials that are currently open and are recruiting can be viewed at [ClinicalTrials.gov](#) [NIH external link](#).

References

[1] American Diabetes Association. Microvascular complications and foot care. *Diabetes Care*. 2016;39(Suppl. 1):S78.

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Related Topics

[English](#)

[Pregnancy if You Have Diabetes](#)

On this page:

[How should I prepare for pregnancy if I have diabetes?](#)

[What should I do if I have diabetes and I am pregnant?](#)

[How can my diabetes affect my baby?](#)

[How can my diabetes affect me during pregnancy?](#)

What health problems could I develop during pregnancy because of my diabetes?

How should I manage my diabetes during pregnancy?

What do I need to know about blood glucose testing during pregnancy?

What tests will check my baby's health during pregnancy?

How can my diabetes affect me after I have my baby?

Clinical Trials on Pregnancy and Diabetes

If you have diabetes and plan to have a baby, you should try to get your blood glucose levels, also called blood sugar levels, close to your target range before getting pregnant. High blood glucose levels can harm your baby during the first weeks of pregnancy. Talk with your primary health care professional or obstetrician about how to prepare for pregnancy when you have diabetes. Your primary health care professional may be a doctor, physician assistant, or nurse practitioner. If you have diabetes and find out you are pregnant, see your health care professional as soon as possible.

How should I prepare for pregnancy if I have diabetes?

Pregnancy is something that you should try to plan if you have diabetes. You should talk with your health care professional about preconception care [External link](#). Your health care professional can help you plan for pregnancy and lower the risk of health problems for you and your baby.

Before you get pregnant, have a complete medical checkup. If you are planning to get pregnant your blood glucose targets may be different. Your health care professional may recommend that you see a registered dietitian. A dietitian can teach you what, how much, and when to eat or drink to manage your blood glucose level and reach or stay at a healthy weight. Your insurance can tell you if this service is covered.

A man and woman talking in a kitchen while cutting up vegetables.

Plan to manage your blood glucose levels before you get pregnant.

Your health care professional may also prescribe different medicines that are safe during pregnancy.

Before getting pregnant, aim to make physical activity a regular part of your life. Try for at least 150 minutes of moderate-intensity physical activity, such as brisk walking, each week.¹

If you and your partner are finding it hard to get pregnant, talk with your health care professional to find out if you need any further evaluation. Diabetes may produce health problems such as erectile dysfunction and infertility [External link](#).

What should I do if I have diabetes and I am pregnant?

If you have diabetes and you find out that you are pregnant, talk with your health care professional as soon as possible about how to manage your diabetes during your pregnancy. Your target blood glucose level range may be different when you are pregnant than before pregnancy. Working with your health care team and following your plan to manage diabetes can help you have a healthy pregnancy and a healthy baby.

Some people may have diabetes and not know it. If you have risk factors for type 2 diabetes, your health care professional may test you for diabetes in the first prenatal visit. Some people develop diabetes during pregnancy that goes away after they have the baby. This type of diabetes is called gestational diabetes. Health care professionals usually diagnose gestational diabetes between weeks 24 and 28 of pregnancy.^{2,3} However, having gestational diabetes increases your risk of getting type 2 diabetes later in life.

Find out if you had diabetes before getting pregnant, or if you have gestational diabetes. These two types of diabetes are managed differently.

How can my diabetes affect my baby?

A baby's organs, such as the brain, heart, kidneys, and lungs, start forming during the first 8 weeks of pregnancy.³ High blood glucose levels can harm the baby during this early stage and increase the chance that your baby will have birth defects of any of these organs.

High blood glucose levels during pregnancy can also increase the chance that your baby will

be born too early

be above average in size

have breathing problems right after birth

have low blood glucose levels, also called hypoglycemia, right after birth

A health care professional may recommend keeping the baby in a special care unit after delivery if there is a chance that the baby may develop low blood glucose levels or other health problems. Babies that are born too early, are very small, or have other serious health problems may also need special care.

High blood glucose levels can also increase the chance that you will have a miscarriage [NIH external link](#) or a stillborn baby. Stillborn means the baby dies in the womb during the second half of pregnancy.

If you have diabetes and your blood glucose levels are high, your baby also has an increased risk of developing some health problems later in life, such as obesity or type 2 diabetes.

How can my diabetes affect me during pregnancy?

Changes to your hormones and your body during pregnancy affect your blood glucose levels. Even if you've had diabetes for years, you may need to adjust your meal plan, physical activity routine, or medicines. If you were not taking insulin before your pregnancy, you may need to add insulin. It is common during pregnancy to need to increase your insulin dose as you get closer to your due date. Your blood glucose levels may change quickly during your delivery and after you have your baby. Your health care team may be checking your blood glucose levels often to adjust your treatment.

What health problems could I develop during pregnancy because of my diabetes?

Pregnancy can make certain diabetes health problems worse, such as eye, heart, and kidney disease, especially if your blood glucose levels are too high.

You also have a greater chance of developing preeclampsia [NIH external link](#). In preeclampsia, your blood pressure goes up and you may have too much protein in your urine during the second half of pregnancy.

People with diabetes may have babies who are above average in size. If your baby is above average in size, your health care team may recommend cesarean delivery [NIH external link](#).

How should I manage my diabetes during pregnancy?

To help you and your baby stay healthy

follow your diabetes care plan

get checkups

take your vitamins

stop smoking

Work with your health care team

You are the most important member of your health care team because you are the one who manages your diabetes every day. You should have regular visits with members of your health care team who are experts in diabetes and pregnancy. Asking for advice and letting your health care team know if you have any concerns will help you and your baby get the best care. Your health care team may include

a health care professional who is an expert in diabetes care, such as an endocrinologist or a diabetologist

an obstetrician with experience treating people with diabetes

a diabetes educator, nurse, or dietitian who can tell you more about how to manage your diabetes, including what to eat and drink

specialists who diagnose and treat diabetes health problems, such as eye, kidney, and heart diseases

nurses, social workers, and psychologists to help you cope with stress, worry, and the extra demands of pregnancy

Health care professional checking a pregnant woman.

You should have regular visits with members of your health care team who are experts in diabetes and pregnancy.

Get checkups

Have a complete checkup as soon as you know you are pregnant.

Pregnancy can make some diabetes health problems worse. To help prevent developing or worsening health problems related to your diabetes, your health care team may recommend adjusting your treatment.

Your health care team will follow your blood pressure closely because diabetes puts you at risk of developing preeclampsia. If you have or are concerned about developing high blood pressure [NIH external link](#), ask your health care professional what your blood pressure goal should be during pregnancy. Your health care professional may recommend changes to your blood pressure medicines before pregnancy or early in pregnancy.

Diabetic eye disease can get worse during your pregnancy, and you may need special care. During your pregnancy, you may have frequent dilated eye exams [NIH external link](#) to check your eye health. After you have your baby, your eye doctor may follow up closely for a year.

Your health care team may check your kidneys at the beginning of your pregnancy. If you already have kidney disease, your team may follow up more closely.

Let your health care team know if you have any changes in the sensation of your legs or arms. Diabetes can affect your nerves, especially in your legs and arms. Pregnancy can worsen nerve damage because the baby can press on nerves already damaged by diabetes.

You may also get other routine pregnancy care, such as appropriate vaccines [External link](#) and screenings for genetic [External link](#), thyroid, or infectious diseases [NIH external link](#) that can affect you or your baby.

Don't smoke

Smoking [External link](#) can increase your chance of having a stillborn baby or a baby born too early. Both smoking and diabetes can make your blood vessels narrow. E-cigarettes [NIH external link](#) aren't a safe option either. If you smoke, vape, or use other tobacco products, it's best to stop as soon as possible. Ask for help [External link](#) so you don't have to do it alone.

See a dietitian

If you don't already see a dietitian, you should start seeing one. Your insurance can tell you if this service is covered.

During pregnancy, a dietitian may help you make changes to your meal plan, such as adding extra calories, protein foods [External link](#), and other nutrients. You may need to see your dietitian more often during pregnancy as your needs change.

Trying to lose weight during pregnancy may not be healthy. However, you should ask your health care professional what amount of weight gain they recommend for you.

Be physically active

Physical activity may help lower your blood glucose, blood pressure, and cholesterol [NIH external link](#) levels. Being active may also help you get better sleep and improve your mood.

Talk with your health care team about what activities are safe for you during your pregnancy. The Physical Activity Guidelines for Americans, 2nd Edition [External link](#) (PDF, 15.6 MB) recommends that during and after pregnancy, most people should try to get at least 150 minutes of moderate-intensity physical activity [External link](#) a week and include strength-training activities twice a week.¹ You may need to adjust your level of activity based on how active you were before your pregnancy.

A man pushing a stroller with a baby outdoors and a pregnant woman walking next to them.

Talk with your health care team about what activities are safe for you during pregnancy.

Avoid alcoholic beverages

You should avoid drinking alcoholic beverages [NIH external link](#) throughout pregnancy. Drinking alcoholic beverages can lead to health problems for you and your baby.

Adjust your medicines

Some medicines [External link](#) are not safe during pregnancy. Tell your health care professional about all the medicines you take, including prescription and over-the-counter medicines and supplements. Your health care professional can tell you which medicines to stop taking or may prescribe different medicines that are safe during pregnancy.

To manage diabetes during pregnancy, health care professionals often prescribe insulin, along with medical nutrition therapy [External link](#). Medical nutrition therapy includes counseling to help you create and follow a meal plan. Ask your health care team or your insurance company if your benefits include medical nutrition therapy.

If you're already taking insulin, you might need to change the type, the amount, or how and when you take it. You may need less insulin during your first trimester, but probably will need more as you go through pregnancy. Your insulin needs may double or even triple as you get closer to your due date and will drop quickly after you have your baby. Your health care team will work with you to create an insulin routine to meet your changing needs. Health care professionals may prescribe oral medicines to manage diabetes in some cases.

Diabetes medicines, technologies, and office visits can be expensive. Tell your health care professional or health care team if you are having trouble affording diabetes care or pregnancy care. A health care professional, diabetes educator, or social worker may be able to tell you about resources that provide financial help for diabetes care and pregnancy care.

Ask your health care team if you should take multivitamins, folic acid [External link](#), iron [External link](#), or calcium [NIH external link](#) supplements. Your health care team may suggest a low dose of aspirin to lower your risk of preeclampsia. Talk with your health care team to learn more about low-dose aspirin.

What do I need to know about blood glucose testing during pregnancy?

Your blood glucose level target ranges, when, and how often you check your blood glucose levels may change during pregnancy. Checking your blood glucose level may help you manage how high or low they get. Ask your health care team how often and at what times you should check your blood glucose levels.

You can check your blood glucose level at home with a blood glucose meter [External link](#). The meter tests a drop of your blood, usually from a prick of your fingertip. The meter will show you how much glucose is in your blood at that moment. If you take insulin to manage your diabetes, you may benefit from using a continuous glucose monitor (CGM).

Your health care team may ask you to check your urine at home for substances called ketones. High levels of ketones in your urine may mean that you need to pay more attention to how you are managing your diabetes. Work with your health care professional to change your diabetes care plan. If high levels of ketones are left untreated, you may develop a dangerous condition called diabetic ketoacidosis [External link](#).

Pregnant woman checking her blood glucose level.

During your pregnancy, you may need to check your blood glucose levels more often.

Target blood glucose levels during pregnancy

Your daily blood glucose targets may be different during pregnancy than before pregnancy. Ask your health care team which targets are right for you.

You can keep track of your blood glucose levels using [My Daily Blood Glucose Record \(PDF, 44 KB\)](#) . You can also use an electronic blood glucose tracking system on your computer or mobile device. Record the results every time you check your blood glucose levels. Your blood glucose

records can help you and your health care team make changes to your meal plan, physical activity level, or diabetes medicines. You also can make notes about your insulin needs and ketones. Take your tracker with you when you visit your health care team.

Health care professionals most often recommend the following blood glucose level targets during pregnancy^{3,4}

fasting, before meals, at bedtime, and overnight: 70 to 95 mg/dL

1 hour after eating: 110 to 140 mg/dL

2 hours after eating: 100 to 120 mg/dL

Ask your health care team what targets are best for you.

If you are using a CGM, the most often recommended target range for pregnancy is 63 to 140 mg/dL. You will need to change this target range in your CGM settings. You can ask your health care team what percentage of time you should spend in this range and above and below this range.

A1C blood test

Another way to check your blood glucose levels is to have an A1C blood test. Results of the A1C test reflect your average blood glucose levels during the last 3 months. Changes in your body during pregnancy affect A1C levels. A1C test is not the best test to monitor your blood glucose level during pregnancy. Most people with diabetes should aim for an A1C of 6.5% or lower just before pregnancy. After the first 3 months of pregnancy, your target may be as low as 6%.^{3,4} These targets may differ from the A1C goals you've had in the past. Your health care team can help you set the A1C targets that are best for you.

What tests will check my baby's health during pregnancy?

You should have prenatal tests throughout your pregnancy. Your health care professional will do ultrasounds [NIH external link](#) and nonstress tests [NIH external link](#) to check your baby's health. Talk with your health care team about other prenatal tests you'll have and when you might have them.

How can my diabetes affect me after I have my baby?

You also need special care after you have your baby.

Blood glucose levels can drop very quickly after delivery. You should ask your health care team what changes should be made to your medicines. You should also check your blood glucose levels often to watch for low blood glucose levels in the first few days.

If you had gestational diabetes, your health care team will check your blood glucose levels in the first 3 months after having your baby to see if your gestational diabetes has completely gone away.⁴ If you have gestational diabetes, you are more likely to develop type 2 diabetes later in life.

You should also check your blood pressure or have it checked by a health care professional during the first 3 to 12 weeks after having your baby.^{5,6}

Diabetic eye disease may get worse after you have a baby. Ask your eye doctor how often you need to have eye exams.

Breastfeeding [External link](#) has benefits for your baby, such as improving nutrition and defenses against some diseases. It also may benefit you by helping you to lose weight after delivery. However, breastfeeding increases your risk of low blood glucose levels at night. If you are planning to breastfeed your baby, talk with your health care team about how to prevent low blood glucose levels.

Ask your health care team about options for family planning and how soon you can begin to use the method of your choice [External link](#).

Clinical Trials on Pregnancy and Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

Find out if clinical studies are right for you [NIH external link](#).

What are clinical trials for pregnancy and diabetes?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

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Related Topics

English

Diabetes, Sexual, & Bladder Problems

In this section:

Can sexual and bladder problems be symptoms of diabetes?

When should I see a doctor about my sexual or bladder problems?

What makes me more likely to develop sexual or bladder problems?

What sexual problems can men with diabetes have?

What sexual problems can women with diabetes have?

What bladder problems can men and women with diabetes have?

How can I prevent and treat my sexual or bladder problems?

Sexual problems and bladder problems are common as people age, but diabetes can make these problems worse. You or your partner may have trouble having or enjoying sex. Or, you may leak urine or have trouble emptying your bladder normally.

Blood vessels and nerves can be damaged by the effects of high blood glucose, also called blood sugar. This damage can lead to sexual and bladder problems. Keeping your blood glucose levels in your target range is an important way to prevent damage to your blood vessels and nerves.

Work with your health care team to help prevent or treat sexual and bladder problems. These problems may be signs that you need to manage your diabetes in a different way. Remember, a healthy sex life and a healthy bladder can improve your quality of life, so take action now if you have concerns.

A couple embraces on park bench.

Sexual and bladder problems may be signs that you need to manage your diabetes in a different way.

Can sexual and bladder problems be symptoms of diabetes?

Yes. Changes in sexual function or bladder habits may be a sign that you have diabetes. Nerve damage caused by diabetes, also called diabetic neuropathy, can damage parts of your body—like your genitals or urinary tract. For example, men with diabetes may develop erectile dysfunction (ED) 10 to 15 years earlier than men without diabetes.¹

Talk with a health care professional if you have any symptoms of diabetes, including sexual and bladder problems.

When should I see a doctor about my sexual or bladder problems?

See a health care professional for problems with sex or your bladder. These problems could be a sign that you need to manage your diabetes differently. You may find it embarrassing and difficult to talk about these things. However, remember that health care professionals are trained to speak with people about every kind of health problem. Everyone deserves to have healthy relationships and enjoy the activities they love.

What makes me more likely to develop sexual or bladder problems

You're more likely to develop sexual or bladder problems if you have diabetes and

have high blood glucose that is not well controlled, also called high blood sugar

have nerve damage, also called neuropathy

have high blood pressure that is not treated

have high cholesterol that is not treated

are overweight or have obesity

are not physically active

are taking certain medicines

drink too many alcoholic drinks

smoke

Research also suggests that certain genes may make people more likely to develop diabetic neuropathy.²

What sexual problems can men with diabetes have?

Changes in your blood vessels, nerves, hormones, and emotional health during diabetes may make it more difficult for you to have satisfactory sex. Diabetes and its related challenges also may make it harder for you to have a child.

Erectile dysfunction

You have ED if you're unable to get or keep an erection firm enough for satisfactory sexual intercourse. More than half of men with diabetes will get ED. Men who have diabetes are more than 3 times more likely to develop ED than men who do not have diabetes.¹ Good diabetes management may help prevent and treat ED caused by nerve damage and circulation problems. A doctor can help treat ED with medicine or a change in your diabetes care plan.

A health care professional talks to a man and a woman about his lab tests.

Managing your diabetes may help prevent and treat ED caused by nerve damage and circulation problems.

Retrograde ejaculation

Rarely, diabetes can cause retrograde ejaculation [NIH external link](#), which is when part or all of your semen goes into your bladder instead of out of your penis during ejaculation. During retrograde ejaculation, semen enters your bladder, mixes with urine, and is safely urinated out.

A urine sample after ejaculation can show if you have retrograde ejaculation. Some men with retrograde ejaculation may not ejaculate at all.

Penile curvature

Men with diabetes are more likely to have Peyronie's disease, also called penile curvature, than men who don't have diabetes.³ Men with Peyronie's disease have scar tissue, called a plaque, in the penis, making it curve when erect. Curves in the penis can make sexual intercourse painful or difficult. Some men with Peyronie's disease may have ED.

Low testosterone

Men's testosterone levels naturally lower with age. However, lower-than-normal testosterone levels may be the cause of some men's ED, or can explain why some men often feel tired, depressed, or have a low sex drive. Men with diabetes, especially those who are older and overweight, are more likely to have low testosterone, or "low T."

If your doctor thinks you might have low T, you will probably be asked to give a blood sample, and a health care professional will give you a physical exam. Your doctor may suggest treating your low testosterone with a prescription gel, injection, or patch.

Several studies show that, along with good diabetes management, testosterone therapy can lessen a man's sexual problems.⁴ However, testosterone therapy may have serious risks and may not be safe for all men. Talk with your doctor about testosterone therapy's side effects and whether it's right for you.

Fertility problems

Some studies show that men with diabetes can have problems with their sperm that make it harder to conceive. Your sperm could be slow or not move well, or your sperm may not be able to fertilize a woman's egg well.⁵ Working closely with your partner and a health care professional trained in fertility issues may help.

If you and your partner want to conceive a child, your doctor may treat retrograde ejaculation caused by diabetes with medicine or by changing your diabetes care plan. Or, talk with a urologist who is a fertility expert. He or she may be able to collect your sperm from your urine and then use it for artificial insemination [External link](#).

What sexual problems can women with diabetes have?

Low sexual desire and response, vaginal dryness, and painful sex can be caused by nerve damage, reduced blood flow to the genitals, and hormonal changes. Other conditions can cause these problems, too, including menopause [External link](#). If you notice a change in your sex life, talk with your health care team. A physical exam, which will include a pelvic exam, and blood and urine tests may help your doctor find the cause of your problems.

Low sexual desire and response

Low sexual desire and sexual response can include

being unable to become or stay aroused

not having enough vaginal lubrication

having little to no feeling in your genitals

being unable to have an orgasm or rarely having one

With diabetes, your body and mind will likely go through many changes. For example, both high and low blood glucose levels can affect how and if you become aroused. Or, you may find yourself more tired than usual or depressed and anxious [NIH external link](#), making you less interested in sex.

Your health care team can help you make changes to your diabetes care plan so that you're back on track. Women who keep blood glucose levels in their target range are less likely to have nerve damage, which can lead to low sexual desire and response.

Painful sex

Some women with diabetes say they have uncomfortable or painful sexual intercourse. The nerves that tell your vagina to lubricate during stimulation can become damaged by diabetes. A prescription or over-the-counter vaginal lubricant may help if you have vaginal dryness. Managing your blood glucose well over many weeks, months, and years can help prevent nerve damage.

Yeast and bladder infections

Women with diabetes are more likely to have yeast infections [External link](#), because yeast organisms can grow more easily when your blood glucose levels are higher. Yeast infections can be uncomfortable or painful and prevent you from enjoying activities, including having sex.

Although some yeast infections can be treated at home, talk with a health care professional first about your symptoms. Some symptoms of yeast infections are similar to other types of infections, including sexually transmitted diseases.

Pregnancy concerns and fertility problems

If you have diabetes and plan to become pregnant, it's important to get your blood glucose levels close to your target range before you get pregnant. High blood glucose can harm your baby during the first weeks of pregnancy, even before you know you're pregnant.

If you have diabetes and are already pregnant, see your doctor as soon as possible to make a plan to manage your diabetes. Working with your health care team and following your diabetes management plan can help you have a healthy pregnancy and a healthy baby.

Health problems such as obesity and polycystic ovary syndrome [External link](#) (PCOS) that are linked to diabetes can make it harder to conceive a child. Talk with a health care professional, such as a gynecologist or a fertility specialist, if you're having problems conceiving a child.

A health care professional speaks to a woman.

If you notice a change in your sex life, talk with your health care team.

What bladder problems can men and women with diabetes have?

Diabetes can cause nerve damage to your urinary tract, causing bladder problems. Overweight and obesity also can increase bladder problems, such as urinary incontinence (UI). Managing diabetes is an important part of preventing problems that can lead to excess urination.

Your health care team may be able to help you manage your blood glucose levels and help you lose weight, if needed. Doctors use blood and urine tests to diagnose bladder problems or conditions with similar symptoms. Doctors also may use urodynamic testing to see what kind of bladder problem you have.

Frequent and urgent urination

Some people with diabetes who regularly have high blood glucose levels may have to urinate too often, also called urinary frequency. Even men and women with diabetes who manage their blood glucose levels within their target range sometimes feel the sudden urge to urinate, called urgency incontinence. This can happen at night, also. Medicines may help reduce the symptoms of bladder control problems.

Trouble “going”

You may find that diabetes causes you to no longer feel when your bladder is full. Many people with diabetes report that they have trouble “going.” Over time, having a too-full bladder can cause damage to your bladder muscles that push urine out. When these muscles don’t work correctly, urine may stay in your bladder too long, also called urinary retention. Urinary retention can cause bladder infections, urine leaks, and the feeling that you always have to go.

Leaking urine

People with diabetes are more likely to have other types of UI, such as stress incontinence. Nerve damage, obesity, and bladder infections, which are linked with diabetes, are often related to bladder control problems. Leaking urine can cause you to avoid activities you once enjoyed, including sex.

If you're overweight, losing weight can help you have fewer leaks. Avoiding weight gain may prevent UI. Studies suggest that, as your body mass index (BMI) increases, you're more likely to leak.⁶ If you're overweight or have obesity, talk with your doctor about how to lose weight.

Work with your health care team to help manage and prevent urine leaks. Bladder control problems are often treatable and are very common, even in people who don't have diabetes. You don't have to accept rushing to the bathroom all the time to avoid leaks.

Bladder infections

People with diabetes are more likely to have urinary tract infections, also called bladder infections, or cystitis. See a doctor right away if you have frequent, urgent urination that may be painful. Bladder infections can develop into kidney infections and can make bladder symptoms, such as leaks and urine retention, worse. Also, bladder infections can get in the way of your everyday life, including intimacy. Managing your blood glucose levels can help prevent bladder infections.

Senior couple walking along suburban street holding hands.

Managing your blood glucose levels can help prevent bladder problems.

How can I prevent and treat my sexual or bladder problems?

Managing your diabetes can help prevent nerve damage and other diabetes problems that can lead to sexual and bladder problems. With your health care team, you can help prevent and treat your sexual or bladder control problems by

keeping your blood glucose, blood pressure, and cholesterol levels close to your target numbers

being physically active

keeping a healthy weight

quitting smoking [External link](#) if you smoke

getting help [NIH external link](#) for any emotional or psychological problems

Sex is a physical activity, so be sure to check your blood glucose level before and after sex, especially if you take insulin. Both high blood glucose levels and low blood glucose levels can cause problems during sex.

Counseling may also be helpful when you notice changes in your sexual function or desire. These types of changes are very common as people age or adjust to health problems.

If you have a partner, he or she also may be an important member of your health care team. You may find it helpful to share your concerns and have that person join you at the doctor's office or at counseling. Your friends and family may also be able to support you if you're having bladder problems.

Related Topics

English

Diabetes, Gum Disease, & Other Dental Problems

On this page:

How can diabetes affect my mouth?

What are the most common mouth problems from diabetes?

How will I know if I have mouth problems from diabetes?

What should I do if I have mouth problems from diabetes?

How can I prepare for dental work?

How can I keep my mouth healthy?

Clinical Trials for Diabetes and Gum Disease

Diabetes affects many parts of your body, including your mouth [NIH external link](#). People with diabetes are more likely to have gum disease [NIH external link](#), cavities, and other problems with their teeth and gums.¹ And some of these mouth problems can make your diabetes worse.

Taking good care of your teeth and gums, including getting regular cleanings and needed treatments, will help you prevent these problems or stop them from getting worse. Keeping your mouth healthy will also make it easier for you to manage your diabetes and prevent diabetes-related health problems, such as heart disease and kidney disease.¹

A person getting a dental cleaning with dental tools inserted into an open mouth.

Good dental care can prevent mouth problems and help you manage your diabetes.

How can diabetes affect my mouth?

Diabetes can affect your mouth by changing your saliva—the fluid that keeps your mouth wet. Saliva helps prevent tooth decay by washing away pieces of food, preventing bacteria from growing, and fighting the acids produced by bacteria. Saliva also has minerals that help protect tissues in your mouth and fight tooth decay.

Diabetes and some medicines used to treat diabetes can cause the salivary glands in your mouth to make less saliva. When less saliva flows, the risk for dental cavities, gum disease, and other mouth problems increases.

Diabetes can also increase the amount of glucose in your saliva. Diabetes occurs when your blood glucose level, also called blood sugar, is too high. High levels of glucose in your blood can also cause glucose to build up in your saliva. This glucose can feed harmful bacteria that combine with food to form a soft, sticky film called plaque, which causes cavities. If you don't remove plaque, it can also build up on your teeth near your gum line and harden into a deposit called tartar, which can cause gum disease.

Untreated, these mouth problems can lead to tooth loss. Almost 25% of U.S. adults with diabetes ages 50 and older have severe tooth loss, compared with about 16% of those without diabetes.²

What are the most common mouth problems from diabetes?

Gum disease

Gum disease [NIH external link](#), also called periodontal gum disease, is the most common and serious mouth problem related to diabetes. Untreated, the disease advances in stages, from inflamed gums to tooth loss. High levels of blood glucose increase the risk that gum disease will progress from mild to severe.¹

View full-sized imageClose-up illustration of tooth and gums, showing four stages of gum disease: (1) healthy gums and tooth, (2) gingivitis, (3) periodontitis, and (4) advanced periodontitis. Labels point out inflammation, calculus, pocket, bone destruction, advanced bone loss, and deep pocket.

Gum disease advances in stages, from inflamed gums to tooth loss.

Gingivitis, or inflamed gums

The first stage of gum disease is gingivitis, a mild inflammation of the soft tissues around your teeth. Gingivitis develops when plaque and tartar build up on your teeth near your gum line, irritating and inflaming your gums. As a result, your gums may become red and swollen, and may bleed easily.

Periodontitis

Untreated, gingivitis can progress to periodontitis, an infection of the gums and bone that hold your teeth in place. Your gums may pull away from your teeth, forming pockets that slowly become infected. The bacteria in your mouth and your body's response to the infection start to break down the bone and tissue that hold your teeth in place. If periodontitis is not treated, the teeth can become loose and may even need to be removed.

Other common mouth problems

In addition to gum disease, diabetes also increases your risk for³

dental cavities

dry mouth [NIH external link](#), a lack of saliva that can cause sores, ulcers, and infections

thrush [External link](#), a fungal infection that causes painful, white patches in your mouth

burning mouth syndrome [NIH external link](#), a burning feeling inside your mouth caused by uncontrolled blood glucose levels

changes in the way foods and beverages taste

These mouth problems, as well as tooth loss, can make it harder for you to follow a healthy meal plan that will help you manage your diabetes.

How will I know if I have mouth problems from diabetes?

Check your mouth for signs and symptoms of problems from diabetes, such as¹

gums that are

red, swollen, or bleeding

receding

pulling away from your teeth

loose teeth

increasing spaces between your teeth

dry mouth, an important sign of diabetes

bad breath that doesn't go away when you brush your teeth

A woman checks her teeth in a mirror.

Check your mouth for signs of problems from diabetes.

What should I do if I have mouth problems from diabetes?

See your dentist right away if you notice any signs or symptoms of mouth problems. Your dentist is an important ally in keeping your mouth healthy and treating mouth problems before they become serious.

If you have diabetes

tell your dentist about your diabetes, including how long you've had the disease, any diabetes-related problems you may have, and any medicines you take

keep your dentist updated on any changes to your health, including any increase in your blood glucose levels

ask how often you should come in for regular cleanings and checkups

follow your dentist's advice on how to prevent and treat mouth problems from diabetes

If you feel nervous about visiting the dentist, tell your dentist and the staff about your feelings. Your dentist can adapt the treatment to your needs.

A dental professional examines an older patient's mouth and teeth using dental tools.

If you notice any mouth problems, see your dentist right away.

How can I prepare for dental work?

High blood glucose increases the risk for infections after mouth surgery and can also take your mouth longer to heal. Tell your doctor if you will be having oral surgery or other extensive dental work. To help you prepare for and heal after the procedure, ask your doctor

how often you should check your blood glucose before and after the procedure

what, if any, changes you should make to your medicines, such as changing the dose or the time you take a medicine

how to best treat any soreness after the procedure

How can I keep my mouth healthy?

Keep your mouth healthy [NIH external link \(PDF, 242.78 KB\)](#) by taking these steps

Keep your blood glucose numbers within your target range. Your doctor can help set your target and explain what to do if your numbers are too high or too low.

Consume healthy meals and beverages and follow the meal plan you and your doctor or registered dietitian have worked out.

Visit your dentist regularly. Ask your dentist how to take care of your teeth at home [NIH external link](#), including the cleaning tools and products you should use. Your dentist can recommend a cleaning routine tailored to your needs.

Taking good care of your teeth is not always easy. Challenges can include finding a dentist nearby who can help you develop a dental routine that works for you and being able to afford dental care and related products. If you are worried about costs, consider free or low-cost dental care options [NIH external link](#), such as

community health centers [External link](#) across the country that provide free or reduced-cost health services, including dental care.

Medicaid [External link](#), a state-run program that provides medical benefits to eligible individuals and families. Most states offer limited dental services for adults, while some provide comprehensive services.

Don't let these concerns stop you from having regular checkups. Waiting too long to take care of your mouth may make things worse.

Smoking raises your risk for many diabetes problems and mouth problems. If you smoke or use other tobacco products, stop. Ask for help so you don't have to do it alone. You can start by calling the National Quitline at 1-800-QUITNOW or 1-800-784-8669. For tips on quitting, visit [Smokefree.gov](#) [External link](#).

Clinical Trials for Diabetes and Gum Disease

The NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for diabetes and gum disease?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of diabetes and gum disease, such as how certain genes may play a role in diabetes, gum disease, heart disease, and bone loss.

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Related Topics

English

Diabetes, Heart Disease, & Stroke

On this page:

What is the link between diabetes, heart disease, and stroke?

What else increases my chances of heart disease or stroke if I have diabetes?

How can I lower my chances of a heart attack or stroke if I have diabetes?

How do doctors diagnose heart disease in people with diabetes?

What are the warning signs of heart attack and stroke?

Clinical Trials for Diabetes, Heart Disease, & Stroke

Having diabetes means you are more likely to develop heart disease. People with diabetes are also more likely to have certain risk factors, such as high blood pressure [NIH external link](#) or high cholesterol, that increase their chances of having a heart attack or a stroke.

If you have diabetes, you can protect your heart and health by managing your blood glucose, also called blood sugar. You can also protect yourself by controlling your high blood pressure and high cholesterol. If you smoke, get help to stop.

What is the link between diabetes, heart disease, and stroke?

High blood glucose from diabetes can damage your blood vessels and the nerves that control your heart and blood vessels. Over time, this damage can lead to heart disease.¹

People with diabetes tend to develop heart disease at a younger age than people without diabetes. Adults with diabetes are nearly twice as likely to have heart disease or stroke as adults without diabetes.^{2,3}

The good news is that the steps you take to manage your diabetes also help lower your chances of having heart disease or stroke.

What else increases my chances of heart disease or stroke if I have diabetes?

Your risk for heart disease is greater if you are male rather than female, whether you have diabetes or not.² If you do have diabetes, other factors add to your chances of developing heart disease or having a stroke.

Smoking

Smoking raises your risk of developing heart disease. If you have diabetes, it is important to stop smoking, because both smoking and diabetes narrow blood vessels. Smoking also increases your chances of developing other long-term problems such as

lung disease [NIH external link](#)

lower leg infections and ulcers

foot or leg amputation

High blood pressure

If you have high blood pressure [NIH external link](#), your heart works harder to pump blood. High blood pressure can strain your heart, damage blood vessels, and increase your risk of heart attack, stroke, and eye or kidney problems. Have your blood pressure checked regularly and work with your doctor to control or lower high blood pressure.

Abnormal cholesterol levels

Cholesterol [NIH external link](#) is a type of fat, produced by your liver and found in your blood. You have two kinds of cholesterol in your blood: LDL and HDL.

LDL, often called “bad” cholesterol, can build up and clog your blood vessels. High levels of LDL cholesterol raise your risk of developing heart disease. HDL is sometimes called “good cholesterol.” Higher levels of HDL is linked to lower risk for heart disease and stroke. To improve LDL and HDL levels, limit the amount of fat in your eating plan, eat more plant-based foods, and get regular physical activity.

Another type of blood fat, triglycerides, also can raise your risk of heart disease when the levels are higher than recommended by your health care team.

Obesity and belly fat

Being overweight or having obesity can make it harder to manage your diabetes and raise your risk for many health problems, including heart disease and high blood pressure. If you are overweight, a healthy eating plan with fewer calories and more physical activity often will lower your blood glucose levels and reduce your need for medicines.

Excess belly fat around your waist, even if you are not overweight, can raise your chances of developing heart disease.

You have excess belly fat if your waist measures

more than 40 inches and you are a man

more than 35 inches and you are a woman

Chronic kidney disease

Heart disease is closely linked with chronic kidney disease, a condition in which your kidneys are damaged and can't filter blood the way they should. Having diabetes is a risk factor for developing kidney disease, which affects about 40% of people with diabetes. Other risk factors for developing kidney disease are high blood pressure and a family history of kidney failure.

If you have risk factors, get tested for kidney disease and protect your kidneys by making healthy food choices, being more active, aiming for a healthy weight, and managing health conditions that cause kidney damage.

Family history of heart disease

A family history of heart disease may add to your chances of developing the condition. If one or more of your family members had a heart attack before age 50, you have double the chance of developing heart disease compared with people who have no family history of the disease.⁴

You can't change whether heart disease runs in your family. But if you have diabetes, it's even more important to take steps to protect yourself from heart disease and decrease your chances of having a stroke.

How can I lower my chances of a heart attack or stroke if I have diabetes?

Taking care of your diabetes is important to help you protect your heart. You can substantially lower your chances of having a heart attack or stroke by taking the following steps to keep your heart and blood vessels healthy.⁵

Manage your diabetes ABCs

Know your diabetes ABCs to help you manage your blood glucose, blood pressure, and cholesterol. Stop smoking if you have diabetes to lower your chances of developing heart disease.

A is for the A1C test

The A1C test shows your average blood glucose level over the past 3 months. This is different from the blood glucose checks you do every day. The higher your A1C number, the higher your blood glucose levels have been during the past 3 months. High levels of blood glucose can harm your heart, blood vessels, kidneys, feet, and eyes.

The A1C goal for many people with diabetes is below 7%. Some people may do better with a slightly higher A1C goal. Your A1C goals may also change as you get older and your lifestyle changes. Ask your health care team what your goal should be.

B is for blood pressure

Blood pressure is the force of your blood against the wall of your blood vessels. If your blood pressure gets too high, it makes your heart work too hard. High blood pressure can cause a heart attack or stroke and damage your kidneys and eyes.

The blood pressure goal for most people with diabetes is below 140/90 mm Hg. Ask what your goal should be.

C is for cholesterol

A buildup of cholesterol, a form of fat found in your blood, can cause a heart attack or stroke. Ask your health care team what your cholesterol numbers should be. If you are over 40 years old, you may need to take medicine, such as a statin [NIH external link](#), to lower your cholesterol levels and protect your heart. Some people with very high LDL (“bad”) cholesterol may need to take medicine starting at a younger age.

Learn about getting your cholesterol checked [External link](#).

S is for stop smoking

Quitting smoking is especially important for people with diabetes because both smoking and diabetes narrow blood vessels, so your heart has to work harder. E-cigarettes aren’t a safe option either.

If you quit smoking

you will lower your risk for heart attack; stroke; nerve, kidney, and eye disease; and amputation

your blood glucose, blood pressure, and cholesterol levels may improve

your blood circulation will improve

you may have an easier time being physically active

If you smoke or use other tobacco products, stop. Ask for help so you don't have to do it alone. You can start by calling the National Quitline at 1-800-QUITNOW or 1-800-784-8669. For tips on quitting, go to [Smokefree.gov](https://www.smokefree.gov) External link.

Ask your health care team about your goals for A1C, blood pressure, and cholesterol, and what you can do to reach these goals. To improve your diabetes self-management skills, you may want a referral to a diabetes educator or a registered dietitian. Medicare, some private insurers, and other organizations will provide financial assistance for some of these services.

You can keep track of your ABCs with a diabetes record form. You can use it when you meet with your health care team in person or remotely. Talk with your team about your goals and how you are doing, and whether you need to make any changes to your diabetes care plan.

Develop or maintain healthy lifestyle habits

An overweight woman jogging.

Develop or maintain healthy lifestyle habits. Physical activity can help you manage your diabetes and may help you cope with stress.

Healthy lifestyle habits that you stick with can help you manage your diabetes and prevent heart disease.

Follow your healthy eating plan.

Make physical activity part of your routine.

Stay at or get to a healthy weight.

Get enough sleep.

Learn more about these tips to manage diabetes.

Watch a video about what you can do to keep your heart healthy [External link](#).

Learn about the Body Weight Planner, which may help you create a personal plan to reach your goal weight.

Learn to manage stress

A man and woman exercising in a park.

Yoga and other forms of exercise can help reduce stress.

Managing diabetes is not always easy. Feeling stressed, sad, lonely, or angry is common when you are living with diabetes. You may know what to do to stay healthy but may have trouble sticking with your plan over time. Long-term stress can raise your blood glucose and blood pressure, but you can learn ways to lower your stress. Try deep breathing, gardening, taking a walk, doing yoga, talking with a loved one, working on a hobby, or listening to your favorite music. Learn more about healthy ways to cope with stress [NIH external link](#).

Take medicine to protect your heart

Medicines may be an important part of your treatment plan. Your doctor will prescribe medicine based on your specific needs. Medicine may help you

meet your A1C (blood glucose), blood pressure, and cholesterol goals.

reduce your risk of blood clots, heart attack, or stroke.

treat angina [NIH external link](#), or chest pain that is often a symptom of heart disease. Angina can also be an early symptom of a heart attack.

treat heart failure, which is a form of heart disease in which your heart cannot pump blood well enough for your body to work properly.

An older woman taking her medicine.

Take medicines as prescribed.

Ask your doctor whether you should take daily aspirin. Aspirin is not safe for everyone. Your doctor can tell you whether taking aspirin is right for you and exactly how much to take.

Statins can reduce the risk of having a heart attack or stroke in some people with diabetes. In addition, certain diabetes medicines have been shown to reduce the risk of heart attacks and death in patients at very high risk of having a heart attack. Talk with your doctor to find out whether taking a statin or a diabetes medicine that reduces heart attack risk is right for you.

Take medicines the way your doctor or health care team tells you to. Talk with your doctor or pharmacist if you have questions about your medicines. Before you start a new medicine, ask your doctor about possible side effects and how you can avoid them. If the side effects of your medicine bother you, tell your doctor. Don't stop taking your medicines without checking with your doctor first.

How do doctors diagnose heart disease in people with diabetes?

Doctors diagnose heart disease in people with diabetes [NIH external link](#) based on their

symptoms

medical and family history

how likely they are to have heart disease

a physical exam

results from tests and procedures

Tests used to monitor your diabetes—A1C, blood pressure, and cholesterol—help your doctor decide whether it is important to do additional tests to check your heart health or to refer you to a specialist such as a cardiologist.

A health professional examines a patient in a doctor's office.

Your health care professional will do a physical exam.

Ask your health care team these questions.

What can we do to lower my chances of getting heart disease?

How can we work together to help me maintain my goals for A1C, blood pressure, and cholesterol?

What medicines, such as aspirin or a statin, will help protect my heart?

Are there diabetes medicines that will help protect my heart?

What are the warning signs of heart attack and stroke?

Call 9-1-1 right away if you have any of these warning signs of a heart attack

pain or pressure in your chest that lasts longer than a few minutes or goes away and comes back

pain or discomfort in one or both of your arms or shoulders, or your back, neck, or jaw

shortness of breath

sweating or light-headedness

indigestion or nausea (feeling sick to your stomach)

feeling very tired

Treatment works best when it is given right away. Warning signs can be different in different people. You may not have all the listed symptoms.

Women may experience chest pain, nausea, and vomiting; feel very tired (sometimes for days); and have pain that spreads to the back, neck, throat, arms, shoulders, or jaw. People with diabetes-related nerve damage may not notice any chest pain.

If you have angina [NIH external link](#), it's important to know how and when to seek medical treatment.

Older man with concerned expression making a phone call.

Call 9-1-1 if you have the warning signs of a heart attack or stroke. Treatment works best when given right away.

Call 9-1-1 right away if you have any of these warning signs of a stroke, including sudden

weakness or numbness of your face, arm, or leg on one side of your body

confusion, or trouble talking or understanding

dizziness, loss of balance, or trouble walking

trouble seeing out of one or both eyes

sudden, severe headache

If you have any one of these warning signs, call 9-1-1. You can help prevent permanent damage by getting to a hospital within an hour of a stroke.

Clinical Trials for Diabetes, Heart Disease, & Stroke

The NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for diabetes, heart disease, and stroke?

Clinical trials—and other types of clinical studies—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of diabetes, heart disease, and stroke, such as

risk factors for heart disease and stroke in specific populations, such as Black Americans with diabetes

improved imaging techniques and tests to help diagnose and treat conditions that lead to heart attacks and stroke

the role of genetics in diabetes, heart disease, and stroke

Find out if clinical studies are right for you [NIH external link](#).

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

What clinical trials for heart disease and stroke are looking for participants?

You can view a filtered list of clinical studies on diabetes and heart disease that are federally funded, open, and recruiting at [ClinicalTrials.gov](https://clinicaltrials.gov) NIH external link. A separate list of clinical studies on diabetes and stroke is also available at [ClinicalTrials.gov](https://clinicaltrials.gov) NIH external link. You can expand or narrow the list to include clinical studies from industry, universities, and individuals; however, the National Institutes of Health does not review these studies and cannot ensure they are safe. Always talk with your health care provider before you participate in a clinical study.

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Related Topics

English

Diabetes Tests & Diagnosis

In this section:

Who should be tested for diabetes?

What tests are used to diagnose diabetes?

What test results tell me if I have diabetes or prediabetes?

Which tests help my doctor know what kind of diabetes I have?

Your doctor can diagnose diabetes, prediabetes, and gestational diabetes using blood tests. The blood tests show if your blood glucose level, also called blood sugar, is higher than the range that is healthy for you. Blood tests can also help identify the type of diabetes you have.

Don't try to diagnose yourself if you think you might have diabetes. Blood testing equipment that you can buy over the counter, such as a blood glucose meter, cannot diagnose diabetes.

Who should be tested for diabetes?

Anyone who has symptoms of diabetes should be tested for the disease. Some people will not have any symptoms but may have risk factors for type 2 diabetes and need to be tested. Most pregnant women should also be tested for gestational diabetes. Testing helps doctors find diabetes sooner, so they can work with their patients to manage the disease and prevent health problems.

If you have Medicare, it may cover the cost of tests if you have certain risk factors [External link](#) for diabetes. If you have different insurance, ask your insurance company if it covers diabetes tests.

A Black health care professional taking blood from the arm of a Black male patient.

Blood tests help doctors diagnose diabetes and prediabetes.

Type 1 diabetes

Doctors often diagnose type 1 diabetes in children and young adults when they have diabetes symptoms, such as feeling thirsty and having to urinate more often, or if they may have diabetic ketoacidosis [External link](#). Symptoms in adults may be less specific.¹

Type 1 diabetes can run in families. A study called TrialNet [External link](#) offers free testing to relatives of people with type 1 diabetes, even if the relatives don't have symptoms.

Type 2 diabetes

Experts recommend routine testing for type 2 diabetes if you have certain risk factors. You may have a higher risk of developing type 2 diabetes if you are

age 35 or older

American Indian, Black or African American, Asian American, Hispanic/Latino, or Pacific Islander
overweight or have obesity and have at least one other risk factor

a woman who had gestational diabetes

Take the American Diabetes Association Diabetes Risk Test to see if you are at risk for type 2 diabetes.

Children can also develop type 2 diabetes. Experts recommend testing children and teens between the ages of 10 and 18 who are overweight or have obesity and have at least one more risk factor, such as a low birth weight, a parent who had diabetes while pregnant, or another factor.¹

Adults and children with normal diabetes test results should be retested every 3 years.¹

If your blood glucose levels are higher than the target range, but not high enough to be diagnosed with diabetes, doctors will diagnose you with prediabetes. Having prediabetes is serious because it raises the chance of developing type 2 diabetes. But developing type 2 diabetes can be prevented. Adults and children diagnosed with prediabetes should be tested for type 2 diabetes every year.

Testing for type 2 diabetes before and during pregnancy

All women who are planning to get pregnant should be tested for type 2 diabetes, especially if they have risk factors.

If you are already pregnant, you should be tested for type 2 diabetes at the first prenatal appointment, or in the first 15 weeks of pregnancy. If you have diabetes, you should receive special care during your pregnancy to avoid problems for you and the baby.

Gestational diabetes

All pregnant women who do not have a previous diagnosis of diabetes should be tested for gestational diabetes between 24 and 28 weeks of pregnancy.¹

If you have gestational diabetes, you should get tested after your baby is born to see if you have type 2 diabetes, usually within 12 weeks after delivery.

What tests are used to diagnose diabetes?

Doctors use a variety of tests to diagnose diabetes and prediabetes. Your doctor may recommend different tests depending on whether you have symptoms or not, or whether you are pregnant.

Fasting plasma glucose test

The fasting plasma glucose (FPG) test measures your blood glucose level at a single point in time. For the most reliable results, your doctor will give you the test in the morning after you have fasted for at least 8 hours. Fasting means having nothing to eat or drink except sips of water.

A1C test

The A1C test is a blood test that provides your average levels of blood glucose over the last 3 months. Other names for the A1C test are hemoglobin A1C, HbA1C, glycated hemoglobin, and glycosylated hemoglobin test. You can eat and drink before this test. Before using the A1C test to diagnose diabetes, your doctor will consider factors, such as whether you are in your second or third trimester of pregnancy or whether you have certain types of anemia [NIH external link](#) or another problem with your blood.¹ The A1C test might not be accurate in those cases.

Certain types of hemoglobin, called hemoglobin variants, can interfere with measuring A1C levels. Most A1C tests used in the United States are not affected by the most common variants. If your A1C test results and blood glucose levels do not match, your doctor should consider that the A1C test may not be a reliable test for you.

Your doctor will report your A1C test result as a percentage, such as an A1C of 7%. The higher the percentage is, the higher your average blood glucose levels are.

Random plasma glucose test

Sometimes doctors use the random plasma glucose test to diagnose diabetes when you have symptoms of diabetes and they do not want to wait until you have fasted for 8 hours. You may have this blood test at any time.

Glucose challenge test

If you are pregnant, your doctor might test you for gestational diabetes with the glucose challenge test. Another name for this test is the glucose screening test. In this test, a health care professional will take a sample of your blood 1 hour after you drink a sweet liquid containing

glucose. You do not need to fast for this test. If your blood glucose level is too high—135 mg/dL to 140 mg/dL or higher—you may need to return for an oral glucose tolerance test while fasting.

Oral glucose tolerance test

The oral glucose tolerance test (OGTT) helps doctors detect type 2 diabetes, prediabetes, and gestational diabetes. However, the OGTT is a more expensive test than the FPG test and the glucose challenge test, and it is not as easy to give.

Before the test, you will need to fast for at least 8 hours. A health care professional will take a blood sample to measure your glucose level after fasting. Next, you will drink a liquid that is high in sugar. Another blood sample is taken 2 hours later to check your blood glucose level. If your blood glucose level is high, you may have diabetes.

If you are pregnant, your blood will be drawn every hour for 2 to 3 hours. If your blood glucose levels are high two or more times during the OGTT, you may have gestational diabetes.

What test results tell me if I have diabetes or prediabetes?

Each test to detect diabetes and prediabetes uses a different measurement. Usually, your doctor will use a second test to confirm you have diabetes.

The table below helps you understand what your test results mean if you are not pregnant.¹ If you are pregnant, some tests use different cutoffs. Ask your doctor what your test results mean.

Test results for diagnosis of prediabetes and diabetes

Previous columnNext column

Diagnosis	A1C	Fasting Plasma Glucose
Normal	below 5.7%	99 mg/dL or below
Prediabetes	5.7% to 6.4%	100 to 125 mg/dL
Diabetes	6.5% or above	126 mg/dL or above

* 2 hours after glucose intake

‡ Used when there are symptoms

Source: American Diabetes Association

Which tests help my doctor know what kind of diabetes I have?

Even though the tests described above can confirm you have diabetes, they can't identify what type you have—type 1, type 2, or a rare type of diabetes called monogenic diabetes. Treatment can depend on the type of diabetes you have, so knowing which type you have is important.

Genetic testing

Genetic testing can diagnose most forms of monogenic diabetes, which is caused by a change—or mutation—in a gene. Anyone can have monogenic diabetes, but it is more common in babies who develop diabetes before they are 6 months old, some teens, and some young adults.

Autoantibodies

If you have diabetes, your doctor may look for certain autoantibodies to see if your diabetes is type 1. Autoantibodies are proteins in your immune system that mistakenly attack your healthy tissues and cells. A health care professional will take a sample of your blood for this test. If autoantibodies for diabetes are found in your blood, you may have type 1 diabetes.¹

If you do not have diabetes but your parent or sibling has type 1 diabetes, your doctor might give you this test to see if you have a higher risk of developing type 1 diabetes. You may want to learn about diabetes symptoms and have follow-up testing as recommended by your doctor.¹

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Related Topics

English

Diabetic Eye Disease

On this page:

What is diabetic eye disease?

How does diabetes affect my eyes?

How common is diabetic eye disease?

Who is more likely to develop diabetic eye disease?

What are the symptoms of diabetic eye disease?

When should I see a doctor right away?

How do doctors diagnose eye problems from diabetes?

How do doctors treat diabetic eye disease?

What can I do to protect my eyes?

What if I already have some vision loss from diabetes?

What is diabetic eye disease?

Diabetic eye disease is a group of eye problems that can affect people with diabetes. These conditions include diabetic retinopathy, diabetic macular edema, cataracts, and glaucoma.

Over time, diabetes can cause damage to your eyes that can lead to poor vision or even blindness. But you can take steps to prevent diabetic eye disease, or keep it from getting worse, by taking care of your diabetes.

The best ways to manage your diabetes and keep your eyes healthy are to

manage your blood glucose, blood pressure, and cholesterol, sometimes called the diabetes ABCs

If you smoke, get help to quit smoking [External link](#)

have a dilated eye exam once a year

Eye doctor examines a man's eyes for signs of eye disease during a full, yearly eye exam.

Photo courtesy of the National Eye Institute, NIH.

Your eyes may seem fine, but having a full, dilated eye exam is the only way to know for sure.

Often, there are no warning signs of diabetic eye disease or vision loss when damage first develops. A full, dilated eye exam helps your doctor find and treat eye problems early—often before much vision loss can occur.

How does diabetes affect my eyes?

Diabetes affects your eyes when your blood glucose, also called blood sugar, is too high.

In the short term, you are not likely to have vision loss from high blood glucose. People sometimes have blurry vision for a few days or weeks when they're changing their diabetes care plan or medicines. High glucose can change fluid levels or cause swelling in the tissues of your eyes [NIH external link](#) that help you to focus, causing blurred vision. This type of blurry vision is temporary and goes away when your glucose level gets closer to normal.

If your blood glucose stays high over time, it can damage the tiny blood vessels in the back of your eyes. This damage can begin during prediabetes, when blood glucose is higher than normal, but not high enough for you to be diagnosed with diabetes. Damaged blood vessels may leak fluid and cause swelling. New, weak blood vessels may also begin to grow. These blood vessels can bleed into the middle part of the eye, lead to scarring, or cause dangerously high pressure inside your eye.

Most serious diabetic eye diseases begin with blood vessel problems. The four eye diseases that can threaten your sight are

Diabetic retinopathy

The retina NIH external link is the inner lining at the back of each eye. The retina senses light and turns it into signals that your brain decodes, so you can see the world around you. Damaged blood vessels can harm the retina, causing a disease called diabetic retinopathy.

In early diabetic retinopathy, blood vessels can weaken, bulge, or leak into the retina. This stage is called nonproliferative diabetic retinopathy.

If the disease gets worse, some blood vessels close off, which causes new blood vessels to grow, or proliferate, on the surface of the retina. This stage is called proliferative diabetic retinopathy. These abnormal new blood vessels can lead to serious vision problems.

Two boys hold rubber balls and smile at the viewer.

Photo courtesy of the National Eye Institute, NIH.

Normal vision of two boys playing.

This photo shows how retinopathy might affect vision. Dark spots hang over a view of two boys with rubber balls. The spots block their faces.

Photo courtesy of the National Eye Institute, NIH.

Same scene, as viewed by a person with diabetic retinopathy.

Diabetic macular edema

The part of your retina that you need for reading, driving, and seeing faces is called the macula. Diabetes can lead to swelling in the macula, which is called diabetic macular edema. Over time, this disease can destroy the sharp vision in this part of the eye, leading to partial vision loss or blindness. Macular edema usually develops in people who already have other signs of diabetic retinopathy.

Glaucoma

Glaucoma is a group of eye diseases that can damage the optic nerve—the bundle of nerves that connects the eye to the brain. Diabetes doubles the chances of having glaucoma, which can lead to vision loss and blindness if not treated early.

Symptoms depend on which type of glaucoma you have. Learn more about glaucoma [NIH external link](#).

This photo shows how glaucoma affects vision. Shadows darken all sides of a view of two boys. Only the center is bright enough to see clearly.

Photo courtesy of the National Eye Institute, NIH.

People with glaucoma slowly lose side vision.

Cataracts

The lenses within our eyes are clear structures that help provide sharp vision—but they tend to become cloudy as we age. People with diabetes are more likely to develop cloudy lenses, called cataracts. People with diabetes can develop cataracts at an earlier age than people without diabetes. Researchers think that high glucose levels cause deposits to build up in the lenses of your eyes.

A blurry photo shows two boys with rubber balls. The photo shows how cloudy lenses in the eye, called cataracts, affect vision.

Photo courtesy of the National Eye Institute, NIH.

Cloudy vision and faded colors are symptoms of cataracts.

How common is diabetic eye disease?

Diabetic retinopathy

About one in three people with diabetes who are older than age 40 already have some signs of diabetic retinopathy.¹ Diabetic retinopathy is the most common cause of vision loss in people with diabetes. Each person's outlook for the future, however, depends in large part on regular care. Finding and treating diabetic retinopathy early can reduce the risk of blindness by 95 percent.

Glaucoma and cataracts

Your chances of developing glaucoma or cataracts are about twice that of someone without diabetes.

Who is more likely to develop diabetic eye disease?

Anyone with diabetes can develop diabetic eye disease. Your risk is greater with

high blood glucose that is not treated

high blood pressure that is not treated

High blood cholesterol and smoking may also raise your risk for diabetic eye disease.

Some groups are affected more than others. African Americans, American Indians and Alaska Natives, Hispanics/Latinos, Pacific Islanders, and older adults are at greater risk of losing vision or going blind from diabetes.

If you have diabetes and become pregnant, you can develop eye problems very quickly during your pregnancy. If you already have some diabetic retinopathy, it can get worse during pregnancy. Changes that help your body support a growing baby may put stress on the blood vessels in your eyes. Your health care team will suggest regular eye exams during pregnancy to catch and treat problems early and protect your vision.

Diabetes that occurs only during pregnancy, called gestational diabetes, does not usually cause eye problems. Researchers aren't sure why this is the case.

Your chances of developing diabetic eye disease increase the longer you have diabetes.

What are the symptoms of diabetic eye disease?

Often there are no early symptoms of diabetic eye disease. You may have no pain and no change in your vision as damage begins to grow inside your eyes, particularly with diabetic retinopathy.

When symptoms do occur, they may include

blurry or wavy vision

frequently changing vision—sometimes from day to day

dark areas or vision loss

poor color vision

spots or dark strings (also called floaters)

flashes of light

Talk with your eye doctor if you have any of these symptoms.

When should I see a doctor right away?

Call a doctor right away if you notice sudden changes to your vision, including flashes of light or many more spots (floaters) than usual. You also should see a doctor right away if it looks like a curtain is pulled over your eyes. These changes in your sight can be symptoms of a detached retina [NIH external link](#), which is a medical emergency.

How do doctors diagnose eye problems from diabetes?

Having a full, dilated eye exam is the best way to check for eye problems from diabetes. Your doctor will place drops in your eyes to widen your pupils. This allows the doctor to examine a larger area at the back of each eye, using a special magnifying lens. Your vision will be blurry for a few hours after a dilated exam.

Your doctor will also

test your vision

measure the pressure in your eyes

Your doctor may suggest other tests, too, depending on your health history.

Most people with diabetes should see an eye care professional once a year for a complete eye exam. Your own health care team may suggest a different plan, based on your type of diabetes and the time since you were first diagnosed.

Eye exam guidelines for diabetes 2,3,4:

Type 1: Yearly eye exams should start within 5 years of diagnosis.

Type 2: Yearly eye exams should start right after diagnosis.

Pregnancy: Women with type 1 and type 2 diabetes need an eye exam before pregnancy or within the first 3 months. Your doctor may want you to repeat the exam later in your pregnancy and until your baby is 1 year old.

Women who develop gestational diabetes don't usually need an eye exam because they don't develop diabetic eye disease during pregnancy. If you have any questions, ask your doctor.

How do doctors treat diabetic eye disease?

Your doctor may recommend having eye exams more often than once a year, along with management of your diabetes. This means managing your diabetes ABCs, which include your A1c, blood pressure, and cholesterol; and quitting smoking. Ask your health care team what you can do to reach your goals.

A woman holds a blood glucose testing meter next to a drop of blood on her finger.

Management of blood glucose can help prevent diabetic eye disease and keep it from getting worse.

Doctors may treat advanced eye problems [NIH external link](#) with medicine, laser treatments, surgery, or a combination of these options.

Medicine

Your doctor may treat your eyes with anti-VEGF medicine, such as aflibercept, bevacizumab, or ranibizumab. These medicines block the growth of abnormal blood vessels in the eye. Anti-VEGF medicines can also stop fluid leaks, which can help treat diabetic macular edema.

The doctor will inject an anti-VEGF medicine into your eyes during office visits. You'll have several treatments during the first few months, then fewer treatments after you finish the first round of therapy. Your doctor will use medicine to numb your eyes so you don't feel pain. The needle is about the thickness of a human hair.

Anti-VEGF treatments can stop further vision loss and may improve vision in some people.

Laser Treatment

Laser treatment, also called photocoagulation, creates tiny burns inside the eye with a beam of light. This method treats leaky blood vessels and extra fluid, called edema. Your doctor usually provides this treatment during several office visits, using medicine to numb your eyes. Laser treatment can keep eye disease from getting worse, which is important to prevent vision loss or blindness. But laser treatment is less likely to bring back vision you've already lost compared with anti-VEGF medicines.

There are two types of laser treatment [NIH external link](#):

Focal/grid laser treatment works on a small area of the retina to treat diabetic macular edema.

Scatter laser treatment, also called panretinal photocoagulation (PRP), covers a larger area of the retina. This method treats the growth of abnormal blood vessels, called proliferative diabetic retinopathy.

Vitrectomy

Vitrectomy is a surgery to remove the clear gel that fills the center of the eye, called the vitreous gel. The procedure treats problems with severe bleeding or scar tissue caused by proliferative diabetic retinopathy. Scar tissue can force the retina to peel away from the tissue beneath it, like wallpaper peeling away from a wall. A retina that comes completely loose, or detaches, can cause blindness.

During vitrectomy, a clear salt solution is gently pumped into the eye to maintain eye pressure during surgery and to replace the removed vitreous. Vitrectomy is done in a surgery center or hospital with pain medicine.

Cataract Lens Surgery

In a surgery center or hospital visit, your doctor can remove the cloudy lens in your eye, where the cataract has grown, and replace it with an artificial lens. People who have cataract surgery generally have better vision afterward. After your eye heals, you may need a new prescription for your glasses. Your vision following cataract surgery may also depend on treating any damage from diabetic retinopathy or macular edema.

What can I do to protect my eyes?

To prevent diabetic eye disease, or to keep it from getting worse, manage your diabetes ABCs: your A1c, blood pressure, and cholesterol; and quit smoking if you smoke. Read more information on how to manage diabetes.

Also, have a dilated eye exam [NIH external link](#) at least once a year—or more often if recommended by your eye care professional. These actions are powerful ways to protect the health of your eyes—and can prevent blindness.

The sooner you work to manage your diabetes and other health conditions, the better. And, even if you've struggled in the past to manage your health, taking better care of yourself now can protect your eyes for the future. It's never too late to begin.

What if I already have some vision loss from diabetes?

Ask your eye care professional to help you find a low vision [NIH external link](#) and rehabilitation clinic. Special eye care professionals can help you manage vision loss that cannot be corrected with glasses, contact lenses, medicine, or surgery. Special devices and training may help you make the most of your remaining vision so that you can continue to be active, enjoy hobbies, visit friends and family members, and live without help from others.

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[3] Fraser CE, D'Amico DJ. Diabetic retinopathy: prevention and treatment. UpToDate website. <https://www.uptodate.com/contents/diabetic-retinopathy-prevention-and-treatment> External link. September 28, 2016. Accessed June 2, 2017.

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Related Topics

English

Diabetic Kidney Disease

In this section:

What is diabetic kidney disease?

How does diabetes cause kidney disease?

What increases my chances of diabetic kidney disease?

How can I tell if I have diabetic kidney disease?

How can I keep my kidneys healthy if I have diabetes?

How can I cope with the stress of managing my diabetes?

Does diabetic kidney disease get worse over time?

What is diabetic kidney disease?

Diabetic kidney disease is a type of kidney disease caused by diabetes.

Diabetes is the leading cause of kidney disease. About 1 out of 3 adults with diabetes has kidney disease.¹

The main job of the kidneys is to filter wastes and extra water out of your blood to make urine. Your kidneys also help control blood pressure and make hormones that your body needs to stay healthy.

Illustration of an upper body showing the location of the kidneys.

Your kidneys are located in the middle of your back, just below your rib cage.

When your kidneys are damaged, they can't filter blood like they should, which can cause wastes to build up in your body. Kidney damage can also cause other health problems.

Kidney damage caused by diabetes usually occurs slowly, over many years. You can take steps to protect your kidneys and to prevent or delay kidney damage.

Watch a video about diabetes and kidney disease [External link](#).

What are other names for diabetic kidney disease?

Diabetic kidney disease is also called DKD, chronic kidney disease, CKD, kidney disease of diabetes, or diabetic nephropathy.

How does diabetes cause kidney disease?

High blood glucose, also called blood sugar, can damage the blood vessels in your kidneys. When the blood vessels are damaged, they don't work as well. Many people with diabetes also develop high blood pressure, which can also damage your kidneys. Learn more about high blood pressure and kidney disease.

What increases my chances of developing diabetic kidney disease?

Having diabetes for a longer time increases the chances that you will have kidney damage. If you have diabetes, you are more likely to develop kidney disease if your

blood glucose is too high

blood pressure is too high

African Americans, American Indians, and Hispanics/Latinos develop diabetes, kidney disease, and kidney failure at a higher rate than Caucasians.

You are also more likely to develop kidney disease if you have diabetes and

smoke

don't follow your diabetes eating plan

eat foods high in salt

are not active

are overweight

have heart disease

have a family history of kidney failure

A woman having her blood pressure checked by a health care professional.

If you have diabetes, you are more likely to develop kidney disease if you have high blood pressure.

How can I tell if I have diabetic kidney disease?

Most people with diabetic kidney disease do not have symptoms. The only way to know if you have diabetic kidney disease is to get your kidneys checked.

Health care professionals use blood and urine tests to check for diabetic kidney disease. Your health care professional will check your urine for albumin and will also do a blood test to see how well your kidneys are filtering your blood.

You should get tested every year for kidney disease if you

have type 2 diabetes

have had type 1 diabetes for more than 5 years

One vial with a blood sample and another vial with a urine sample.

Health care professionals use blood and urine tests to check for kidney disease.

How can I keep my kidneys healthy if I have diabetes?

The best way to slow or prevent diabetes-related kidney disease is to try to reach your blood glucose and blood pressure goals. Healthy lifestyle habits and taking your medicines as prescribed can help you achieve these goals and improve your health overall.

Reach your blood glucose goals

Your health care professional will test your A1C. The A1C is a blood test that shows your average blood glucose level over the past 3 months. This is different from the blood glucose checks that you may do yourself. The higher your A1C number, the higher your blood glucose levels have been during the past 3 months.

The A1C goal for many people with diabetes is below 7 percent. Ask your health care team what your goal should be. Reaching your goal numbers will help you protect your kidneys.

To reach your A1C goal, your health care professional may ask you to check your blood glucose levels. Work with your health care team to use the results to guide decisions about food, physical activity, and medicines. Ask your health care team how often you should check your blood glucose level.

Man measuring the level of glucose in his blood.

Protect your kidneys by keeping your blood glucose under control.

Control your blood pressure

Blood pressure is the force of your blood against the wall of your blood vessels. High blood pressure makes your heart work too hard. It can cause heart attack, stroke, and kidney disease.

Your health care team will also work with you to help you set and reach your blood pressure goal. The blood pressure goal for most people with diabetes is below 140/90 mm Hg. Ask your health care team what your goal should be.

Man having his blood pressure checked by a health care professional.

Protect your kidneys by keeping your blood pressure under control.

Medicines that lower blood pressure can also help slow kidney damage. Two types of blood pressure medicines, ACE inhibitors and ARBs, play a special role in protecting your kidneys. Each has been found to slow kidney damage in people with diabetes who have high blood pressure and DKD. The names of these medicines end in –pril or –sartan. ACE inhibitors and ARBs are not safe for women who are pregnant.

Develop or maintain healthy lifestyle habits

Healthy lifestyle habits can help you reach your blood glucose and blood pressure goals.

Following the steps below will also help you keep your kidneys healthy

Stop smoking.

Work with a dietitian to develop a diabetes meal plan and limit salt and sodium.

Make physical activity part of your routine.

Stay at or get to a healthy weight.

Get enough sleep. Aim for 7 to 8 hours of sleep each night.

Learn more about these tips to manage diabetes.

Take medicines as prescribed

Medicines may be an important part of your treatment plan. Your health care professional will prescribe medicine based on your specific needs. Medicine can help you meet your blood glucose and blood pressure goals. You may need to take more than one kind of medicine to control your blood pressure.

Couple talking with a pharmacist.

Talk to your health care professional or pharmacist about all of the medicines you take, including over-the-counter medicines.

How can I cope with the stress of managing my diabetes?

Managing diabetes isn't always easy. Feeling stressed, sad, or angry is common when you are living with diabetes. You may know what to do to stay healthy but may have trouble sticking with your plan over time. Long-term stress can raise your blood glucose and blood pressure, but you can learn ways to lower your stress. Try deep breathing, gardening, taking a walk, doing yoga, meditating, doing a hobby, or listening to your favorite music. Learn more about healthy ways to cope with stress [NIH external link](#).

Does diabetic kidney disease get worse over time?

Kidney damage from diabetes can get worse over time. However, you can take steps to keep your kidneys healthy and help slow kidney damage to prevent or delay kidney failure. Kidney failure means that your kidneys have lost most of their ability to function—less than 15 percent of normal kidney function. However, most people with diabetes and kidney disease don't end up with kidney failure.

If your kidneys become damaged as a result of diabetes, learn how to manage kidney disease.

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Related Topics

English

Financial Help for Diabetes Care

On this page:

What health insurance plans cover people with diabetes?

Local Resources for Care and Costs

Help Paying for Diabetes Medicines and Supplies

Help Paying for Prosthetic Care

Help Paying for Kidney Dialysis and Transplantation

Help for College Students with Diabetes

Assistive Technology for Living with Diabetes

Key Terms

Diabetes management is expensive. According to the American Diabetes Association (ADA), the average cost of health care for a person with diabetes is \$16,752 a year—more than twice the cost of health care for a person without diabetes.¹

Many people who have diabetes need help paying for their care. You can find help through private or government health insurance, local programs, patient support groups, and medicine-assistance programs. You can also find ways to save money on diabetes medicines and supplies.

What health insurance plans cover people with diabetes?

Health insurance helps pay for medical care, including the cost of diabetes care. Your health insurance options are

private health insurance, which includes group and individual health insurance

government health insurance, such as Medicare [External link](#), Medicaid [External link](#), the Children’s Health Insurance Program [External link](#) (CHIP), TRICARE [External link](#), and veterans’ health care programs [External link](#)

Under current law, health insurance companies can’t refuse to cover you or charge you more just because you have a “pre-existing condition [External link](#)”—that is, a health problem you had before the date that new health coverage starts.

Changes to insurance rules happen often and may affect your health insurance choices. Learn more or get help with your health insurance questions through

HealthCare.gov [External link](#) , the federal marketplace, 1-800-318-2596, TTY: 1-855-889-4325
the Center for Consumer Information & Insurance Oversight Program [External link](#) , which offers help by phone, by email, or at walk-in locations

the state health insurance resources tool [External link](#) of the American Diabetes Association

the National Association of Insurance Commissioners [External link](#)

Learn key terms about diabetes costs, insurance, and financial aid.

Compare coverage for diabetes care

When choosing a health insurance plan, ask about every service and supply you’ll need to manage your diabetes. Does the plan cover the medicines, supplies, and devices that you are using, and, if not, what are the alternatives? What does the plan pay? How much will you need to pay out of pocket? Consider

each prescription medicine you need

diabetes supplies and devices, such as an insulin pump

diabetes lab tests and screening exams

medical office visits, nutrition counseling, and specialists, such as eye doctors

copayments and deductibles

health care providers—do your providers participate in the plan, or will you have to change providers or pay more to see your current providers because they are out of network?

Ask for a Summary of Benefits and Coverage, which explains the plan in plain language, including free preventive services. This summary may include a coverage example for managing diabetes.

A couple reviewing paperwork and typing into a calculator.

When looking for health insurance, determine if each plan pays for the services and supplies you need to manage your diabetes.

Private health insurance

Insurance companies sell private health insurance plans. Two types of private health insurance are

Group health insurance. People buy this kind of health insurance through their employer, union, or a family member's employer or union. Professional or alumni groups may also offer group health insurance, which may be helpful for people who are self-employed.

Individual health insurance. People buy this coverage for themselves and their families.

HealthCare.gov is one way to find, compare, and buy individual insurance plans in most states—or connect to your state's insurance marketplace. HealthCare.gov also

tells you whether you qualify for a premium tax credit [External link](#) that lowers your monthly premiums and cost-sharing reductions [External link](#) that lower your out-of-pocket costs. Only people who buy plans through HealthCare.gov or a state exchange can get premium tax credits or cost-sharing reductions.

lists the yearly open enrollment period when you can choose or change your plan

lists the life events that allow you to buy insurance any time of year

Learn more about Healthcare.gov online [External link](#) or by calling 1-800-318-2596, TTY: 1-855-889-4325.

Insurance companies, agents, brokers, and online insurance stores are other ways to find and buy individual health insurance plans. Learn how to find a trustworthy insurance company at USA.gov [External link](#) or by calling 1-844-USA-GOV1 (1-844-872-4681).

Preventive care

You may be eligible for some important preventative services [External link](#) at no additional cost to you. These services can help find and treat health problems early. For example, adults who are overweight or who have obesity or high blood pressure may be able to get diabetes screening tests at no out-of-pocket cost. Adults and children may be able to get obesity screening and weight-management counseling at no out-of-pocket cost.

Keeping health insurance after leaving a job

A federal law called COBRA allows you to stay on your employer's health plan for 18 to 36 months after leaving a job. You'll pay both your own monthly premiums and the employer's portion, so your cost is likely to be higher than before.

People with a disability may be able to extend COBRA coverage for an extra 11 months.

COBRA may also cover young adults who "age out" of a parent's policy when they reach the age limit of 26.

Learn more about COBRA online [External link](#) or call the U.S. Department of Labor at 1-866-4-USA-DOL (1-866-487-2365).

Federal Government health insurance

What is Medicare?

Medicare is a federal health insurance program that pays health care costs for eligible people who are

age 65 or older

under age 65 with certain disabilities

of any age with end-stage renal disease—total and permanent kidney failure that requires a kidney transplant or blood-filtering treatments called dialysis [External link](#)

What health plans does Medicare offer?

Medicare has four parts:

Part A [External link](#) , or hospital insurance, covers hospital stays, skilled nursing homes, hospice care, and some home health care. Part A has no premium for those who have paid enough Medicare taxes. Part A has a deductible, which is an amount you pay for your care each year before the plan begins to pay.

Part B [External link](#), or medical insurance, covers doctor visits, outpatient care, some home health care, medical equipment including insulin pumps, and some preventive services. You pay a monthly premium based on your income. After you pay the deductible each year, Part B pays 80 percent for most covered services, and you pay 20 percent.

Part B covers the Medicare Diabetes Prevention Program [External link](#) (MDPP) for members with prediabetes. The MDPP provides healthy eating and physical activity training sessions and support over 12 months, which research shows can sharply lower diabetes risk.

Medicare Advantage Plans [External link](#) are private insurance plans that combine Medicare Part A and B benefits with extras such as medicine coverage. The Federal Government sets the basic rules for Medicare Advantage plans. However, you'll find many options for extra services, along with different premiums and copayments. Covered services [External link](#) can change from year to year. Ask about

whether you need a referral to see a specialist

whether you must use doctors, clinics, or suppliers that belong to one plan, also called "in network" providers

Part D [External link](#), or prescription drug coverage, is optional coverage offered by private companies, with approval from Medicare. Your premiums, deductible, additional costs, and covered medicines depend on which Part D plan you choose. You may qualify for lower Part D premiums and lower medicine costs, depending on your income and resources. Learn more about the Medicare Extra Help Program [online External link](#).

Other Medicare health plans are available with special rules or for specific groups of people. Medicare offers more information [online External link](#). These alternate health plans include

Medicare Cost Plans, which often include payments for out-of-network providers

Medicare Demonstrations and Pilot Projects, which test improvements in Medicare coverage and care

Programs of All-Inclusive Care for the Elderly (PACE), which combine medical, social, and long-term care services for frail people who live in the community

Medication Therapy Management Programs, which can help you learn how to manage your medications if you're in a Medicare drug plan

Learn more about benefits, rules, and coverage details at [Medicare.gov](https://www.medicare.gov) [External link](#) or by calling 1-800-MEDICARE (1-800-633-4227).

Which diabetes services and supplies does Medicare cover?

Medicare Part B and Part D cover many diabetes services and supplies. You may first need to pay a deductible, copayment, or coinsurance. Some services are free if your doctor orders them.

Medicare Part B helps pay for

diabetes screening tests for people at risk of developing diabetes

diabetes self-management training [External link](#)

diabetes supplies such as glucose monitors, test strips, and lancets

certain continuous glucose monitors and supplies

counseling to help people who have obesity lose weight

eye exams to check for glaucoma and diabetic retinopathy

flu and pneumonia shots

foot exams and treatment for people with diabetes

insulin pumps and insulin that will be used with an insulin pump

medical nutrition therapy services for people with diabetes or kidney disease

therapeutic shoes or inserts

Medicare Part D helps pay for

diabetes medications

diabetes supplies such as needles and syringes for injecting insulin

insulin, except for insulin used with an insulin pump, which is covered by Medicare Part B

Check your own plan to learn exactly what is covered. Each plan, especially Medicare Advantage Plans, may offer different coverage for diabetes care. Read more about Medicare's diabetes coverage rules [External link \(PDF, 671 KB\)](#) online or call 1-800-MEDICARE (1-800-633-4227).

Diabetes supplies on a table, including a lancet, insulin syringe, glucose meter, and test strips.

Medicare pays for many diabetes supplies and drugs under Part B and Part D, which is the optional, extra coverage for medicines.

What is Medigap?

Medigap plans help pay costs that are not covered by Medicare, such as deductibles, copayments, and coinsurance. You can buy a Medigap policy from a licensed insurance company in your home state. Learn more online about Medigap rules, benefits, deadlines, and premiums [External link](#).

What other federal programs can help?

The following federal programs can provide more resources for people with diabetes

U.S. Department of Veterans Affairs [External link](#)

TRICARE [External link](#)

Indian Health Service [External link](#)

Hill-Burton Free and Reduced-cost Health Care Program [External link](#)

Bureau of Primary Health Care [External link](#)

Social Security Administration

Social Security Disability Insurance (SSDI)

Supplemental Security Income (SSI)

Women, Infants, and Children (WIC)

The Social Security Administration can provide financial help through two programs

SSDI pays a monthly amount to people who can't work and have paid enough Social Security taxes. If an illness or injury keeps you out of work for at least a year, SSDI payments may be an option.

SSI pays a monthly amount to disabled children and adults who earn little and don't have many financial assets. A person who gets SSI may be able to get Medicaid and food stamps External link through the Supplemental Nutrition Assistance Program (SNAP), too.

Read more about both SSDI and SSI and how to apply at ssa.gov External link or by calling 1-800-772-1213, TTY: 1-800-325-0778.

WIC serves mothers and children who are at risk for poor nutrition or hunger. This includes low-income women who are pregnant, breastfeeding, or recently had a baby, as well as infants and children up to age 5. Pregnant women who have gestational diabetes may qualify for help through the WIC program. WIC provides

supplemental foods

health care referrals

nutrition education

breastfeeding information

Learn more about WIC online External link, or call the WIC's headquarters at 703-305-2062.

State government health insurance

What is Medicaid?

Medicaid is a health insurance program funded by the Federal Government and state governments for people with low incomes and few assets. Each state runs its own program based on federal rules. Some states cover more services than federal rules require. Many states expanded Medicaid in recent years to cover more people, including low-income adults without dependent children.

Find out whether you qualify for Medicaid in your state, or apply for coverage through

HealthCare.gov External link, 1-800-318-2596, TTY: 1-855-889-4325

National Association of Medicaid Directors state agency locator External link

Learn more at Medicaid.gov External link or call 1-877-267-2323 (TTY: 1-866-226-1819).

What is the Children's Health Insurance Program (CHIP)?

CHIP offers free or low-cost Medicaid to children whose parents earn too much for Medicaid but not enough to pay for a private health plan. In some states, CHIP may also cover pregnant women and parents. Learn whether your family members qualify for CHIP through HealthCare.gov External link or your state's Medicaid or CHIP agency External link.

Learn more about CHIP at www.insurekidsnow.gov External link or call 1-877-543-7669.

What other state programs can help?

Medicare Savings Programs in certain states may pay Medicare premiums, deductibles, and coinsurance if a person has a low income and few assets. Learn more online External link.

State Health Insurance Assistance Programs (SHIPs) give advice about health insurance in every state to people who have Medicare. SHIP counselors can help you choose the best plan for your needs. Find a SHIP counselor at ShiptaCenter.org External link or call 1-877-839-2675.

State Pharmaceutical Assistance Programs help pay for prescription medicines in certain states. Learn what programs are available in your state. External link

Local Resources for Care and Costs

Local health departments and clinics can help people with diabetes find medical care at little or no cost. Search for help near you at FindaHealthCenter.hrsa.gov External link, or call 1-877-464-4772; TTY: 1-877-897-991.

Charitable groups in your community may offer financial help for diabetes care

Lions Clubs International can help with vision care: [LionsClubs.org](https://lionsclubs.org) External link

Shriners Hospitals offer free treatment for children: [ShrinersHospitalsforChildren.org](https://shrinersHospitalsforChildren.org) External link

Kiwanis clubs run service projects to help children and communities: [Kiwanis.org](https://kiwanis.org) External link

Help Paying for Diabetes Medicines and Supplies

Don't hesitate to speak up and talk with your health care team if you have trouble paying for your diabetes medicines and supplies. Ask if there are generic versions of the medicines you need for managing diabetes, blood pressure, and cholesterol. Generic medicines are cheaper than brand-name medicines.

Even when generic versions of medicines aren't available, there may be different, less expensive medicines available to treat diabetes, blood pressure, and cholesterol. If you can't afford a medicine, ask your team if you can switch to a different, less expensive medicine.

Your team may suggest other ways to help you save money, such as using free samples of glucose test strips or other supplies or contacting local programs that may provide financial support.

A health care professional talks about diabetes equipment with a man seated across a desk.

Talk with your health care team if you have trouble paying for your diabetes medicines and supplies.

Drug discount programs may help you find the medicines you need for free or a lower cost. Try these resources

[BenefitsCheckUp.org](https://benefitscheckup.org) External link. Seniors with limited incomes can search for help with medicines, health care, rent, and other needs through this service of the National Council on Aging.

NeedyMeds.org [External link](#). This organization lists programs that help pay for medicines and supplies. You can search by medicine or manufacturer name.

Partnership for Prescription Assistance ([helpingpatients.org](#) [External link](#)). People who don't have insurance coverage for prescriptions may find their medicines and supplies for free or at low cost through [helpingpatients.org](#).

RxAssist.org [External link](#) lists drug-company assistance programs, state programs, discount drug cards, copay help, and more.

Rx Outreach is a nonprofit, mail-order pharmacy that provides affordable medicine to people in need through its website, [RxOutreach.org](#) [External link](#) , or by phone at 1-888-RX0-1234 (1-888-796-1234).

Websites can make it easier to find the exact medicine or supply you need at the lowest cost. Look for reputable websites that sell the same high-quality medicines as your local pharmacy.

Providers strongly advise against skipping doses of medicine or leaving some prescriptions unfilled to save money.

Help Paying for Prosthetic Care

People who have had an amputation may get help paying for rehabilitation and a prosthesis.

Organizations that may help include

Amputee Coalition [External link](#)

Phone: 1-888-AMP-KNOW (1-888-267-5669)

Limbs for Life Foundation [External link](#)

Phone: 1-888-235-5462 or 405-843-5171

Help Paying for Kidney Dialysis and Transplantation

Kidney failure may be a complication of diabetes. People of any age with kidney failure can get Medicare if they meet certain criteria. Learn more about [Financial Help for Treatment of Kidney Failure](#).

Help for College Students with Diabetes

College students who have diabetes-related disabilities may face both the costs of tuition and medical expenses. These expenses may include special equipment and health costs not covered by insurance. Special equipment and support services may be available at your college, through community organizations, your state vocational rehabilitation agency, or specific disability organizations.

The [HEATH Resource Center External link](#) offers information about financial aid and other types of support for college students with disabilities.

Female college student looks at a laptop computer screen while writing in a notebook.

College students with diabetes may qualify for financial help with tuition and medical expenses.

Assistive Technology for Living with Diabetes

Assistive technology is any device that assists or helps someone with a disability function more safely, effectively, and independently at home, at work, and in the community. Assistive technology may include

computers with features that let people with disabilities use them more easily

low-vision devices, such as magnifiers

adaptive equipment, such as wheelchairs

bathroom changes, such as adding grab bars or shower seats

The following organizations offer information, awareness, resources, and training in assistive technology

Able Data

Phone: 1-800-227-0216; TTY: 703-992-8313

[United Cerebral Palsy External link](#)

Phone: 1-800-872-5827 or 202-776-0406

Key Terms

Some terms listed here have many meanings; only those meanings that relate to the financial and medical aspects of diabetes and its management and treatment are included.

coinsurance: an amount a person may still need to pay after a deductible for health care. The amount is most often a percentage, such as 20 percent.

copay (or copayment): an amount a person may have.

Related Topics

[Gestational Diabetes](#)

[View or Print All Sections](#)

Definition & Facts

Gestational diabetes is a type of diabetes that occurs only during pregnancy. Gestational diabetes can cause health problems in both mother and baby. Managing your diabetes can help protect you and your baby.

[Pregnant woman preparing meal with son from fresh vegetables](#)

[Symptoms & Causes](#)

Gestational diabetes often has no symptoms, or they may be mild, such as being thirstier than normal or having to urinate more often. Gestational diabetes is sometimes related to the hormonal changes of pregnancy that make your body less able to use insulin. Genes and extra weight may also play a role.

Tests & Diagnosis

Your doctor will test you for gestational diabetes between 24 and 28 weeks of pregnancy. Tests include the glucose challenge test and the oral glucose tolerance test (OGTT). If the results of the glucose challenge test show high blood glucose, you will return for an OGTT test to confirm the diagnosis of gestational diabetes.

Management & Treatment

Managing gestational diabetes includes following a healthy eating plan and being physically active. If your eating plan and physical activity aren't enough to keep your blood glucose in your target range, you may need insulin.

Prevention

You can lower your chance of getting gestational diabetes by losing extra weight before you get pregnant if you are overweight. Being physically active before and during pregnancy also may help prevent gestational diabetes.

After Your Baby is Born

If you had gestational diabetes, you are more likely to develop type 2 diabetes. Your child is more likely to have obesity or develop type 2 diabetes. You may be able to lower your and your child's chances of developing these problems by reaching a healthy weight, making healthy food choices, and being physically active.

Related Conditions & Diseases

Health Tips for Pregnant Women

Type 2 Diabetes

Clinical Trials

The NIDDK and other components of the National Institutes of Health (NIH) support and conduct research into many diseases and conditions.

View clinical trials that are currently recruiting volunteers.

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Related Topics

English

Insulin, Medicines, & Other Diabetes Treatments

On this page:

What medicines might I take for diabetes?

What type of diabetes do I have?

What are the different types of insulin?

What are the different ways to take insulin?

What oral medicines treat type 2 diabetes?

What other injectable medicines treat diabetes?

What should I know about side effects of diabetes medicines?

What questions should I ask about my diabetes medicines?

Do I have other treatment options for my diabetes?

Clinical Trials for Insulin, Medicines, & Other Diabetes Treatments

Taking insulin or other diabetes medicines is often part of treating diabetes. In addition to making healthy food and beverage choices, getting physical activity, getting enough sleep, and

managing stress, medicines can help you manage the disease. Some other treatment options are also available.

What medicines might I take for diabetes?

The medicine you take depends on the type of diabetes you have and how well the medicine controls your blood glucose levels, also called blood sugar levels. Other factors, such as any other health conditions you may have, medication costs, your insurance coverage and copays, access to care, and your lifestyle, may affect what diabetes medicine you take.

What type of diabetes do I have?

Type 1 diabetes

If you have type 1 diabetes, you must take insulin because your pancreas does not make it. You will need to take insulin several times during the day, including when you eat and drink, to control your blood glucose level.

There are different ways to take insulin. You can use a needle and syringe, an insulin pen, or an insulin pump. An artificial pancreas—also called an automated insulin delivery system—may be another option for some people.

Type 2 diabetes

Some people with type 2 diabetes can control their blood glucose level by making lifestyle changes. These lifestyle changes include consuming healthy meals and beverages, limiting calories if they have overweight or obesity, and getting physical activity.

Many people with type 2 diabetes need to take diabetes medicines as well. These medicines may include diabetes pills or medicines you inject, such as insulin. Over time, you may need more than one diabetes medicine to control your blood glucose level. Even if you do not take insulin, you may need it at special times, such as if you are pregnant or if you are in the hospital for treatment.

Gestational diabetes

If you have gestational diabetes, you can manage your blood glucose level by following a healthy eating plan and doing a moderate-intensity physical activity, such as brisk walking for 150 minutes, each week. If consuming healthy food and beverages and getting regular physical activity aren't enough to keep your blood glucose level in your target range, a doctor will work with you and may recommend you take insulin. Insulin is safe to take while you are pregnant.

No matter what type of diabetes you have, taking diabetes medicines every day can feel like a burden sometimes. New medications and improved delivery systems can help make it easier to manage your blood glucose levels. Talk with your doctor to find out which medications and delivery systems will work best for you and fit into your lifestyle.

What are the different types of insulin?

Several types of insulin are available. Each type starts to work at a different speed, known as "onset," and its effects last a different length of time, known as "duration." Most types of insulin reach a peak, which is when they have the strongest effect. After the peak, the effects of the insulin wear off over the next few hours or so. Table 1 lists the different types of insulin, how fast they start to work, when they peak, and how long they last.

Table 1. Types of insulin and how they work^{1,2}

Insulin Type	How Fast It Starts to Work (onset)	When It Peaks	How Long It Lasts (duration)
rapid-acting/ ultra rapid-acting	15 minutes	1 hour	2 to 4 hours (rapid) 5 to 7 hours (ultra)
rapid-acting, inhaled	10 to 15 minutes	30 minutes	3 hours
regular, also called short-acting	30 minutes	2 to 3 hours	3 to 6 hours
intermediate-acting	2 to 4 hours	4 to 12 hours	12 to 18 hours
long-acting	2 hours	does not peak	24 hours

ultra long-acting 6 hours does not peak 36 hours or longer

Another type of insulin, called premixed insulin, is a combination of insulins listed in Table 1. Premixed insulin starts to work in 15 to 60 minutes and can last from 10 to 16 hours. The peak time varies depending on which insulins are mixed.

Your doctor will work with you to review your medication options. Talk with your doctor about your activity level, what you eat and drink, how well you manage your blood glucose levels, your age and lifestyle, and how long your body takes to absorb insulin.

Follow your doctor's advice on when and how to take your insulin. If you're worried about the cost, talk with your doctor. Some types of insulin cost more than others. You can also find resources to get financial help for diabetes care.

What are the different ways to take insulin?

The way you take insulin may depend on your lifestyle, insurance plan, and preferences. Talk with your doctor about the options and which one is best for you. Most people with diabetes take insulin using a needle and syringe, insulin pen, or insulin pump. Inhalers and insulin jet injectors are less common ways to take insulin. Artificial pancreas systems are now approved by the U.S. Food and Drug Administration (FDA). Talk with your doctor to see if an artificial pancreas is an option for you.

Needle and syringe

You can give yourself insulin shots using a needle and syringe. You draw up your dose of insulin from the vial—or bottle—through the needle into the syringe. Insulin works fastest when you inject it in your belly, but your doctor may recommend alternating the spot where you inject it. Injecting insulin in the same spot repeatedly could cause the tissue to harden, making it harder to take shots in that area over time. Other spots you can inject insulin include your thigh, buttocks, or upper arm, but it may take longer for the insulin to work from those areas. Some people with diabetes who take insulin need 2 to 4 shots a day to reach their blood glucose targets. Others can take a single shot. Injection aids can help you give yourself the shots.

Two syringes and a vial containing insulin.

Insulin shots involve drawing insulin from a vial into a syringe and then injecting it.

Pen

An insulin pen looks like a writing pen but has a needle for its point. Some insulin pens come filled with insulin and are disposable. Others have room for an insulin cartridge that you insert and replace after use. Many people find insulin pens easier to use, but they cost more than needles and syringes. You may want to consider using an insulin pen if you find it hard to fill the syringe while holding the vial or cannot read the markings on the syringe. Different pen types have features that can help with your injections. Some reusable pens have a memory function, which can recall dose amounts and timing. Other types of “connected” insulin pens can be programmed to calculate insulin doses and provide downloadable data reports, which can help you and your doctor adjust your insulin doses.

An insulin pen

An insulin pen is a convenient way to take insulin.

Pump

An insulin pump is a small machine that gives you steady doses of insulin throughout the day. You wear one type of pump outside your body on a belt or in a pocket or pouch. The insulin pump connects to a small plastic tube and a very small needle. You insert the plastic tube with a needle under your skin, then take out the needle. The plastic tube will stay inserted for several days while attached to the insulin pump. The machine pumps insulin through the tube into your body 24 hours a day and can be programmed to give you more or less insulin based on your needs. You can also give yourself doses of insulin through the pump at mealtimes.

Another type of pump has no tubes. This pump attaches directly to your skin with a self-adhesive pad and is controlled by a hand-held device. The plastic tube and pump device are changed every several days.

A woman holds an insulin pump with the tube connected to a patch on her stomach, where the needle is inserted.

Insulin pumps can be programmed to deliver insulin directly into your body, 24 hours a day or at certain times.

Inhaler

Another way to take insulin is by breathing powdered insulin into your mouth from an inhaler device. The insulin goes into your lungs and moves quickly into your blood. You may want to use an insulin inhaler [NIH external link](#) to avoid using needles. Inhaled insulin is only for adults with type 1 or type 2 diabetes. Taking insulin with an inhaler is less common than using a needle and syringe.

Jet injector

A jet injector is a device that sends a fine spray of insulin into the skin at high pressure instead of using a needle to deliver the insulin. It is used less commonly than a needle and syringe or a pen.

Artificial pancreas

An artificial pancreas is a system of three devices that work together to mimic how a healthy pancreas controls blood glucose in the body. A continuous glucose monitor (CGM) tracks blood glucose levels every few minutes using a small sensor inserted under the skin that is held in place with an adhesive pad. The CGM wirelessly sends the information to a program on a smartphone or an insulin infusion pump. The program calculates how much insulin you need. The insulin infusion pump will adjust how much insulin is given from minute to minute to help keep your blood glucose level in your target range. An artificial pancreas is mainly used to help people with type 1 diabetes.

What oral medicines treat type 2 diabetes?

You may need to take medicines to manage your type 2 diabetes, in addition to consuming healthy foods and beverages and being physically active. You can take many diabetes medicines by mouth. These medicines are called oral medicines.

Most people with type 2 diabetes start with metformin [NIH external link](#) pills. Metformin also comes as a liquid. Metformin helps your liver make less glucose and helps your body use insulin better. This drug may help you lose a small amount of weight.

Other oral medicines act in different ways to lower blood glucose levels. Combining two or three kinds of diabetes medicines can lower blood glucose levels better than taking just one medicine.

Read about different kinds of diabetes medicines [External link \(PDF, 2.8 MB\)](#) from the FDA.

What other injectable medicines treat diabetes?

Type 1 diabetes

If you have type 1 diabetes, your doctor may recommend you take other medicines, in addition to insulin, to help control your blood glucose. Some of these medicines work to slow how fast food and beverages move through your stomach. These medicines also slow down how quickly and how high your blood glucose levels rise after eating. Other medicines work to block certain hormones in your digestive system that raise blood glucose levels after meals or help the kidneys to remove more glucose from your blood.

Type 2 diabetes

Besides insulin, other types of injected medicines [External link \(PDF, 2.8 MB\)](#) are available that will keep your blood glucose level from rising too high after you eat or drink. These medicines, known as glucagon-like peptide-1 (GLP-1) receptor agonists,³ may make you feel less hungry and help you lose some weight. GLP-1 medicines are not substitutes for insulin.

What should I know about side effects of diabetes medicines?

Side effects are problems that result from taking a medicine. Some diabetes medicines can cause hypoglycemia, also called low blood glucose, if you don't balance your medicines with food and activity.

Ask your doctor whether your diabetes medicine can cause hypoglycemia or other side effects, such as upset stomach and weight gain. Aim to take your diabetes medicines as your doctor instructs you, to help prevent side effects and diabetes problems.

What questions should I ask about my diabetes medicines?

Ask your doctor these questions when you get a prescription for a medicine. You may want to make copies of this list and fill it out for each of your medicines.

What are the names of my medicine?

Brand name: _____

Generic name: _____

What does my medicine do?

When should I start this medicine?

This medicine is prescribed by: _____

How long will this medicine take to work?

What is the strength (for example, how many milligrams, written as mg)?

How much should I take for each dose?

How many times a day should I take my medicine?

At what times should I take my medicine?

Should I take it before, with, or after a meal?

Should I avoid any foods or medicines when I take it?

Should I avoid alcoholic beverages when I take it?

Are there any times when I should change the amount of medicine I take?

What should I do if I forget to take it?

If I'm sick and can't keep food down, should I still take my medicine?

Can my diabetes medicine cause low blood glucose?

What should I do if my blood glucose is too low?

What side effects can this medicine cause?

What should I do if I have side effects?

How should I store this medicine?

Do I have other treatment options for my diabetes?

If medicines and lifestyle changes are not enough to manage your diabetes, there are other treatments that might help you. These treatments include weight-loss (bariatric) surgery for

certain people with type 1 or type 2 diabetes, or pancreatic islet transplantation for some people with type 1 diabetes.

Weight-loss surgery

Weight-loss surgery are operations that help you lose weight by making changes to your digestive system. Weight-loss surgery is also called bariatric or metabolic surgery.

This type of surgery may help some people who have obesity and type 2 diabetes lose a large amount of weight and bring their blood glucose levels back to a healthy range. How long the improved response lasts can vary by patient, type of weight-loss surgery, and the amount of weight the person lost. Other factors include how long a person had diabetes and whether the person used insulin. Some people with type 2 diabetes may no longer need to use diabetes medicines after weight-loss surgery.⁴

Researchers are studying whether weight-loss surgery can help control blood glucose levels in people with type 1 diabetes who have obesity.⁵

Pancreatic islet transplantation

Pancreatic islet transplantation is a treatment for type 1 diabetes in people who struggle to manage their blood glucose levels. Pancreatic islets are clusters of cells in the pancreas that make the hormone insulin. In type 1 diabetes, the body's immune system attacks these cells. A pancreatic islet transplantation replaces destroyed islets with new islets from a deceased donor. The new islets make and release insulin.

Clinical Trials for Insulin, Medicines, & Other Diabetes Treatments

The NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for insulin, medicines, and other diabetes treatments?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help health care professionals and researchers learn more about disease and improve health care for people in the future.

Find out if clinical trials are right for you [NIH external link](#).

Researchers are studying many aspects of diabetes medicines, including

new types of insulin

the most effective times to take diabetes medicines

new types of monitoring devices and delivery systems

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

What clinical trials for insulin, medicines, and other diabetes treatments are looking for participants?

You can view a filtered list of clinical studies on insulin, medicines, and other diabetes treatments covered in this health topic that are federally funded, open, and recruiting at

[ClinicalTrials.gov NIH external link](#). You can expand or narrow the list to include clinical studies from industry, universities, and individuals; however, the National Institutes of Health does not review these studies and cannot ensure they are safe. Always talk with your health care provider before you participate in a clinical study.

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Related Topics

English

Low Blood Glucose (Hypoglycemia)

On this page:

What is low blood glucose?

How common is low blood glucose?

Who is more likely to develop low blood glucose?

What are the symptoms of low blood glucose?

What are the complications of low blood glucose?

What causes low blood glucose in people with diabetes?

How can I prevent low blood glucose if I have diabetes?

How do I treat low blood glucose?

What if I have severe low blood glucose and can't treat myself?

Clinical Trials for Low Blood Glucose

What is low blood glucose?

Low blood glucose, also called low blood sugar or hypoglycemia, occurs when the level of glucose in your blood drops below what is healthy for you. For many people with diabetes, this means a blood glucose reading lower than 70 milligrams per deciliter (mg/dL).¹ Your number might be different, so check with your doctor or health care team to find out what blood glucose level is low for you.

How common is low blood glucose?

Low blood glucose is common among people with type 1 diabetes and among people with type 2 diabetes who take insulin or some other diabetes medicines. In a large global study of people with diabetes who take insulin, 4 in 5 people with type 1 diabetes and nearly half of those with type 2 diabetes reported a low blood sugar event at least once over a 4-week period.²

Severely low blood glucose, defined as when your blood glucose level drops so low you can't treat it yourself, is less common. Among U.S. adults with diabetes who take insulin or some diabetes medicines that help the pancreas release insulin into the blood, 2 in 100 may develop severely low blood glucose each year.³

Who is more likely to develop low blood glucose?

You are more likely to develop low blood glucose if you⁴

have type 1 diabetes

take insulin or some other diabetes medicines

are age 65 or older⁵

had low blood glucose before

have other health problems, such as kidney disease, heart disease, or cognitive impairment [NIH external link](#)

What are the symptoms of low blood glucose?

Symptoms of low blood glucose tend to come on quickly and can vary from person to person. As Table 1 below shows, symptoms can range from mild to severe.

Table 1. Symptoms of low blood glucose⁶

Mild-to-Moderate	Severe
------------------	--------

If your blood glucose level is low, you may feel

shaky or jittery	
------------------	--

hungry	
--------	--

tired	
-------	--

dizzy, lightheaded, confused, or irritable	
--	--

your heart is beating too fast or not steadily

you have a headache

you can't see or speak clearly

If your blood glucose level is very low, your brain may stop working as it should. You may

lose consciousness

have a seizure

Severe hypoglycemia is dangerous and needs to be treated right away.

Low blood glucose during sleep

Your blood glucose level can drop while you sleep and stay low for several hours, causing serious problems.⁷ Symptoms of low blood glucose while you sleep can include

crying out or having nightmares

sweating enough to make your pajamas or sheets damp

feeling tired, irritable, or confused after waking up

Although you may not wake up or notice any symptoms, low blood glucose can interfere with your sleep, which may affect your quality of life, mood, and ability to work. Having low blood glucose during sleep can also make you less likely to notice and respond to symptoms of low blood glucose during the day.

What are the complications of low blood glucose?

Mild-to-moderate low blood glucose can be easily treated. But severely low blood glucose can cause serious complications, including passing out, coma, or death.

Repeated episodes of low blood glucose can lead to

high blood glucose levels, if worry or fear of low blood glucose keeps you from taking the medicines you need to manage your diabetes⁸

hypoglycemia unawareness, a condition in which you don't notice any symptoms of low blood glucose until your blood glucose level has dropped very low

What causes low blood glucose in people with diabetes?

Low blood glucose levels can be a side effect of insulin or some other medicines that help your pancreas release insulin into your blood. Taking these can lower your blood glucose level.

Diabetes medicines in a pill bottle, a vial of insulin, and a syringe.

Insulin and some other diabetes medicines can lower your blood glucose level.

Two types of diabetes pills can cause low blood glucose

sulfonylureas, usually taken once or twice per day, which increase insulin over several hours

meglitinides, taken before meals to promote a short-term increase in insulin

The following may also lower your blood glucose level

Not eating or drinking enough carbohydrates (carbs). When you eat foods or drink beverages that contain carbohydrates, your digestive system breaks down the sugars and starches into glucose. Glucose then enters your bloodstream and raises your blood glucose level. If you don't eat enough carbohydrates or skip or delay any meals, your blood glucose level could drop lower than what is healthy for you.

Fasting. Continuing to take medicines that lower your glucose level while fasting for a medical procedure or for other purposes can also increase the risk for low blood glucose.

Increasing physical activity. Increasing your physical activity level beyond your usual routine can lower your blood glucose level for up to 24 hours after the activity. Learn more about diabetes, eating, and physical activity.

Drinking too much alcohol without enough food. Alcohol makes it harder for your body to keep your blood glucose level steady, especially if you haven't eaten in a while. Alcohol can also keep you from feeling the first symptoms of low blood glucose, which can lead to severe symptoms.

Being sick. When you're sick, you may not be able to eat as much or keep food down, which can lower blood glucose.

How can I prevent low blood glucose if I have diabetes?

If you take insulin or other medicines that lower blood glucose, the following actions may help you prevent low blood glucose levels.

Ask your doctor or health care team how to check your blood glucose level. The most common way to do so is by using a blood glucose meter. If you have hypoglycemia unawareness or have low blood glucose often, a continuous glucose monitor (CGM) may be a good option. The CGM measures your blood glucose level at regular times and can sound an alarm if it drops below your target range.

Make sure your regular eating plan includes meals, snacks, and beverages with enough carbohydrates to help keep your blood glucose level in your target range. Carry a source of fast-acting carbohydrate, such as glucose tablets or a juice box, with you. Also, if you drink alcoholic beverages, it's safer to eat some food at the same time.

Be safe during exercise or physical activity. Physical activity can lower your blood glucose during the activity and for hours afterward. You may need to check your blood glucose before, during, and after physical activity and adjust your medicine or carbohydrate intake to prevent low blood glucose. For example, you might eat a snack before physical activity to prevent low blood glucose.

Work with your doctor or health care team to prevent low blood glucose. Ask your team if any of your diabetes medicines can cause low blood glucose and how you can prevent and treat symptoms. Work with your doctor or health care team to adjust your diabetes management plan as needed.

A woman with a smart sensor attached to her arm looks at a mobile phone that shows her blood glucose reading.

Some CGMs let you track your blood glucose levels using a small sensor attached to your arm or abdomen and view the information on your smartphone.

How do I treat low blood glucose?

A woman holds a glass of fruit juice..

If your blood glucose is below your target, take 15 to 20 grams of glucose or carbohydrates right away.

If you begin to feel one or more symptoms of low blood glucose, check your blood glucose level. If your blood glucose level is below your target or less than 70 mg/dL, follow these steps

Eat or drink 15 to 20 grams of glucose or carbohydrates right away. Examples include four glucose tablets or one tube of glucose gel.

1/2 cup (4 ounces) of fruit juice—not low-calorie or reduced-sugar juice. If you have kidney disease, don't drink orange juice because it has a lot of potassium. Apple, grape, or cranberry juice are good options.

1/2 can (4 to 6 ounces) of soda—not low-calorie or reduced-sugar soda.

1 tablespoon of sugar, honey, or corn syrup.

Wait 15 minutes and check your blood glucose level again. If your glucose level is still low, eat or drink another 15 to 20 grams of glucose or carbohydrates.

Check your blood glucose level again after another 15 minutes. Repeat these steps until your glucose level is back to your target range.

If your next meal is more than 1 hour away, have a snack to keep your blood glucose level within a range that is healthy for you. Try crackers or a piece of fruit.

Treating low blood glucose if you take medicines that slow down digestion

Some diabetes medicines slow down the digestion of carbohydrates to keep blood glucose levels from rising too high after you eat. If you develop low blood glucose while taking these medicines, you will need to take glucose tablets or glucose gel right away. Eating or drinking other sources of carbohydrates won't raise your blood glucose level quickly enough.

What if I have severe low blood glucose and can't treat myself?

Glucagon—a hormone that raises blood glucose levels—is the best way to treat severely low blood glucose. Available as an injection or a nasal spray, glucagon will quickly raise your blood glucose level. Your doctor can prescribe you a glucagon kit for use in case of an emergency.

If your blood glucose level drops very low, you won't be able to treat it by yourself. Be prepared to address severely low blood glucose by

talking with your doctor or health care team about when and how to use a glucagon emergency kit. If you have an emergency kit, regularly check the date on the package to make sure it hasn't expired.

teaching your family, friends, and coworkers when and how to give you glucagon. Tell them to call 911 right away after giving you glucagon or if you don't have a glucagon emergency kit with you.

wearing a medical alert identification bracelet or pendant. A medical alert ID tells other people that you have diabetes and need care right away. Getting prompt care can help prevent the serious problems that low blood glucose levels can cause.

Clinical Trials for Low Blood Glucose

The NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for low blood glucose?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of low blood glucose levels in diabetes, such as

how to diagnose and treat low blood glucose among people with diabetes

medicines that can treat symptoms of low blood glucose in people with hypoglycemia
unawareness

educational approaches to reduce fear of low blood glucose, which can make it harder for you to control your diabetes

Find out if clinical studies are right for you [NIH external link](#).

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

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Related Topics

English

Managing Diabetes

On this page:

What are the diabetes ABCs?

How can a healthy lifestyle help manage my diabetes?

What types of medicines should I take?

How can I check my blood glucose level at home?

How can I keep track of my progress?

What special situations should I plan for?

Who should be on my health care team?

Clinical Trials for Managing Diabetes

Diabetes can affect almost every part of your body, including your heart, eyes, kidneys, and nerves. Diabetes is also linked to some types of cancer. You may be able to prevent or delay diabetes health problems by leading a healthy lifestyle, taking your medicines, and managing your blood glucose level, also called blood sugar level. Work with your health care team to create a diabetes care plan that works for you.

What are the diabetes ABCs?

The diabetes ABCs can help you manage your blood glucose, blood pressure, and cholesterol levels. If you smoke, quitting is also important. Managing your ABCs may lower your chances of having health problems from diabetes, such as a heart attack, stroke, kidney disease, blindness, and foot or leg amputations.

A is for the A1C blood glucose test

The A1C test shows your average blood glucose level over the last 3 months. This test is done at a health care professional's office or lab. For most people with diabetes, their goal is an A1C level below 7%.¹ Your A1C goal may be different if you have other health problems.² Ask your health care team how often you should have the test and what your goal should be.

B is for blood pressure

High blood pressure [NIH external link](#) can damage your heart, kidneys, brain, and eyes. Some people with diabetes have a blood pressure goal below 130/80 mm Hg.^{2,3} If you have heart disease or are at high risk for diabetes health problems, your goal may be lower. Ask your health care team what your goal should be.

C is for cholesterol

Cholesterol [NIH external link](#) is a fat, also called lipid, that is produced by your liver. Unhealthy levels of cholesterol in your blood can build up and clog your blood vessels, which may result in a heart attack or a stroke. Ask your health care team how often you need a cholesterol test [NIH external link](#), also called a lipid panel, and what your cholesterol level should be. Some people may need to take a medicine called a statin [External link](#), or another medicine, to lower their cholesterol for heart health.

S is for stop smoking

Not smoking is important for your health, especially if you have diabetes. Both smoking and diabetes can make your blood vessels narrow. E-cigarettes aren't a safe option either. If you smoke, vape, or use other tobacco products, stop. Ask for help [External link](#) so you don't have to do it alone.

Child writing “A+B+C+S = Heart” on a school chalkboard.

Follow your ABCs to keep your heart healthy. A is for the A1C blood glucose test, B is for blood pressure, C is for cholesterol, and S is for stop smoking.

How can a healthy lifestyle help manage my diabetes?

Staying healthy when you have diabetes can be a challenge. But leading a healthy lifestyle may help keep your blood glucose, blood pressure, and cholesterol levels in the ranges recommended by your health care team.

Create a healthy meal plan

You may worry that having diabetes means going without foods you enjoy. The good news is that you can still eat your favorite foods, but you might need to eat smaller portions or enjoy them less often. Work with your health care team, which may include a registered dietitian or a diabetes educator, to create a meal plan that works for you.

Talk with your health care team about any challenges you may have finding healthy food [External link](#). A doctor, registered dietitian, diabetes educator, or social worker may be able to recommend resources that can help you. The U.S. Department of Agriculture also provides a list of resources [External link](#) about affording healthy food and finding help from food assistance programs.

Get physical activity

Physical activity may help you lower blood glucose, blood pressure, and cholesterol levels. Being active may also help you get better sleep and improve your mood. Try to get at least 150 minutes of moderate-intensity [External link](#) physical activity, such as brisk walking, each week. If you can, try to do muscle-strengthening activities, such as wall push-ups or seated arm raises, two days a week.⁴

Reach and maintain a healthy weight

If you are overweight or have obesity, ask your health care team how you can manage your weight. To lose weight, you may need to consume fewer calories or get more physical activity. Your health care team may also recommend medicines or surgery to help manage your weight.

Get enough sleep

Getting enough sleep may improve your mood, energy level, and blood glucose level. Most adults should aim for about 7 to 8 hours each night.⁵ Children and adolescents may need more sleep.

Take care of your mental health

Feeling stressed, sad, or angry can be common for people with diabetes. Many people with chronic, or long-term, illnesses such as diabetes develop anxiety or other mental health conditions [NIH external link](#). Learn healthy ways to lower your stress [NIH external link](#), and ask for help from your health care team or a mental health professional.

Grandfather playing soccer with his grandson on the beach.

Being physically active can help you manage your diabetes.

What types of medicines should I take?

Taking diabetes medicines is often part of managing diabetes. The medicines you take depend on your type of diabetes, how well medicines control your blood glucose level, and other health conditions you may have. You may also need medicines to help manage blood pressure or cholesterol levels. Other factors such as medicine costs, insurance coverage, who is providing your diabetes care, and your lifestyle can also affect what medicines you take.

Take your medicines as they were prescribed, even if you feel good or have reached your blood glucose, blood pressure, or cholesterol goals. Avoid rationing your medicines or skipping taking your medicines if you have trouble paying for them. Talk with your health care team if you can't afford your medicines or if you experience any side effects.

How can I check my blood glucose level at home?

If you take insulin or other diabetes medicines, checking your blood glucose level at home may help you avoid high or low blood glucose levels. Even if you do not use insulin, checking your blood glucose level at home can help you manage your diabetes. The test results may help you and your health care team make changes to your meal plan, physical activity level, or diabetes

medicines. It may help you understand how certain types of food affect your blood glucose level. You do not get the same level of detail from blood tests—such as the A1C test—that you take at a health care professional’s office.

If you have type 1 diabetes, your health care team may ask you to check your urine at home for substances called ketones. A large amount of ketones in your urine may mean that you need to manage your diabetes more carefully. Work with your primary health care professional—who may be a doctor, physician assistant, or nurse practitioner—to change your diabetes care plan. If high levels of ketones are left untreated, you may develop a dangerous condition called diabetic ketoacidosis [External link](#).

A woman checking her blood glucose level with a blood glucose meter.

Checking and recording your blood glucose level is an important part of managing diabetes.

Devices to help manage diabetes

You can check your blood glucose level at home with a blood glucose meter [External link](#). The meter tests a drop of your blood, usually from a prick of your fingertip. The meter will show you how much glucose is in your blood at that moment.

People who take insulin to manage their diabetes may benefit from using a continuous glucose monitor (CGM). A CGM uses a sensor in the tissue under your skin and shows changes in glucose levels throughout the day and night. Many CGM systems have alarms and warnings that let you know if your glucose level is too low or too high.

For safety, you may sometimes need to compare your CGM glucose readings with results from a standard blood glucose meter. This could be needed if you doubt the accuracy of your CGM readings, if you are changing your insulin dose, or if your CGM gives a warning alert.

Some people use a CGM as part of an artificial pancreas. An artificial pancreas is a system that mimics how a healthy pancreas controls blood glucose in the body. A CGM, an insulin pump, and a software program that shares information between the CGM and insulin pump make up the artificial pancreas.

Ask your health care team about the best way to check your blood glucose level and how often you should check it. Make sure to keep a record of your blood glucose self-checks, including the date and time you checked it, the results, and other information, such as what you ate or what physical activity you did that day. You can use a chart or a smartphone app to help you keep track. Take this information with you when you visit your health care team.

Recommended targets for blood glucose levels

Talk with your health care team about the blood glucose range that is healthy for you. Be sure to tell your health care team if your glucose levels frequently go above or below your target range.

Many people with diabetes aim to keep their blood glucose at these levels¹

Before a meal: 80 to 130 mg/dL

About 2 hours after a meal starts: less than 180 mg/dL

Your blood glucose goals may be different if you are an adult over age 65 and have had diabetes for a long time. Your blood glucose goals may also be different if you have other health problems, like heart disease, or if your blood glucose often gets too low.

If you use a CGM, your health care team may ask you to keep track of your time in range. Time in range is the percentage of time that your glucose level stays in your target range. For many people with diabetes, the target range is between 70 and 180 mg/dL. Your health care team may adjust your diabetes care plan to help you stay in your target range at least 70% of the time.

Low blood glucose

When your blood glucose level is too low, it is called hypoglycemia. For most people with diabetes, the blood glucose level is too low when it is below 70 mg/dL.^{1,2} Talk with your primary care professional or health care team if you often have low blood glucose levels. Some people with low blood glucose experience symptoms such as

feeling shaky or jittery

headache

cold sweat

If you often have low blood glucose levels, you may need to change your diabetes meal plan, physical activity plan, or medicines. Very low blood glucose may lead to confusion or fainting [NIH external link](#), and it can become a serious medical emergency that must be treated right away.

Glucagon—a hormone that raises blood glucose levels—is the best way to treat severely low blood glucose. Ask your health care team if you need a prescription for glucagon. If you have a glucagon prescription, you should teach your family, friends, and coworkers when and how to give you glucagon in case you are not able to do it yourself.

High blood glucose

When your blood glucose level is too high, it is called hyperglycemia. For most people with diabetes, the blood glucose level is too high when it is above 180 mg/dL. Talk with your health care team if you often have high blood glucose levels. Some people with high blood glucose have symptoms such as

feeling tired

feeling thirsty

having blurry vision

urinating too often

If you often have high blood glucose levels, you may need to make a change to your diabetes meal plan, physical activity plan, or medicines. Very high blood glucose may lead to confusion or fainting, and it may become a serious medical emergency that must be treated right away.

Checking urine for ketones

Your health care team may want you to check your urine for ketones if you have symptoms of diabetic ketoacidosis. Diabetic ketoacidosis is a medical emergency that needs to be treated right away.

Symptoms of diabetic ketoacidosis include

feeling very tired

having trouble breathing

having fruity-smelling breath

fainting

Most often, ketoacidosis affects people with type 1 diabetes. However, some people with type 2 diabetes can develop ketoacidosis if they do not produce enough insulin.

How can I keep track of my progress?

You may find it helpful to keep track of your ABCs and other important health information that you get during checkups with your primary care professional or other health care professionals. Download the Health Checkups Tracker (DOCX, 67.97 KB) , which includes information you may want to track. You can take this information with you when you visit your health care team. Talk with your primary care professional about your goals and whether you need to make any changes to your diabetes care plan.

What special situations should I plan for?

If you have diabetes, it is important to take care of your health no matter what is going on in your life. Ask your health care team if you should adjust your meal plan, activity level, or medicines in special situations, such as when you are sick [External link](#), fasting, at school [External link](#), planning to travel [External link](#), or experiencing an emergency or natural disaster [External link](#).

Who should be on my health care team?

You are the most important member of your health care team, but working with your health care team may help you improve your self-care. Most people with diabetes get help from a primary care professional, such as an internal medicine doctor, family doctor, pediatrician, physician assistant, or certified nurse practitioner trained in diabetes care. Ask your primary care professional about adding other members to your team, including

a diabetes educator, nurse, or registered dietitian who can tell you more about your disease
External link and how to improve your self-care

a dentist to check for gum diseases and other dental problems

an eye doctor to check for diabetic eye disease

a pharmacist to help you with your medicines

other specialists if you develop health problems such as heart disease, kidney disease, erectile dysfunction, or foot problems

a social worker who may be able to help you find financial help for diabetes care and community resources

Write a list of questions External link before your visit with a health care professional so you don't forget what you want to ask.

Woman with obesity talking with her doctor.

When you see your health care professional, discuss your diabetes self-care plan, review your blood glucose chart or app, and ask questions.

Clinical Trials for Managing Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat diseases and improve quality of life.

What are clinical trials for managing diabetes?

Clinical trials—and other types of clinical studies NIH external link—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help health care professionals and researchers learn more about diseases and improve health care for people in the future.

Researchers are studying many aspects of managing diabetes, such as

what are the best medicines to add to a diabetes care plan if a person with type 2 diabetes isn't meeting their blood glucose targets using their current diabetes medicines

how to train adolescents to support or care for family members with diabetes

how treating obesity with weight-loss medicines or weight-loss surgery, also called metabolic and bariatric surgery, may help manage type 2 diabetes

what different devices can help people with diabetes check their blood glucose level and inject insulin

Find out if clinical trials are right for you [NIH external link](#).

Watch a video of NIDDK Director Dr. Griffin P. Rodgers explaining the importance of participating in clinical trials.

What clinical trials for managing diabetes are looking for participants?

You can view a filtered list of clinical studies on managing diabetes that are federally funded, open, and recruiting at [ClinicalTrials.gov](#) [NIH external link](#). You can expand or narrow the list to include clinical studies from industry, universities, and individuals; however, the National Institutes of Health does not review these studies and cannot ensure they are safe for you. Always talk with your primary health care professional before you participate in a clinical study.

What have we learned about managing diabetes from NIDDK-funded research?

The research funded by NIDDK has improved quality of life for people with diabetes by finding better ways to diagnose and treat the disease.

Artificial pancreas

For decades, NIDDK has supported research that helped create artificial pancreas technology used by some people who need insulin to manage their diabetes.

NIDDK supported research that helped scientists develop the first continuous glucose monitor (CGM) that was approved by the U.S. Food and Drug Administration (FDA) in 1999. CGMs, which are part of artificial pancreas systems, help people track glucose levels in real-time with fewer finger pricks. Another major milestone came in 2016, when the FDA approved a commercial “hybrid” artificial pancreas system. Other artificial pancreas systems have been approved since 2016. NIDDK continues to support research into artificial pancreas systems and other technologies that may improve diabetes management.

Blood glucose management studies

NIDDK funded the landmark Diabetes Control and Complications Trial (DCCT). The study showed that participants with type 1 diabetes who kept their blood glucose levels close to normal greatly lowered their chances of developing eye, kidney, and nerve disease. A follow-up study, the ongoing Epidemiology of Diabetes Interventions and Complications (EDIC) study [External link](#), has continued to follow DCCT participants since 1994. EDIC has shown that there are long-term benefits of early and intensive blood glucose management on slowing or preventing the development of health problems such as heart, eye, kidney, and nerve diseases. Findings from the DCCT and EDIC studies changed the way diabetes is treated worldwide, and they show that blood glucose control can help people with diabetes lead longer and healthier lives.

NIDDK also funded the Diabetes & Aging Study to research the effects of early blood glucose management and the development of diabetes health problems in people with type 2 diabetes. The study found that people with less effective diabetes management during the first year after being diagnosed with diabetes had a higher risk of diabetes complications and death.

The Treatment Options for Type 2 Diabetes in Adolescents and Youth (TODAY) study and a follow-up study, TODAY2, found that type 2 diabetes can be more difficult to treat and manage when it occurs in younger people. The TODAY studies, both funded by NIDDK, show the importance of early, intensive treatment and blood glucose management for children or adolescents diagnosed with type 2 diabetes.

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Related Topics

English

Monogenic Diabetes (MODY & Neonatal Diabetes Mellitus)

On this page:

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What is maturity-onset diabetes of the young (MODY)?

What is neonatal diabetes mellitus (NDM)?

How common is monogenic diabetes?

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What health problems can people with monogenic diabetes develop?

What are the symptoms of monogenic diabetes?

What causes monogenic diabetes?

How do health care professionals diagnose monogenic diabetes?

What do I need to know about genetic counseling?

How can I manage monogenic diabetes?

Can I prevent monogenic diabetes?

Clinical Trials for Monogenic Diabetes

What is monogenic diabetes?

Monogenic diabetes is a group of uncommon forms of diabetes that are caused by a variant NIH external link, or change, in a single gene. Researchers have found variants in more than 20 genes that cause monogenic diabetes.¹ Monogenic forms of diabetes are part of the group of atypical diabetes. People with atypical forms of diabetes have symptoms and signs that are different than those of type 1 or type 2 diabetes.

If you have monogenic diabetes, you may have high levels of blood glucose, also called blood sugar. Diabetes may raise your risk of eye, kidney, nerve, and heart damage. Diabetes is also linked to some types of cancer. You may be able to prevent or delay diabetes health problems by managing your blood glucose level.

Maturity-onset diabetes of the young (MODY) and neonatal diabetes mellitus (NDM) are the two main forms of monogenic diabetes.

What is maturity-onset diabetes of the young (MODY)?

MODY forms of diabetes usually develop in teens or young adults. People with some forms of MODY have blood glucose levels that are only a little higher than normal and do not change much during their lives. The levels of blood glucose are typically not high enough to damage their organs, so they may not need treatment. However, people with other forms of MODY have higher blood glucose levels and need treatment.

What is neonatal diabetes mellitus (NDM)?

NDM forms of diabetes develop in the first 6 to 12 months of life.^{2,3} Babies with forms of NDM do not produce enough insulin, which allows their blood glucose levels to go up more than normal. Babies who develop diabetes in the first 6 months of life almost always have a form of NDM.²

In most forms of NDM, babies will have lifelong diabetes. These forms of NDM are called permanent NDM. Babies with some forms of NDM have diabetes as infants, but their diabetes goes away during childhood. For these babies, diabetes often shows up again later in life. These forms of NDM are called transient NDM.

How common is monogenic diabetes?

Some research, mainly from the United States and Europe, has found that

about 1 to 5 of every 100 people with diabetes have monogenic diabetes^{4,5}

NDM is a rare disease, and only 1 in every 90,000 babies has the disease⁴

People with monogenic diabetes may be misdiagnosed with type 1 or type 2 diabetes because the symptoms may be similar. Therefore, it is difficult to know how many people have monogenic diabetes.

Who is more likely to have monogenic diabetes?

People with MODY forms of diabetes usually have other family members with diabetes. MODY forms of diabetes most often occur in people ages 30 or younger.⁶ Forms of NDM most often occur in babies ages 6 months or younger.^{1,2} More studies are needed to understand which populations, based on race, are more affected by forms of MODY and NDM.

What health problems can people with monogenic diabetes develop?

Some people with forms of monogenic diabetes may develop the same diabetes health problems as people with other types of diabetes. High blood glucose levels from diabetes can damage every part of your body, including your

heart

kidneys

nerves

eyes

feet

teeth

If you have a form of monogenic diabetes, work with your primary health care professional, who may be a doctor, physician assistant, or nurse practitioner, to manage your diabetes. Your health care team can help you lower your chances of developing other health problems.

What are the symptoms of monogenic diabetes?

The symptoms of monogenic diabetes depend on which gene variant you have. Some people have no symptoms. Other people may have general symptoms of diabetes that include

increased urination

feeling thirsty

feeling very hungry, even after you have eaten

blurred vision

fatigue

numbness or tingling in the feet or hands

sores that do not heal

What causes monogenic diabetes?

A variant, or change, in a single gene causes monogenic diabetes.

You inherited genes—which determine things like your hair color, eye color, and height—from each of your parents. In most cases of MODY forms of diabetes, you inherit a gene from one or both parents that causes the disease.

Babies with NDM forms of diabetes rarely have a family history of the disease. Most cases of NDM are caused by a gene change that happens while a baby develops in the womb.

Most people with MODY forms of diabetes have inherited a gene for the disease from a parent who also had the disease. Diseases caused by a gene from one parent are called autosomal dominant NIH external link. If you have a form of NDM or MODY that is autosomal dominant, your child has a 50% chance of having the disease.

Some forms of NDM or MODY occur only if a child inherits a gene for the disease from both parents. In most cases, neither of the child's parents had the disease. Diseases that occur only if you inherit a gene from both parents who don't have the disease are called autosomal recessive

NIH external link. If you inherit a gene for a recessive form of NDM or MODY from only one parent, you won't have the disease. However, you can pass that gene to future generations.

Different gene variants cause monogenic diabetes in different ways. For example, gene variants may

affect how the pancreas develops

cause the pancreas to produce insulin that does not work well

destroy cells in the pancreas that produce insulin

How do health care professionals diagnose monogenic diabetes?

Your health care professional will usually test you for forms of monogenic diabetes if you have diabetes, are younger than 30 years old, don't have overweight or obesity, and have family members with these forms of diabetes.

Health care professionals will diagnose diabetes using your symptoms, medical and family history, and diabetes tests. Your health care professional may also order tests to find out if your diabetes is type 1, type 2, or monogenic. The type of diabetes you have will help your health care professional determine how to treat the disease.

Blood glucose tests

Health care professionals can diagnose diabetes using blood tests that measure your glucose level to see if it is higher than normal. People with some forms of monogenic diabetes have blood glucose levels that are a little higher than normal.²

Blood glucose tests can help your health care professional tell if you have diabetes. However, these tests do not show the type of diabetes you have.

Autoantibodies

Your health care professional will order a test for autoantibodies to find out if you have type 1 diabetes. Autoantibodies are proteins made by your immune system that attack your healthy tissues and cells by mistake. In rare cases, people with monogenic diabetes may also have autoantibodies in their blood.

Genetic testing

Genetic testing [NIH external link](#) looks for gene variants. Health care professionals use genetic tests to diagnose monogenic diabetes.

Health care professionals may request a genetic test for

babies ages 6 months or younger who develop diabetes

babies who develop diabetes between ages 6 months and 1 year and have some other medical problems

people who have diabetes at a young age, do not have overweight or obesity, and have a strong family history of diabetes

Young couple with a baby.

If you think you or a member of your family may have a monogenic form of diabetes, talk with your health care professional.

What do I need to know about genetic counseling?

If you, your partner, or family members have monogenic diabetes, you may want to consider genetic counseling [NIH external link](#). Genetic counseling can help you decide if you should get tested for monogenic diabetes. Genetic counseling can also help you understand the risk of your children developing this disease. Talk with your health care professional for more information about how genetic counseling could help you, what it may cost, and whether it's covered by your insurance.

How can I manage monogenic diabetes?

How you manage monogenic diabetes will depend on the specific gene variant you have. Your health care professional and diabetes care team will work with you to develop a plan to manage

your diabetes. Your health care professional may refer you to, or work with, a medical center that focuses on monogenic diabetes to help manage the disease.

Medicines

Some people with monogenic diabetes may need diabetes medicines. These medicines may include

sulfonylureas, oral diabetes medicines that help your body make more insulin. These medicines can treat certain forms of MODY and NDM.

insulin. Some people with monogenic diabetes need to take insulin all their lives. Others may need to take insulin early in life and may be able to switch to sulfonylureas later in life.

other medicines. Some medicines, called sodium-glucose cotransporter-2 (SGLT2) inhibitors, work in your kidneys to get rid of glucose. Other medicines, called glucagon-like peptide-1 receptor (GLP-1) agonists, increase insulin production and decrease your appetite.

Diabetes medicines and office visits can be expensive. Talk with your health care professional or health care team if you are having trouble affording diabetes care. A health care professional, diabetes educator, or social worker may be able to tell you about resources that provide financial help for diabetes care.

Can I prevent monogenic diabetes?

At this time, monogenic diabetes can't be prevented. However, researchers are working to find possible risk factors and ways to prevent the disease.

Clinical Trials for Monogenic Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for monogenic diabetes?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying different aspects of monogenic diabetes, such as ways to improve diagnosis.

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Expand Health Information

English

Related Topics

Diabetic Neuropathy

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Diabetic neuropathy is nerve damage that can occur in people with diabetes. Different types of nerve damage cause different symptoms. Symptoms can range from pain and numbness in your feet to problems with the functions of your internal organs, such as your heart and bladder.

senior woman with glucometer checking blood sugar level at home

What Is Diabetic Neuropathy?

Diabetic neuropathy is nerve damage that is caused by diabetes. Over time, high blood glucose levels, also called blood sugar, and high levels of fats, such as triglycerides, in the blood from diabetes can damage your nerves. Symptoms depend on which type of diabetic neuropathy you have.

Peripheral Neuropathy

Peripheral neuropathy is a type of nerve damage that typically affects the feet and legs and sometimes affects the hands and arms. This type of neuropathy is very common. About one-third to one-half of people with diabetes have peripheral neuropathy.

Autonomic Neuropathy

Autonomic neuropathy is damage to nerves that control your internal organs, leading to problems with your heart rate and blood pressure, digestive system, bladder, sex organs, sweat glands, and eyes. The damage can also lead to hypoglycemia unawareness.

Focal Neuropathies

Focal neuropathies are conditions in which you typically have damage to single nerves, most often in your hand, head, torso, or leg. The most common types of focal neuropathy are entrapment syndromes, such as carpal tunnel syndrome. Other types of focal neuropathy are much less common.

Proximal Neuropathy

Proximal neuropathy is a rare and disabling type of nerve damage in your hip, buttock, or thigh. The damage typically affects one side of your body and may rarely spread to the other side. Symptoms gradually improve over a period of months or years.

Related Conditions & Diseases

Diabetes

Diabetes and Foot Problems

Erectile Dysfunction

Gastroparesis

Diabetes and Sexual and Urologic Problems

Constipation

Diarrhea

Bowel Control Problems

Clinical Trials

The NIDDK and other components of the National Institutes of Health (NIH) support and conduct research into many diseases and conditions. View clinical trials that are currently recruiting volunteers [NIH external link](#).

Additional Languages

This content is also available in:

Spanish

Related Research

See more about diabetes research at the NIDDK.

Related Topics

English

Pancreatic Islet Transplantation

On this page:

What are islets?

What is pancreatic islet transplantation and how can it treat type 1 diabetes?

Who are candidates for islet transplantation?

How do doctors perform islet transplantation?

What are the benefits of islet transplantation?

How successful is islet transplantation?

What are the risks of islet transplantation?

How common is islet transplantation?

Why isn't islet transplantation used more often to treat type 1 diabetes?

Is islet transplantation used to treat any other conditions?

What are islets?

Pancreatic islets, also called islets of Langerhans, are groups of cells in your pancreas. The pancreas is an organ that makes hormones to help your body break down and use food. Islets contain several types of cells, including beta cells that make the hormone insulin. Insulin helps your body use glucose for energy and helps control your blood glucose levels, also called blood sugar.

What is pancreatic islet transplantation and how can it treat type 1 diabetes?

In people with type 1 diabetes, the body's immune system attacks and destroys the beta cells. People with type 1 diabetes must take insulin because their bodies no longer make this hormone.

Illustration of the pancreas with insets showing a normal islet, with many healthy beta cells, and an islet from a person with type 1 diabetes, with many destroyed beta cells.

Pancreatic islets contain beta cells that produce the hormone insulin.

In the type of islet transplantation used to treat type 1 diabetes, also called islet allo-transplantation, doctors take islets with healthy beta cells from the pancreas of a deceased organ donor. Doctors then inject the healthy islet cells taken from the donor into a vein that carries blood to the liver of a person with type 1 diabetes. A person receiving a transplant is called a recipient. These islets begin to make and release insulin in the recipient's body. More than one injection of transplanted islet cells is often needed to stop using insulin.

For many years, islet transplantation was considered experimental and was only performed in clinical trials. However, in 2023, the U.S. Food and Drug Administration (FDA) approved the first cell therapy made from islets of deceased donors for use in islet transplantation. This cell therapy is approved for adults with type 1 diabetes who have a hard time managing their blood glucose levels and continue to have repeated severely low blood glucose, also called severe hypoglycemia, despite intensive diabetes management and education.

Islet transplantation may help people with type 1 diabetes

improve their blood glucose levels

lower or remove the need for insulin injections

better recognize symptoms of low blood glucose, also called hypoglycemia

prevent severe hypoglycemia, which is when a person's blood glucose level becomes so low that he or she needs help from another person to treat the hypoglycemia

A whole pancreas transplant [NIH external link](#) is another procedure that can give a person with type 1 diabetes healthy beta cells. However, a pancreas transplant is a major surgery that carries a greater risk of complications than an islet transplant.

Who are candidates for islet transplantation?

Not all people with type 1 diabetes are good candidates for islet transplantation. Certain people with type 1 diabetes who have blood glucose levels that are difficult to manage, experience severe hypoglycemia, and have hypoglycemia unawareness—a dangerous condition in which a person can't feel or recognize the symptoms of hypoglycemia—may be candidates.

Doctors consider people for islet transplantation if the possible benefits, such as being better able to reach blood glucose targets without problems such as hypoglycemia, outweigh the risks, such as the possible side effects of immunosuppressants. Immunosuppressants are medicines that recipients must take to prevent their immune system from attacking and destroying the transplanted islets.

People who have type 1 diabetes and have had, or are planning to have, a kidney transplant to treat kidney failure may be candidates for islet transplantation. Islet transplantation may be performed at the same time as or after a kidney transplant. Kidney transplant recipients will already be taking immunosuppressants to prevent rejection of the transplanted kidney. Therefore, the islet transplant does not add much more risk.

How do doctors perform islet transplantation?

Special enzymes are used to remove islets from the pancreas of a deceased donor. The islets are purified and counted in a lab. On average, about 400,000 islets are transplanted in each procedure.¹

The transplant recipient will most often receive a local anesthetic and an external link and a sedative—medicine to help you relax—for the procedure. In some cases, the recipient may receive general anesthesia.

The islet transplant infusion procedure involves inserting a thin, flexible tube called a catheter through a small cut in the recipient's upper abdomen. A radiologist uses x-rays and ultrasound to guide the catheter into the portal vein of the liver. The islets are slowly infused through the catheter and into the liver by gravity. Alternatively, a minimally invasive open procedure can be used to directly visualize a vein near the liver to insert the catheter.

Illustration of the liver and pancreas.

The islets are transplanted into the recipient's liver.

Over the next 2 weeks, new blood vessels form and connect the islets with the blood vessels of the recipient.² The beta cells in the islets begin to make and release insulin into the bloodstream immediately after transplant.

What are the benefits of islet transplantation?

Recipients may see the following benefits

improved blood glucose levels

less need or no need for insulin injections to manage diabetes

fewer or no episodes of severe hypoglycemia

improved awareness of hypoglycemia, which helps prevent episodes of severe hypoglycemia

Research also suggests that islet transplantation may prevent or slow the development of diabetes complications such as heart disease, kidney disease, and nerve or eye damage.³

Insulin pen

Islet transplant recipients may be able to keep blood glucose levels in their target range while taking less insulin or taking no insulin.

How successful is islet transplantation?

A Phase 3 clinical trial [External link](#) conducted by the National Institutes of Health (NIH)-sponsored Clinical Islet Transplantation Consortium [External link](#) looked at outcomes associated with islet transplantation. Phase 3 trials test new treatments in large groups of people to confirm that treatments are effective and monitor side effects. Participants in this NIH-sponsored study included people with type 1 diabetes with problems managing their blood glucose levels, such as severe hypoglycemia and hypoglycemia unawareness.

The study found that, 1 year after islet transplantation, nearly 9 out of 10 transplant recipients had an A1C level below 7 percent and did not have episodes of severe hypoglycemia. The A1C goal for many people with diabetes is below 7 percent. About half of the recipients did not need to take any insulin. Two years after islet transplantation, about 7 out of 10 recipients had an A1C level of less than 7 percent and did not have episodes of severe hypoglycemia, and about 4 out of 10 did not need insulin.⁴

Ongoing research from this study also found that islet transplant recipients experienced significant improvements in their diabetes-related quality of life and reported better overall health status after the transplant. Even transplant recipients who still needed to take insulin to manage their diabetes experienced these improvements.⁵

What are the risks of islet transplantation?

Risks of islet transplantation include

bleeding, blood clots, and pain after the procedure

the chance that the transplanted islets may not work well or may stop working

side effects of anti-rejection medicines, also called immunosuppressants, which are described below

development of antibodies against the donor cells that may make it more difficult to find an appropriate organ donor if another transplant is needed in the future

Immunosuppressants

After an islet transplant, recipients will take medicines, called immunosuppressants, for as long as the transplanted islets are working. These medicines help prevent the body from rejecting the transplanted islets. Rejection occurs when the body's immune system sees the islets as "foreign" and tries to destroy them. If the recipient stops taking immunosuppressants, the recipient's body will reject the transplanted islets, and the islets will stop working.

Immunosuppressants can have many serious side effects. Possible side effects include

a higher chance of getting infections

a higher chance of getting cancer [NIH external link](#)

digestive side effects, such as vomiting, nausea, or diarrhea

headaches, tremors [NIH external link](#), or confusion

high blood pressure

high blood glucose levels

high levels of cholesterol and triglycerides in the blood

kidney damage

How common is islet transplantation?

Islet transplantation is not a common treatment for type 1 diabetes. In the United States, islet transplantation was considered an experimental treatment for many years and was only performed in clinical trials. With the FDA approval of an islet cell therapy in 2023, islet transplantation is available outside of clinical trials to treat certain adults with type 1 diabetes.

The Collaborative Islet Transplantation Registry (CITR) [External link](#) has reported that, between 1999 and 2015, 1,086 people worldwide received islet transplants for type 1 diabetes.¹ The CITR collects information on islet transplants performed at research centers in North America, Europe, Australia, and Asia. The National Institute of Diabetes and Digestive and Kidney

Diseases established and supports the CITR. Breakthrough T1D, formerly known as JDRF or the Juvenile Diabetes Research Foundation, also helped establish CITR.

Why isn't islet transplantation used more often to treat type 1 diabetes?

Islet transplantation is not often used to treat type 1 diabetes because recipients must take immunosuppressants, and donor islets are in short supply.

Islet transplant recipients must take immunosuppressants

Islet transplant recipients must take long-term immunosuppressants, and these medicines may cause serious side effects. Researchers are looking for ways to prevent islet rejection without long-term immunosuppressants. In one approach, called encapsulation, islets are coated with a material that protects them from being attacked by the recipient's immune system.

Donor islets are in short supply

Only a small number of donor pancreases are available for islet transplantation each year. According to the Organ Procurement and Transplantation Network, 1,315 pancreases were recovered from deceased donors in 2017. Many donated pancreases are not suitable for islet isolation. Also, some donor islets may be damaged or destroyed during the transplant process.

Researchers are studying different ways to overcome the short supply of donor islets. For example, scientists are studying ways to transplant islets from pigs or create new human islets from stem cells [NIH external link](#).

Is islet transplantation used to treat any other conditions?

Doctors may perform a different type of islet transplantation, called islet auto-transplantation, in people who have their whole pancreas removed to treat severe and chronic pancreatitis. People with type 1 diabetes cannot receive islet auto-transplantation.

In islet auto-transplantation, doctors remove a patient's pancreas, remove the islets from the pancreas, and transplant the islets into the patient's liver. The goal is to give the body enough

healthy islets that make insulin. Patients don't need to take immunosuppressants after islet auto-transplantation because they receive islets from their own body.

Islet auto-transplantation is not considered experimental.

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Related Topics

Preventing Diabetes Problems

View or Print All Sections

Heart Disease & Stroke

Diabetes can damage blood vessels and lead to heart disease and stroke. You can do a lot to prevent heart disease and stroke by managing your blood glucose, blood pressure, and cholesterol levels; and by not smoking.

Low Blood Glucose (Hypoglycemia)

Hypoglycemia occurs when your blood glucose drops too low. Certain diabetes medicines make low blood glucose more likely. You can prevent hypoglycemia by following your meal plan and balancing your physical activity, food, and medicines. Testing your blood glucose regularly can also help prevent hypoglycemia.

Diabetic Neuropathy

Diabetic neuropathy is nerve damage that can result from diabetes. Different types of nerve damage affect different parts of your body. Managing your diabetes can help prevent nerve damage that affects your feet and limbs, and organs such as your heart.

Kidney Disease

Diabetic kidney disease, also called diabetic nephropathy, is kidney disease caused by diabetes. You can help protect your kidneys by managing your diabetes and meeting your blood pressure goals.

Foot Problems

Diabetes can cause nerve damage and poor blood flow, which can lead to serious foot problems. Common foot problems, such as a callus, can lead to pain or an infection that makes it hard to walk. Get a foot checkup at each visit with your health care team.

Eye Disease

Diabetes can damage your eyes and lead to low vision and blindness. The best way to prevent eye disease is to manage your blood glucose, blood pressure, and cholesterol; and to not smoke. Also, have a dilated eye exam at least once a year.

Gum Disease & Other Dental Problems

Diabetes can lead to problems in your mouth, such as infection, gum disease, or dry mouth. To help keep your mouth healthy, manage your blood glucose, brush your teeth twice a day, see your dentist at least once a year, and don't smoke.

Sexual & Bladder Problems

Sexual and bladder problems are more common in people with diabetes. Problems like erectile dysfunction, loss of interest in sex, bladder leaks, and retained urine can happen if diabetes damages your blood vessels and nerves. Treatments are available to help control symptoms and restore intimacy.

Depression & Diabetes

Depression [NIH external link](#) is common among people with a chronic, or long-term, illness such as diabetes. Depression can be treated so tell your doctor if you feel sad, hopeless, or anxious.

Cancer & Diabetes

Diabetes is linked to some types of cancer [External link](#). Many risk factors for cancer and for diabetes are the same. Not smoking and getting recommended cancer screenings can help prevent cancer.

Dementia & Diabetes

High blood glucose increases the chance of developing dementia [NIH external link](#). Tell your doctor if you are forgetful because dementia can make it hard to manage your diabetes.

Sleep Apnea & Diabetes

People who have sleep apnea [NIH external link](#)—when you stop breathing for short periods during sleep—are more likely to develop type 2 diabetes. Sleep apnea also can make diabetes worse. Treatment for sleep apnea can help.

Related Topics

English

Preventing Type 2 Diabetes

Perhaps you have learned that you have a high chance of developing type 2 diabetes, the most common type of diabetes. You might be overweight or have a parent, brother, or sister with type 2 diabetes. Maybe you had gestational diabetes, which is diabetes that develops during pregnancy. These are just a few examples of factors that can raise your chances of developing type 2 diabetes.

Diabetes can cause serious health problems, such as heart disease, stroke, and eye and foot problems. Prediabetes also can cause health problems. The good news is that type 2 diabetes can be delayed or even prevented. The longer you have diabetes, the more likely you are to develop health problems, so delaying diabetes by even a few years will benefit your health. You can help prevent or delay type 2 diabetes by losing a modest amount of weight by following a reduced-calorie eating plan and being physically active most days of the week. Ask your doctor if you should take the diabetes drug metformin [NIH external link](#) to help prevent or delay type 2 diabetes.¹

How can I lower my chances of developing type 2 diabetes?

Research such as the Diabetes Prevention Program [External link](#) shows that you can do a lot to reduce your chances of developing type 2 diabetes. Here are some things you can change to lower your risk:

Lose weight and keep it off. You may be able to prevent or delay diabetes by losing 5 to 7 percent of your starting weight.¹ For instance, if you weigh 200 pounds, your goal would be to lose about 10 to 14 pounds.

Losing 5% to 7% of body weight may reduce the risk of developing diabetes.

Supported by a study from

Move more. Get at least 30 minutes of physical activity 5 days a week. If you have not been active, talk with your health care professional about which activities are best. Start slowly to build up to your goal.

Eat healthy foods most of the time. Eat smaller portions to reduce the amount of calories you eat each day and help you lose weight. Choosing foods with less fat is another way to reduce calories. Drink water instead of sweetened beverages.

Ask your health care professional about what other changes you can make to prevent or delay type 2 diabetes.

Most often, your best chance for preventing type 2 diabetes is to make lifestyle changes that work for you long term. Get started with [Your Game Plan to Prevent Type 2 Diabetes](#).

Photo of a young man weighing himself

Losing weight through healthy eating and regular physical activity can help you prevent type 2 diabetes.

What should I do if my health care professional told me I have prediabetes?

Prediabetes is when your blood glucose, also called blood sugar, levels are higher than normal, but not high enough to be called diabetes. Having prediabetes is serious because it raises your

chance of developing type 2 diabetes. Many of the same factors that raise your chance of developing type 2 diabetes put you at risk for prediabetes.

Other names for prediabetes include impaired fasting glucose or impaired glucose tolerance. Some people call prediabetes “borderline diabetes.”

About 1 in 3 Americans has prediabetes, according to recent diabetes statistics [External link](#) from the Centers for Disease Control and Prevention. You won’t know if you have prediabetes unless you are tested.

If you have prediabetes, you can lower your chance of developing type 2 diabetes. Lose weight if you need to, become more physically active, and follow a reduced-calorie eating plan.

Get started with [Your Game Plan to Prevent Type 2 Diabetes](#). For more support, you can find a lifestyle change program near you through the National Diabetes Prevention Program [External link](#).

Photo of two smiling middle-aged women on exercise bikes

Being physically active is one way to help prevent prediabetes from progressing to type 2 diabetes.

If I had gestational diabetes when I was pregnant, how can I lower my chances of developing type 2 diabetes?

Gestational diabetes is a type of diabetes that develops during pregnancy. Most of the time, gestational diabetes goes away after your baby is born. Even if your gestational diabetes goes away, you still have a greater chance of developing type 2 diabetes within 5 to 10 years. Your child may also be more likely to become obese and develop type 2 diabetes later in life. Making healthy choices helps the whole family and may protect your child from becoming obese or developing diabetes.

Photo of mother and young daughter riding bikes

Being physically active together is a great way to lower your own and your child's chance of developing type 2 diabetes.

Here are steps you should take for yourself and your child if you had gestational diabetes:

Get tested for diabetes 6 to 12 weeks after your baby is born. If your blood glucose is still high, you may have type 2 diabetes. If your blood glucose is normal, you should get tested every 3 years to see if you have developed type 2 diabetes.

Be more active and make healthy food choices to get back to a healthy weight.

Breastfeed your baby. Breastfeeding gives your baby the right balance of nutrients and helps you burn calories.

Ask your doctor if you should take the diabetes drug metformin to help prevent type 2 diabetes.¹

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Related Topics

English

Risk Factors for Type 2 Diabetes

Your chances of developing type 2 diabetes depend on a combination of risk factors. Although you can't change risk factors related to family history, age, race, or ethnicity, you may be able to avoid some risk factors by maintaining a healthy weight and being physically active.¹

Read about risk factors for type 2 diabetes below, and see which ones apply to you. Acting on the factors you can change may help delay or prevent type 2 diabetes.

What are the risk factors for type 2 diabetes?

Black grandfather and grandson playing American football.

Diabetes type 2 can affect people at any age.

You can develop type 2 diabetes at any age, even during childhood. You are more likely to develop type 2 diabetes if you¹

are overweight or have obesity.

are age 35 or older. Children and teens can also develop type 2 diabetes, but the risk increases as a person gets older.

have a family history of diabetes.

are African American, American Indian, Asian American, Hispanic/Latino, or Pacific Islander.

are not physically active, because of physical limitations, a sedentary lifestyle, or a job that requires sitting for long periods of time.

have prediabetes.

have a history of gestational diabetes, a type of diabetes that develops during pregnancy, or gave birth to a baby weighing 9 pounds or more.

Children and teens are also at higher risk of developing type 2 diabetes if, in addition to the previous risk factors, they were born with a low birth weight or if their parent had gestational diabetes while pregnant with the child.¹

You are encouraged to talk with a doctor about any of the health conditions listed above that may require medical treatment. Managing health problems may help reduce your chances of developing type 2 diabetes.² Also, ask your doctor about any medicines you or your child take that might increase your risk. You can also take the Diabetes Risk Test to learn more about your risk for type 2 diabetes.

Does your weight put you at risk for type 2 diabetes?

If you have overweight or obesity, you may be able to prevent or delay type 2 diabetes by losing weight and being more physically active.¹

To see if your weight may put you at risk for type 2 diabetes, learn your body mass index (BMI) and waist circumference [External link](#).

Body mass index

Your BMI can help you tell if you are at a healthy weight or have overweight or obesity.

Most adults with a BMI of 25 or higher are overweight and have a higher risk of developing type 2 diabetes.² Asian Americans are overweight if their BMI is 23 or higher,¹ while Pacific Islanders are overweight if their BMI is 26 or higher.^{3,4,5}

If you are age 20 or older, use the BMI calculator for adults [NIH external link](#) to learn your body mass index.

You can use a different BMI calculator for children or teens [External link](#) ages 2–19 years. This calculator uses BMI, sex, and age to estimate if a child or teen has overweight or obesity.

Waist circumference

Another way to estimate your risk of developing diabetes is to measure your waist circumference. Men have a higher risk of developing diabetes if their waist circumference is more than 40 inches, while women who are not pregnant have a higher risk if their waist circumference is more than 35 inches.^{6,7}

Waist circumference is an indirect measurement of the amount of fat in your abdomen. Having a large waist circumference is a risk factor for diabetes and heart disease, even if you have a normal BMI.

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Related Topics

English

Symptoms & Causes of Diabetes

In this section:

What are the symptoms of diabetes?

What causes type 1 diabetes?

What causes type 2 diabetes?

What causes gestational diabetes?

What else can cause diabetes?

Clinical Trials for Symptoms and Causes of Diabetes

What are the symptoms of diabetes?

Common symptoms of diabetes include

increased urination

feeling very thirsty

feeling very hungry, even after you have eaten

blurred vision

fatigue

sores that do not heal

frequent infections, such as urinary tract infections, skin infections [NIH external link](#), or yeast infections [NIH external link](#)

There are different types of diabetes. People with any type of diabetes can have these symptoms. Some other symptoms, and how quickly the symptoms develop, depend on the type of diabetes that you have. Talk with your primary health care professional about any symptoms you are having and ask if you should be tested for diabetes. Your primary health care professional may be a doctor, physician assistant, or nurse practitioner.

Type 1 diabetes symptoms

In addition to some of the common diabetes symptoms listed above, people with type 1 diabetes may also have an unexplained weight loss. Most people don't know they have type 1 diabetes until they develop diabetes symptoms.

Symptoms of type 1 diabetes usually develop quickly, over a few days or weeks. Type 1 diabetes is often diagnosed in children and young adults, but it can begin at any age. Children with type 1 diabetes typically develop symptoms over a short period of time. Common symptoms of type diabetes in children include

having to urinate more often

eating more but losing weight

feeling very thirsty

These symptoms are less common in adults with type 1 diabetes. For some adults, symptoms of type 1 diabetes may develop slowly and seem more like type 2 diabetes. Some adults who develop diabetes may need special tests to find out what type of diabetes they have. Talk with your health care professional about the type of diabetes you may have. They can help you understand its causes and treatments.

Sometimes, people find out they have type 1 diabetes when they have signs and symptoms of a condition called diabetic ketoacidosis (DKA) [External link](#). Symptoms of DKA may include

feeling very tired

having trouble breathing

having fruity-smelling breath

fainting from dehydration

having pain in your abdomen, nausea, or vomiting

If you have type 1 diabetes, your body doesn't have enough insulin to use blood glucose—also called blood sugar—for energy. Your body will begin to use fat as a source of energy instead. This process produces substances called ketones. High levels of ketones can cause DKA, which is a medical emergency that needs to be treated right away.

DKA usually affects people with type 1 diabetes. People with type 2 diabetes may also develop DKA if their body doesn't produce enough insulin.

Type 2 diabetes symptoms

Symptoms of type 2 diabetes can develop slowly, over several years. Many people with type 2 diabetes have no symptoms, or symptoms can be so mild that people might not even notice them. In addition to the common symptoms of diabetes listed above, people may find out they have type 2 diabetes when they develop symptoms of other health problems from diabetes. These include

pain, numbness, or tingling in the feet or hands

sexual problems

chest pain [NIH external link](#)

vision loss

An older adult receiving an eye exam.

Vision loss can be a symptom of diabetes.

Gestational diabetes symptoms

Gestational diabetes is a type of diabetes that may develop when you are pregnant. Most people with this type of diabetes have no symptoms. If you do have symptoms of gestational diabetes, they may be mild, such as being thirstier than usual or having to urinate more often.

People with a history of gestational diabetes are at higher risk of developing type 2 diabetes later in life. If you have a history of gestational diabetes, your health care professional may recommend regular testing for type 2 diabetes.

What causes type 1 diabetes?

Type 1 diabetes develops when the body's immune system [NIH external link](#) destroys the cells in the pancreas that make insulin. These cells are called beta cells. Genes and factors in the environment—the places where people live, play, work, study, and gather—may trigger the immune system to destroy beta cells in type 1 diabetes.

Insulin is a hormone that helps blood glucose get into the body's cells to be used as energy. When your body doesn't have enough insulin, glucose in your blood can't get into the cells. As a result, your cells lack energy, your blood glucose level rises, and you develop diabetes.

Studies such as [The Environmental Determinants of Diabetes in the Young \(TEDDY\) External link](#) are trying to learn more about the causes of type 1 diabetes. [Type 1 Diabetes TrialNet External link](#) and other groups are working to find treatments that may prevent or slow the development of type 1 diabetes in people who are at risk for getting the disease.

What causes type 2 diabetes?

Type 2 diabetes is the most common form of diabetes. Type 2 diabetes develops when your pancreas doesn't produce enough insulin, and your body has trouble using insulin, a condition called insulin resistance. Insulin resistance is when the cells in your muscles, fat, and liver don't respond well to insulin. When there isn't enough insulin to balance blood glucose levels, they begin to rise.

When blood glucose levels are higher than normal but not high enough to be diagnosed as type 2 diabetes, you have a condition called prediabetes. You have type 2 diabetes when health care professionals diagnose your blood glucose level as being too high.

Researchers don't fully know what causes problems with insulin production and insulin resistance in type 2 diabetes. However, people are more likely to develop type 2 diabetes if they

have overweight, obesity, or a large waist size.

are age 35 or older. Children and teens can also develop type 2 diabetes, but the risk increases as a person gets older.

have a family history of type 2 diabetes.

are African American, American Indian, Asian American, Hispanic, Latino, or Pacific Islander.

are not physically active, because of a job that requires sitting for long periods of time or because of difficulty moving their body.

smoke [External link](#), including tobacco products and e-cigarettes [External link](#), or are exposed to secondhand smoke [External link](#).

have prediabetes.

have a history of gestational diabetes, a type of diabetes that develops during pregnancy, or gave birth to a baby weighing 9 pounds or more.

You can't change some of these risk factors, such as your age, family history, or race. However, you may be able to change some risk factors and prevent or delay type 2 diabetes by maintaining a healthy weight, being physically active, and stopping smoking.

Father and daughter dancing together.

You may be able to prevent type 2 diabetes by maintaining a healthy weight and being physically active.

What causes gestational diabetes?

Gestational diabetes occurs when your body can't make the extra insulin that you need during pregnancy. Scientists believe that substances made in your body, called hormones, may play a role in causing this form of diabetes.

Changes to hormones during pregnancy can make it harder for blood glucose to enter the cells in your body. This causes insulin resistance. In most pregnancies, the pancreas can make enough insulin to overcome these changes. If the pancreas can't make enough insulin, you'll develop gestational diabetes.

You're more likely to develop gestational diabetes if you have certain risk factors, such as

having overweight or obesity

gaining too much weight during pregnancy

having a family history of gestational diabetes, type 2 diabetes, or both

What else can cause diabetes?

Other causes of diabetes may include

diseases caused by changes in your genes, also called gene variants

endocrine diseases

damage to or removal of the pancreas

taking certain medicines

Diseases caused by changes in your genes

Gene variants [NIH external link](#) are changes to certain genes. Some gene variants can cause diseases that can affect the pancreas and lead to diabetes.

Monogenic diabetes

Monogenic diabetes is a group of rare forms of diabetes caused by a variant, or change, in a single gene. Parents may pass these gene variants to their children. In some cases, a person may be the first one with the variant in the family. Most forms of monogenic diabetes reduce the amount of insulin your pancreas can make.

Cystic fibrosis

Cystic fibrosis [NIH external link](#) is a genetic disease that leads to swelling and scarring in the pancreas. This scarring can prevent the pancreas from making enough insulin.

Hemochromatosis

Hemochromatosis is a genetic condition that causes your body and organs to build up harmful levels of iron [NIH external link](#). If the disease is not treated, high levels of iron may damage your pancreas and other organs.

Endocrine diseases

Some endocrine diseases make the body produce too much of certain hormones. High levels of these hormones may cause insulin resistance or diabetes. Endocrine diseases that may cause insulin resistance or diabetes include

Cushing's syndrome, which can cause the body to produce too much cortisol—often called the “stress hormone”

acromegaly, which can cause the body to produce too much growth hormone

hyperthyroidism, which can cause the body to produce too much thyroid hormone

hypothyroidism, which can cause the body to produce too little thyroid hormone

Damage to or removal of the pancreas

The pancreas can be damaged by injury or diseases such as inflammation of the pancreas, also called pancreatitis. Diabetes may be the first sign of pancreatic cancer [NIH external link](#) in older people without diabetes risk factors who develop diabetes. A damaged pancreas may make less insulin and lead to diabetes. In some cases, your pancreas may need to be removed. Without a pancreas, your body can't produce insulin. You'll develop diabetes and will have to take insulin.

Taking certain medicines

Certain medicines may harm the pancreas or affect the way insulin works in the body. Some of these medicines may include

antipsychotics [NIH external link](#) that are used to treat certain mental health problems

antiseizure medicines that are used to treat epilepsy [NIH external link](#)

beta-blockers and diuretics [NIH external link](#) that are used to treat high blood pressure

glucocorticoids [NIH external link](#) that are used to treat certain inflammatory diseases

immunosuppressants that are used after an organ transplant

niacin [NIH external link](#), a type of vitamin B that is sometimes used to treat high blood cholesterol

Statins, which are medicines to reduce LDL (“bad”) cholesterol levels, can slightly increase the chance that you’ll develop diabetes. However, statins help protect you from heart disease [NIH external link](#) and stroke [NIH external link](#). For this reason, the strong benefits of taking statins outweigh the small chance that you could develop diabetes.

If you take medicines for these conditions, you may want to ask your primary health care professional if your medicines could affect your risk of developing diabetes.

Clinical Trials for Symptoms and Causes of Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including diabetes. The trials look to find new ways to prevent, detect, or treat diseases and improve quality of life.

What clinical trials study symptoms and causes of diabetes?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of symptoms and causes of diabetes, such as

tests and treatments for rare and uncommon types of diabetes

body changes in people who develop diabetes after having pancreatitis

Find out if clinical trials are right for you [NIH external link](#).

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Related Topics

English

Type 1 Diabetes

On this page:

What is type 1 diabetes?

How common is type 1 diabetes?

Who is more likely to have type 1 diabetes?

What health problems can people with type 1 diabetes develop?

What are the symptoms of type 1 diabetes?

What causes type 1 diabetes?

How do health care professionals diagnose type 1 diabetes?

How can I manage type 1 diabetes?

Can I delay or prevent type 1 diabetes in me or in my family members who may be at risk?

Clinical Trials for Type 1 Diabetes

What is type 1 diabetes?

Type 1 diabetes is a disease that occurs when your body makes little or no insulin. Insulin is a hormone that helps blood glucose, also called blood sugar, get into the body's cells to be used as energy. When your body doesn't have enough insulin, glucose in your blood can't get into the cells. As a result, your cells lack energy, your blood glucose level goes up, and you develop diabetes.

A problem with the body's immune system causes type 1 diabetes. Your immune system normally fights infections. Type 1 diabetes develops when the body's immune system destroys the cells in the pancreas that make insulin. When the immune system attacks healthy cells in your body, it's called autoimmunity. Type 1 diabetes is an autoimmune disease.

Most people with type 1 diabetes need to take insulin every day to manage their blood glucose level and stay alive. You may be able to prevent or delay health problems that can result from type 1 diabetes by learning to manage your blood glucose level, consuming healthy foods and drinks, and getting regular physical activity.

A young girl checking her blood glucose level with a blood glucose meter.

People with type 1 diabetes need to manage the disease to prevent health problems.

How common is type 1 diabetes?

In 2021, about 1.7 million U.S. adults ages 20 years or older—or 5.7% of adults with diagnosed diabetes—had type 1 diabetes and were taking insulin.¹ Overall, 5% to 10% of people with diabetes have type 1 diabetes.²

The number of young people with type 1 diabetes has been increasing over the last two decades. In 2021, about 304,000 children and teens younger than age 20 had type 1 diabetes.¹

Who is more likely to have type 1 diabetes?

Type 1 diabetes occurs more commonly in children and young adults, but it can develop at any age. Having a parent or sibling with type 1 diabetes increases your chance of developing the disease. People diagnosed with type 1 diabetes have the disease for the rest of their lives. Some people with type 1 diabetes can get a pancreatic islet or pancreas transplant and make insulin again. However, these transplants don't cure type 1 diabetes. People still need to take medicines called immunosuppressants to prevent the immune system from attacking and destroying the new islets or pancreas.

Type 1 diabetes is increasing among youth from certain racial and ethnic groups. However, new cases of type 1 diabetes are most common among White youth. Adults with diabetes who are younger than age 35 and have a healthy body weight are more likely to have type 1 diabetes than type 2 diabetes.²

What health problems can people with type 1 diabetes develop?

People with type 1 diabetes can develop the same diabetes health problems as people with other types of diabetes. High blood glucose levels from diabetes can damage many parts of your body, including your

heart

kidneys

nerves

eyes

feet

teeth

People with diabetes may also have high blood pressure [NIH external link](#) or cholesterol [NIH external link](#) problems. Diabetes is also linked to some types of cancer [NIH external link](#).

If you have type 1 diabetes, managing your diabetes and having a healthy lifestyle may lower your chances of developing health problems.

Diabetic ketoacidosis

People with type 1 diabetes also have a higher risk of developing a condition called diabetic ketoacidosis (DKA) [External link](#). When you have type 1 diabetes, your body doesn't have enough insulin to use blood glucose for energy. Your body will begin to use fat as a source of energy instead. This process produces substances called ketones. High levels of ketones can cause DKA. People have a higher risk of developing DKA when they have an infection or when they're under stress [External link](#). DKA is a medical emergency that needs to be treated right away.

Hypoglycemia unawareness

Some people develop a serious condition called hypoglycemia unawareness. People with this condition don't feel their blood glucose level dropping, so they aren't aware they should treat low blood glucose, also called hypoglycemia.

If you have type 1 diabetes, you'll need to carefully manage the amount of insulin you take every day to keep your blood glucose level in a healthy range. If you take too much insulin, you may develop low blood glucose.

Hypoglycemia unawareness is a medical emergency that needs to be treated right away. That's because hypoglycemia unawareness can cause problems with thinking and other brain functions and, if not treated, may lead to a coma.

Mental health conditions

The stress of managing type 1 diabetes can make some people more likely to develop sleep problems [NIH external link](#), anxiety [NIH external link](#), or depression [NIH external link](#). You may find it helpful to speak with a counselor [NIH external link](#) or your primary health care professional about your mental health. Your primary health care professional may be a doctor, physician assistant, or nurse practitioner.

Other health problems

People with type 1 diabetes may also have a higher risk of other health problems, including

eating disorders, such as anorexia [External link](#), bulimia [External link](#), and binge eating disorder

other autoimmune diseases [NIH external link](#)

hip fractures in older adults due to osteoporosis [NIH external link](#)

If you're pregnant and have diabetes, changes to your hormones may affect how your body uses insulin. Ask your health care professional if you should adjust your medicines during pregnancy to prevent health problems for you and your baby.

What are the symptoms of type 1 diabetes?

Symptoms of type 1 diabetes may include

increased urination

feeling very thirsty

feeling very hungry, even after you've eaten

blurred vision

fatigue

sores that don't heal

frequent infections, such as urinary tract infections, skin infections [NIH external link](#), or yeast infections [NIH external link](#)

unexplained weight loss

Symptoms of type 1 diabetes usually develop quickly, over a few days or weeks. Children with type 1 diabetes typically develop symptoms over a short period of time. Common symptoms in children include

having to urinate more often

eating more but losing weight

feeling very thirsty

A young boy drinking water from a fountain outdoors.

Feeling thirsty is a common symptom of type 1 diabetes in children.

These symptoms are less common in adults with type 1 diabetes. For some adults, symptoms of type 1 diabetes may develop slowly and seem more like type 2 diabetes. Some adults who develop diabetes may need special tests to find out what type of diabetes they have.

The first symptoms of type 1 diabetes can be symptoms of DKA, especially in children. Symptoms of DKA include

feeling very tired

having trouble breathing

having fruity-smelling breath

fainting from dehydration

having pain in your abdomen, nausea, or vomiting

What causes type 1 diabetes?

Type 1 diabetes develops when the body's immune system destroys the cells in the pancreas that make insulin. These cells are called beta cells. Genes and factors in the environment—the place where people live, play, work, study, and gather—may trigger the immune system to destroy beta cells in type 1 diabetes.

Studies such as The Environmental Determinants of Diabetes in the Young (TEDDY) [External link](#) are trying to learn more about the causes of type 1 diabetes. Type 1 Diabetes TrialNet [External link](#) and other groups are working to find treatments that may prevent or slow the development of type 1 diabetes in people who are at risk for getting the disease.

How do health care professionals diagnose type 1 diabetes?

Your health care professional can diagnose diabetes by reviewing your symptoms, asking about your medical history, and using diabetes tests. Your health care professional may also have you take diabetes tests to find out if your diabetes is type 1, type 2, or another type of diabetes. Your treatment will depend on the type of diabetes you have.

Health care professionals often test for type 1 diabetes in people who have common diabetes symptoms. Since type 1 diabetes can run in families, your health care professional may test you for type 1 diabetes if you have no symptoms but have a parent or sibling with type 1 diabetes.

Blood glucose tests

Health care professionals can diagnose diabetes using blood tests that measure your blood glucose level to see if it's higher than normal. Health care professionals may use

the random plasma glucose test to diagnose diabetes in people who have diabetes symptoms

the fasting plasma glucose test

the A1C test

The random plasma glucose test and fasting plasma glucose test measure your blood glucose level at the time your blood is drawn. The A1C test shows your average blood glucose level over the past 3 months.

Although these blood glucose tests can help your health care professional tell if you have diabetes, the tests don't show the type of diabetes you have.

Autoantibodies test

To find out if your diabetes is type 1, your health care professional may test your blood for certain autoantibodies.

Autoantibodies and type 1 diabetes

Autoantibodies are proteins made by your immune system that attack healthy tissues and cells by mistake. If you have certain autoantibodies in your blood, you may have type 1 diabetes.

Autoantibodies and latent autoimmune diabetes in adults

Some adults with diabetes have autoantibodies in their blood but develop symptoms slower than most people with type 1 diabetes. In this type of diabetes, your immune system damages your pancreas more slowly. You may not need to take insulin for several months after being diagnosed. This type of diabetes is sometimes called latent autoimmune diabetes in adults (LADA).

Autoantibodies and family members of people with type 1 diabetes

If you have a parent or sibling with type 1 diabetes, your health care professional may suggest a test for autoantibodies, even if you don't have any symptoms. The test may show if you have a higher risk of developing symptoms of type 1 diabetes and if you should be treated to delay the disease. Talk with your health care professional about the symptoms of diabetes and ask if they want you to get more testing in the future.

TrialNet is a research network that offers autoantibody testing [External link to some family members of people with type 1 diabetes](#). Some people at risk of developing symptoms of type 1 diabetes may be able to participate in TrialNet studies.

Other tests

If it's not clear what type of diabetes you have, your health care professional may ask you to do

a C-peptide test [NIH external link to see how much insulin your pancreas makes](#)

genetic testing [NIH external link to rule out less common types of diabetes, such as monogenic diabetes](#)

How can I manage type 1 diabetes?

If you have type 1 diabetes, you'll need to take insulin to manage your blood glucose level. Using devices to check your blood glucose level and inject insulin may help keep your blood glucose level in a healthy range. Your health care professional may also prescribe other diabetes medicines.

Managing your type 1 diabetes may help prevent or delay diabetes health problems.

Having a healthy lifestyle can help manage your type 1 diabetes and prevent other health problems. With the help of your health care team, you can create a diabetes care plan that works for you.

Diabetes medicines, technologies, and office visits can be expensive. Tell your health care professional or health care team if you're having trouble affording diabetes care. A health care professional, diabetes educator, or social worker may be able to tell you about resources that provide financial help for diabetes care.

Insulin and other diabetes medicines

If you have type 1 diabetes, you need to take insulin to stay alive. Different types of insulin are available. Some types start to work quickly while others start to work more slowly. The effects of

each type of insulin last for varying lengths of time. You may need to take more than one type of insulin. You can take insulin in a few ways, including

needle and syringe

insulin pen

insulin pump

If you take insulin, you need to check your blood glucose level regularly. You should match your insulin dose with your food, physical activity, and current blood glucose level. If you don't, your blood glucose level may drop too low, which is called hypoglycemia. Hypoglycemia is a medical emergency that needs to be treated right away.

Some people have trouble managing their blood glucose level with just insulin. Your health care professional may recommend other medicines to take with insulin to help manage your blood glucose level.

Diabetes devices and technology

Diabetes devices and technology can help people with type 1 diabetes take insulin and check their blood glucose level.

Insulin pump

An insulin pump is a small, portable machine that gives you frequent, small doses of insulin using a plastic tube under the skin. The pump is refilled with insulin every 2 to 3 days. Insulin pumps can be programmed to deliver insulin directly into your body 24 hours a day or at certain times.

Blood glucose meter

People with diabetes can check their blood glucose level with a blood glucose meter. After drawing a small amount of blood from your finger, you can use the meter to measure how much glucose is in your blood at that moment.

Continuous glucose monitor

A continuous glucose monitor (CGM) is a device that automatically tracks your blood glucose level throughout the day and night. Researchers have found that using a CGM at home may help you manage your blood glucose level and reduce your risk of developing health problems from diabetes.

CGMs use sensors placed under your skin that estimate your blood glucose level. You can see information about your blood glucose level at any time in a software program stored on a smartphone, on an insulin pump, or on a separate device called a receiver. These results may help you and your health care team make changes to your meal plan, physical activity level, or diabetes medicines. Many CGM systems have alarms and warnings that let you know if your blood glucose level is too low or too high. For safety, it's important to act quickly if a CGM alarm sounds.

A woman sits on a couch and checks her blood glucose level with a CGM and mobile phone.

Diabetes technologies such as CGMs can help you keep your blood glucose level in a healthy range.

Artificial pancreas

An artificial pancreas is a system that mimics how a healthy pancreas controls blood glucose levels in the body. The system is made up of three parts. A CGM automatically estimates your blood glucose level and sends the information, without using wires, to a software program on your smartphone or on an insulin pump. The software program calculates how much insulin your body needs, and the insulin pump automatically delivers that amount of insulin to your body.

Researchers have found that artificial pancreas systems may control blood glucose levels better than other methods in adults and children as young as age 2 years. This is especially true at night, when managing blood glucose levels can be hard for many people with type 1 diabetes.

Ask your health care professional which diabetes devices will work for you.

Pancreatic islet and pancreas transplant

Pancreatic islet transplant

A pancreatic islet transplant is used to treat type 1 diabetes in people who struggle to manage their blood glucose levels. Pancreatic islets are clusters of cells in the pancreas that make insulin. In type 1 diabetes, the body's immune system attacks these cells. A pancreatic islet transplant replaces destroyed islets with new islets from a deceased donor. The new islets make and release insulin. After a pancreatic islet transplant, you'll take immunosuppressants to prevent the immune system from attacking and destroying the new islets.

Pancreas transplant

Health care professionals may consider a pancreas transplant [NIH external link](#) for people with type 1 diabetes who need a kidney transplant or who haven't been able to manage their blood glucose levels in any other way. A pancreas transplant may restore normal blood glucose levels and prevent other diabetes-related health problems. After a pancreas transplant, you'll take immunosuppressants to prevent the immune system from attacking and destroying the new pancreas.

Can I delay or prevent type 1 diabetes in me or in my family members who may be at risk?

Teplizumab [NIH external link](#) is a medicine that can delay type 1 diabetes symptoms from developing and delay the need for insulin injections. If you or a family member has two or more autoantibodies for type 1 diabetes and increasing blood glucose levels, your health care professional may recommend this medicine.

Through studies such as those conducted by TrialNet, researchers are working to identify more risk factors for type 1 diabetes and find new ways to delay and prevent the disease.

Clinical Trials for Type 1 Diabetes

NIDDK conducts and supports clinical trials in many diseases and conditions, including type 1 diabetes. The trials look to find new ways to prevent, detect, or treat disease and improve quality of life.

What are clinical trials for type 1 diabetes?

Clinical trials—and other types of clinical studies [NIH external link](#)—are part of medical research and involve people like you. When you volunteer to take part in a clinical study, you help doctors and researchers learn more about disease and improve health care for people in the future.

Researchers are studying many aspects of type 1 diabetes, such as

improving the safety and success of pancreatic islet transplantation

reducing the immune system response that destroys cells that produce insulin

looking at insulin resistance in adolescents with type 1 diabetes

Find out if clinical trials are right for you.

Related Topics

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How can I lower my chances of developing type 2 diabetes?

What is type 2 diabetes?

Type 2 diabetes, the most common type of diabetes, is a disease that occurs when your blood glucose, also called blood sugar, is too high. Blood glucose is your main source of energy and comes mainly from the food you eat. Insulin, a hormone made by the pancreas, helps glucose get into your cells to be used for energy. In type 2 diabetes, your body doesn't make enough insulin or doesn't use insulin well. Too much glucose then stays in your blood, and not enough reaches your cells.

The good news is that you can take steps to prevent or delay the development of type 2 diabetes.

Who is more likely to develop type 2 diabetes?

You can develop type 2 diabetes at any age, even during childhood. However, type 2 diabetes occurs most often in middle-aged and older people. You are more likely to develop type 2 diabetes if you are age 45 or older, have a family history of diabetes, or are overweight or have obesity. Diabetes is more common in people who are African American, Hispanic/Latino, American Indian, Asian American, or Pacific Islander.

Physical inactivity and certain health problems such as high blood pressure affect your chances of developing type 2 diabetes. You are also more likely to develop type 2 diabetes if you have prediabetes or had gestational diabetes when you were pregnant. Learn more about risk factors for type 2 diabetes.

A smiling overweight older man

Type 2 diabetes occurs most often in middle-aged and older people.

What are the symptoms of diabetes?

Symptoms of diabetes include

increased thirst and urination

increased hunger

feeling tired

blurred vision

numbness or tingling in the feet or hands

sores that do not heal

unexplained weight loss

Symptoms of type 2 diabetes often develop slowly—over the course of several years—and can be so mild that you might not even notice them. Many people have no symptoms. Some people do not find out they have the disease until they have diabetes-related health problems, such as blurred vision or heart disease.

What causes type 2 diabetes?

Type 2 diabetes is caused by several factors, including

overweight and obesity

not being physically active

insulin resistance

genes

Learn more about the causes of type 2 diabetes.

How do health care professionals diagnose type 2 diabetes?

Your health care professional can diagnose type 2 diabetes based on blood tests. Learn more about blood tests for diabetes and what the results mean.

How can I manage my type 2 diabetes?

Managing your blood glucose, blood pressure, and cholesterol, and quitting smoking if you smoke, are important ways to manage your type 2 diabetes. Lifestyle changes that include planning healthy meals, limiting calories if you are overweight, and being physically active are

also part of managing your diabetes. So is taking any prescribed medicines. Work with your health care team to create a diabetes care plan that works for you.

A smiling young woman chopping vegetables in her kitchen

Following your meal plan helps you manage your diabetes.

What medicines do I need to treat my type 2 diabetes?

Along with following your diabetes care plan, you may need diabetes medicines, which may include pills or medicines you inject under your skin, such as insulin. Over time, you may need more than one diabetes medicine to manage your blood glucose. Even if you don't take insulin, you may need it at special times, such as during pregnancy or if you are in the hospital. You also may need medicines for high blood pressure, high cholesterol, or other conditions.

Learn more about medicines, insulin, and other diabetes treatments.

What health problems can people with diabetes develop?

Following a good diabetes care plan can help protect against many diabetes-related health problems. However, if not managed, diabetes can lead to problems such as

heart disease and stroke

nerve damage

kidney disease

foot problems

eye disease

gum disease and other dental problems

sexual and bladder problems

Many people with type 2 diabetes also have nonalcoholic fatty liver disease (NAFLD). Losing weight if you are overweight or have obesity can improve NAFLD. Diabetes is also linked to other health problems such as sleep apnea, depression, some types of cancer, and dementia [NIH external link](#).

You can take steps to lower your chances of developing these diabetes-related health problems.

Related Topics

English

What Is Diabetes?

Diabetes is a disease that occurs when your blood glucose, also called blood sugar, is too high. Glucose is your body's main source of energy. Your body can make glucose, but glucose also comes from the food you eat.

Insulin is a hormone made by the pancreas that helps glucose get into your cells to be used for energy. If you have diabetes, your body doesn't make enough—or any—insulin, or doesn't use insulin properly. Glucose then stays in your blood and doesn't reach your cells.

Diabetes raises the risk for damage to the eyes, kidneys, nerves, and heart. Diabetes is also linked to some types of cancer. Taking steps to prevent or manage diabetes may lower your risk of developing diabetes health problems.

View full-sized imageOn the left, a diagram of a blood vessel that has a normal blood glucose level and contains fewer glucose molecules. On the right, a diagram of a blood vessel that has a high blood glucose level and contains more glucose molecules.

Diabetes is a disease that occurs when your blood glucose, also called blood sugar, is too high.

What are the different types of diabetes?

The most common types of diabetes are type 1, type 2, and gestational diabetes.

Type 1 diabetes

If you have type 1 diabetes, your body makes little or no insulin. Your immune system attacks and destroys the cells in your pancreas that make insulin. Type 1 diabetes is usually diagnosed in children and young adults, although it can appear at any age. People with type 1 diabetes need to take insulin every day to stay alive.

Type 2 diabetes

If you have type 2 diabetes, the cells in your body don't use insulin properly. The pancreas may be making insulin but is not making enough insulin to keep your blood glucose level in the normal range. Type 2 diabetes is the most common type of diabetes. You are more likely to develop type 2 diabetes if you have risk factors, such as overweight or obesity, and a family history of the disease. You can develop type 2 diabetes at any age, even during childhood.

You can help delay or prevent type 2 diabetes by knowing the risk factors and taking steps toward a healthier lifestyle, such as losing weight or preventing weight gain.

Gestational diabetes

Gestational diabetes is a type of diabetes that develops during pregnancy. Most of the time, this type of diabetes goes away after the baby is born. However, if you've had gestational diabetes, you have a higher chance of developing type 2 diabetes later in life. Sometimes diabetes diagnosed during pregnancy is type 2 diabetes.

Prediabetes

People with prediabetes have blood glucose levels that are higher than normal but not high enough to be diagnosed with type 2 diabetes. If you have prediabetes, you have a higher risk of developing type 2 diabetes in the future. You also have a higher risk for heart disease than people with normal glucose levels.

Other types of diabetes

A less common type of diabetes, called monogenic diabetes, is caused by a change in a single gene. Diabetes can also come from having surgery to remove the pancreas, or from damage to the pancreas due to conditions such as cystic fibrosis [NIH external link](#) or pancreatitis.

How common are diabetes and prediabetes?

More than 133 million Americans have diabetes or prediabetes.¹

As of 2019, 37.3 million people—or 11.3% of the U.S. population—had diabetes.¹ More than 1 in 4 people over the age of 65 had diabetes. Nearly 1 in 4 adults with diabetes didn't know they had the disease.²

About 90% to 95% of diabetes cases are type 2 diabetes.³

In 2019, 96 million adults—38% of U.S. adults—had prediabetes.⁴

What other health problems can people with diabetes develop?

Over time, high blood glucose can damage your heart, kidneys, feet, and eyes. If you have diabetes, you can take steps to lower your chances of developing diabetes health problems by taking steps to improve your health and learning how to manage the disease. Managing your blood glucose, blood pressure, and cholesterol levels can help prevent future health problems.

Doctor using a special device to check the inside of a patient's eye.

Managing your blood glucose levels can help prevent future health problems, such as damage to your eyes.